IMAGE FORMATION & DIGITAL PHOTOGRAPHY

Computational Imaging – Spring 2024

Ana Serrano, Albert Redó, Julio Marco

Master in Robotics, Graphics and Computer Vision – Universidad de Zaragoza

Slide credits: Julio Marco

Slide credits

Many of these slides are based on the following content:

- **Computational Photography (15-463). Carnegie Mellon University.**
- Computational aspects of digital photography (CS 89/189) by Wojciech Jarosz, Dartmouth College.

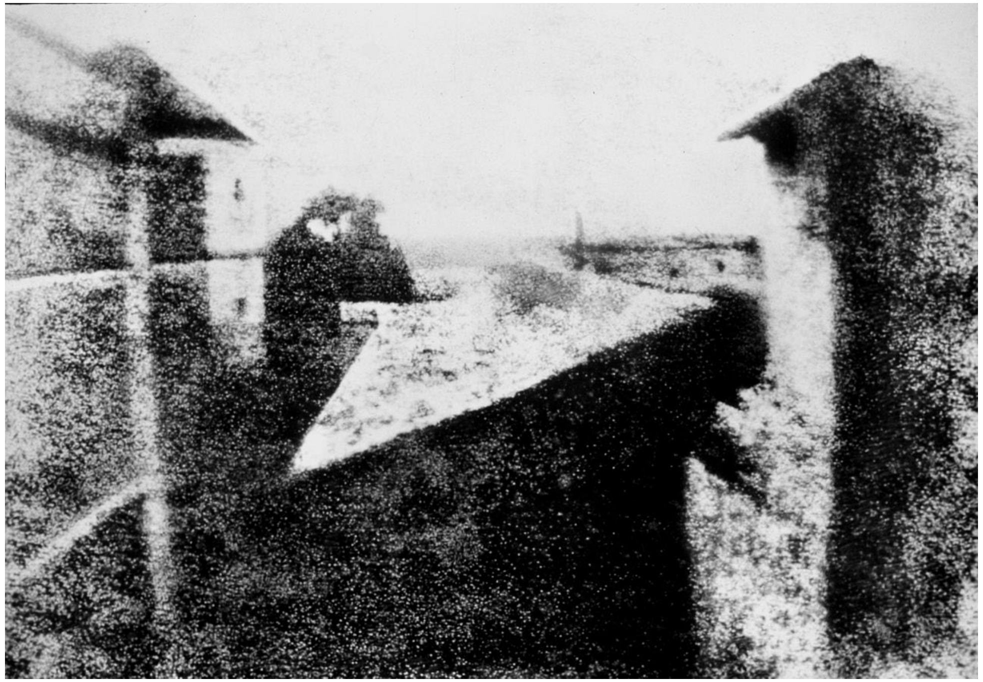
Image formation pipeline

What is an image?

Sources: CMU 15-463, Dartmouth College CS89/189.

What is an image?

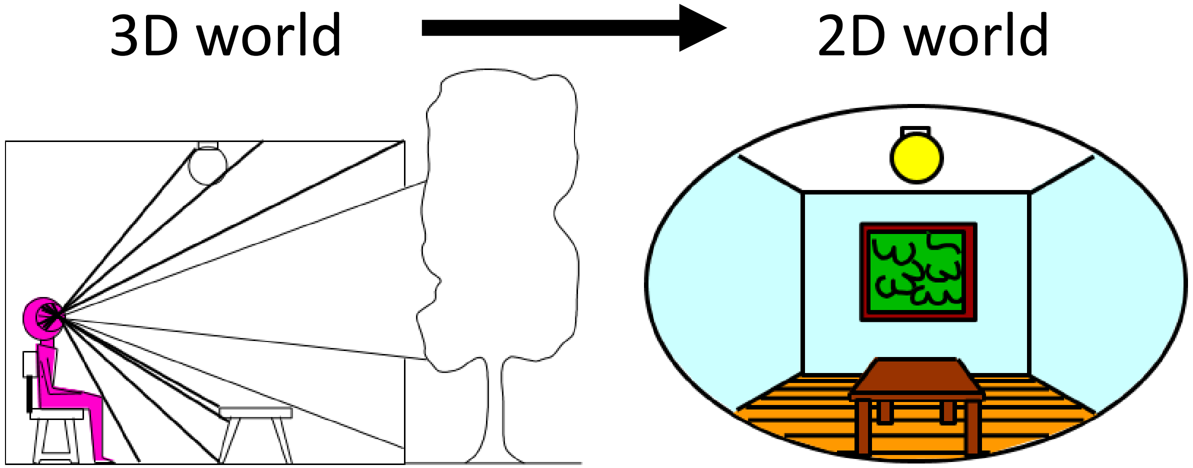
An image is a 2D realization of a 3D world:



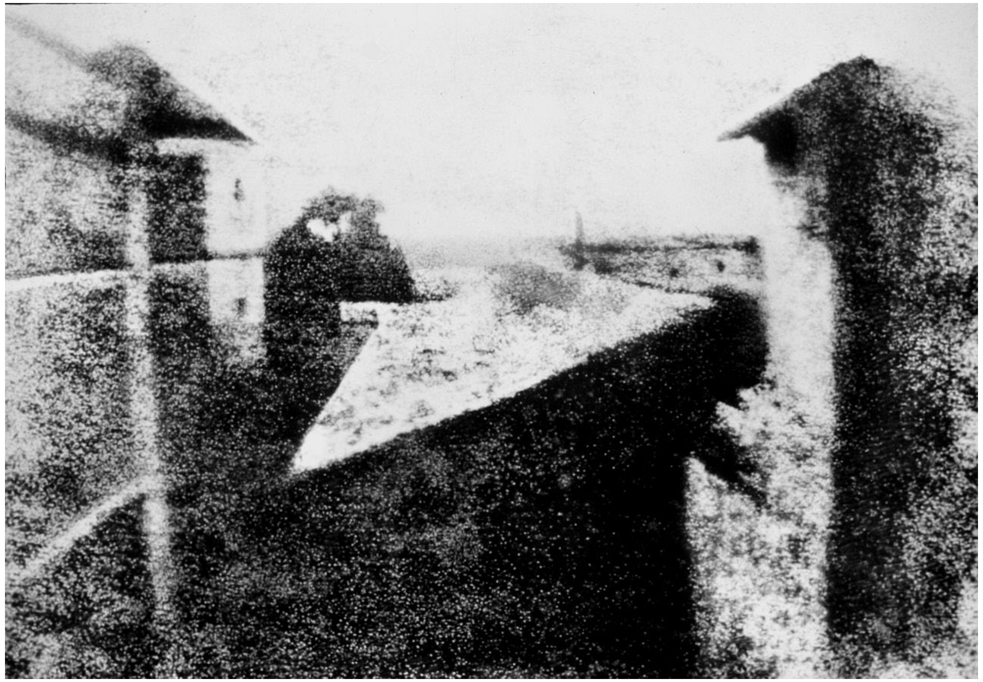
View from the Window at Le Gras (1826)

What is an image?

An image is a 2D realization of a 3D world:



What do we lose?



View from the Window at Le Gras (1826)

What is an image?

An image is a 2D realization of a 3D world: What do we lose?

Luminance :: HDR Imaging

Direction :: LightFields

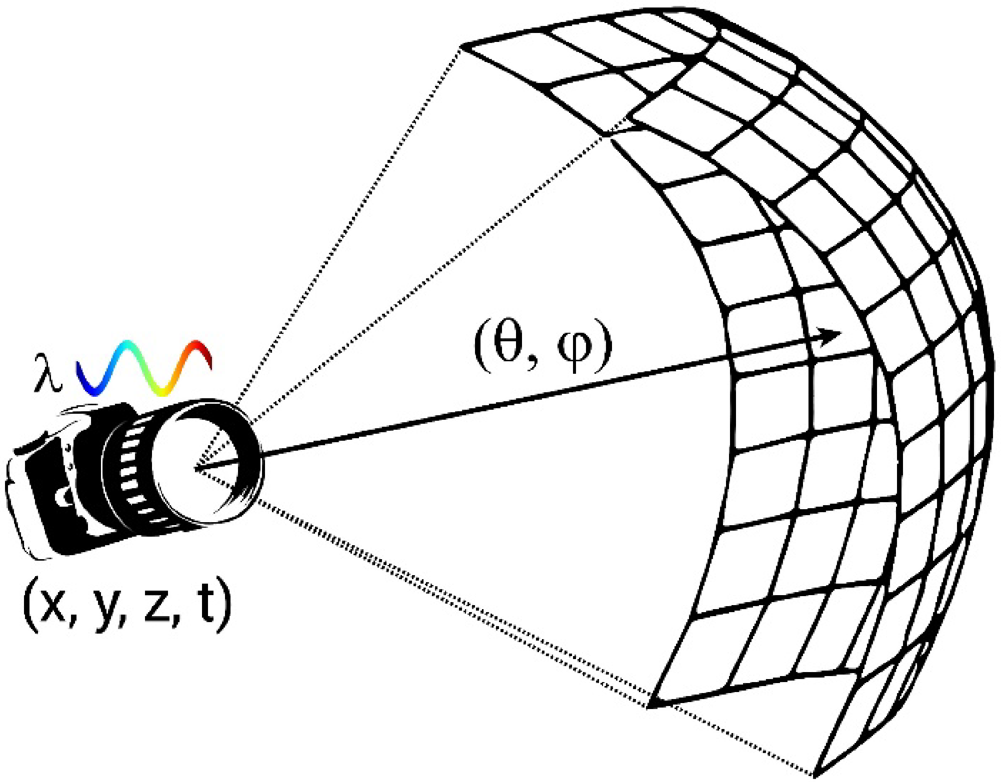
Time :: Video

Wavelength (Color) :: Multi-Sprectral Imaging

And more….

What is an image?

An image is a 2D realization of a 3D world: ***The Plenoptic function***



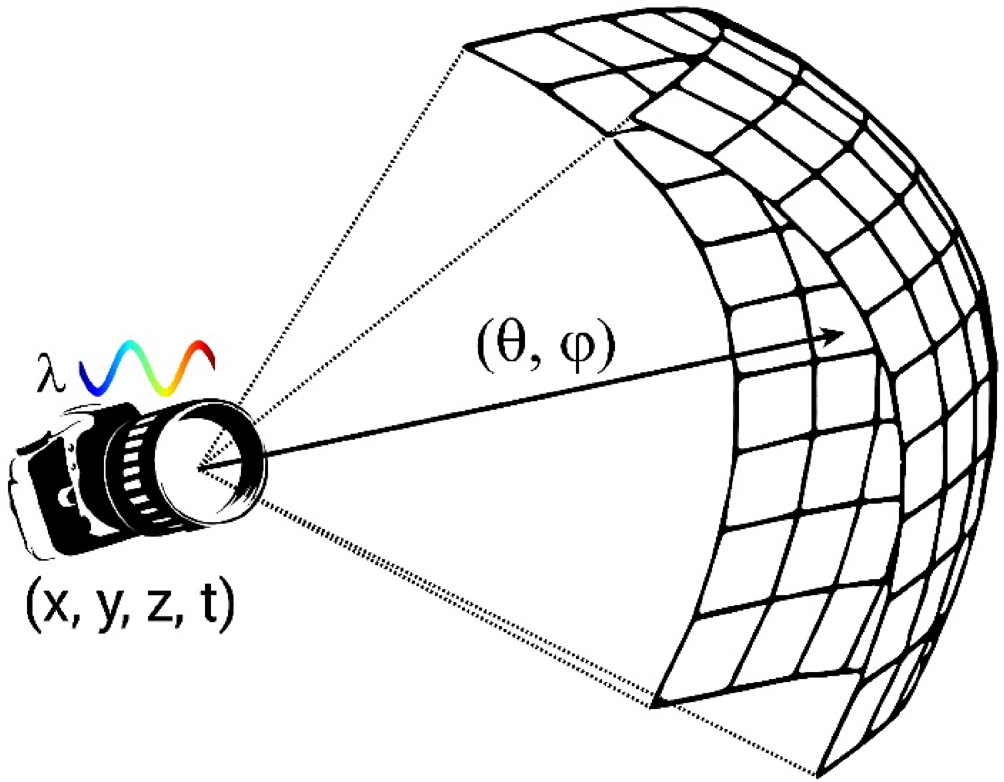
$$L = f(x, y, z, \theta, \varphi, \lambda, t)$$

- Position (x, y, z)
- Direction (θ, φ)
- Wavelenght (λ)
- Time (t)

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

What is an image?

An image is a 2D realization of a 3D world: ***The Plenoptic function***



$$L = f\,(x, y, z, \theta, \varphi, \lambda, t)$$

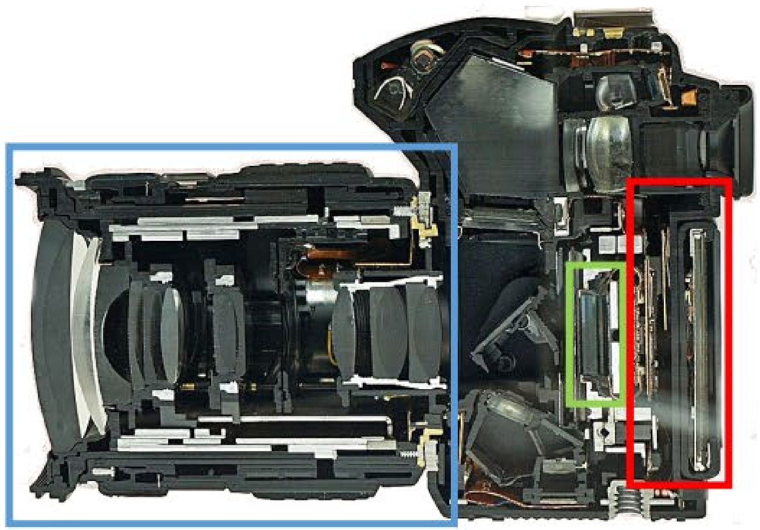
Conventional photograph

- Position (x, y, z)
- Direction (θ, φ)
- Wavelenght (λ)
- Time (t)

Sources: CMU 15-463, Dartmouth College CS89/189.

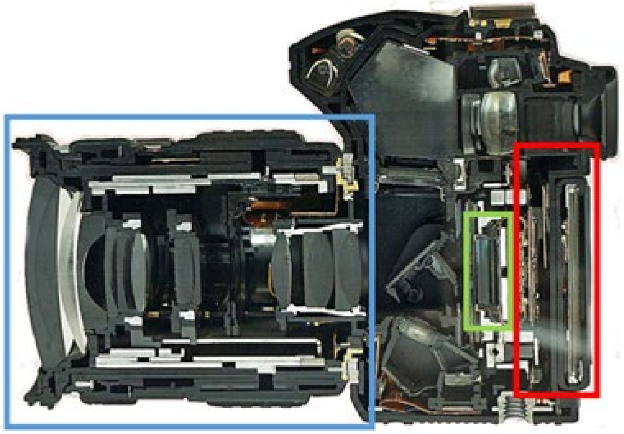
The modern photography pipeline

post-capture processing
(lectures 5-10)



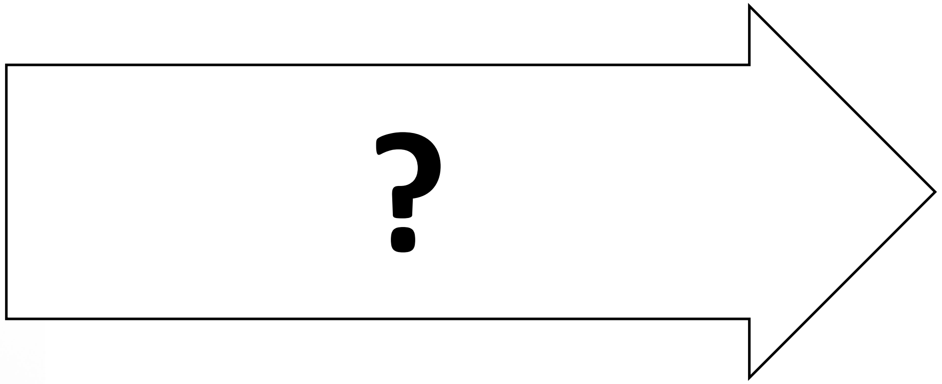
optics and optical controls → sensor, analog front-end, and color filter array → in-camera image processing pipeline

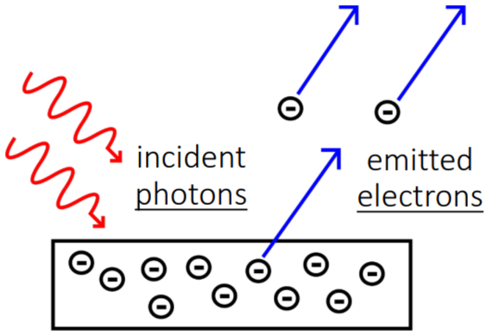
Sources: CMU 15-463, Dartmouth College CS89/189.

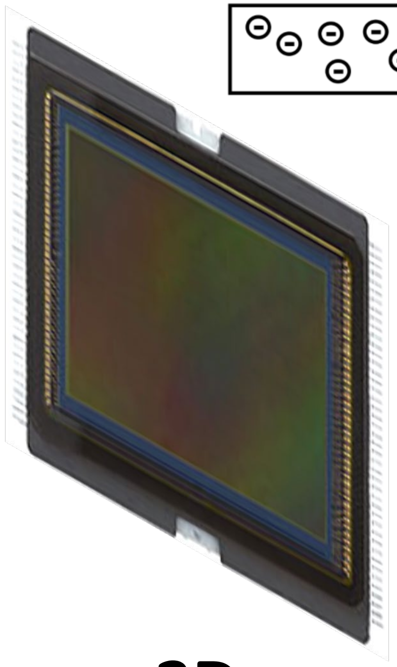


optics and optical controls

3D







2D

Let’s say we have a sensor…

digital sensor
(CCD or CMOS)

… and an object we’d like to photograph

real-world
object

digital sensor
(CCD or CMOS)

What would an image taken like this look like?

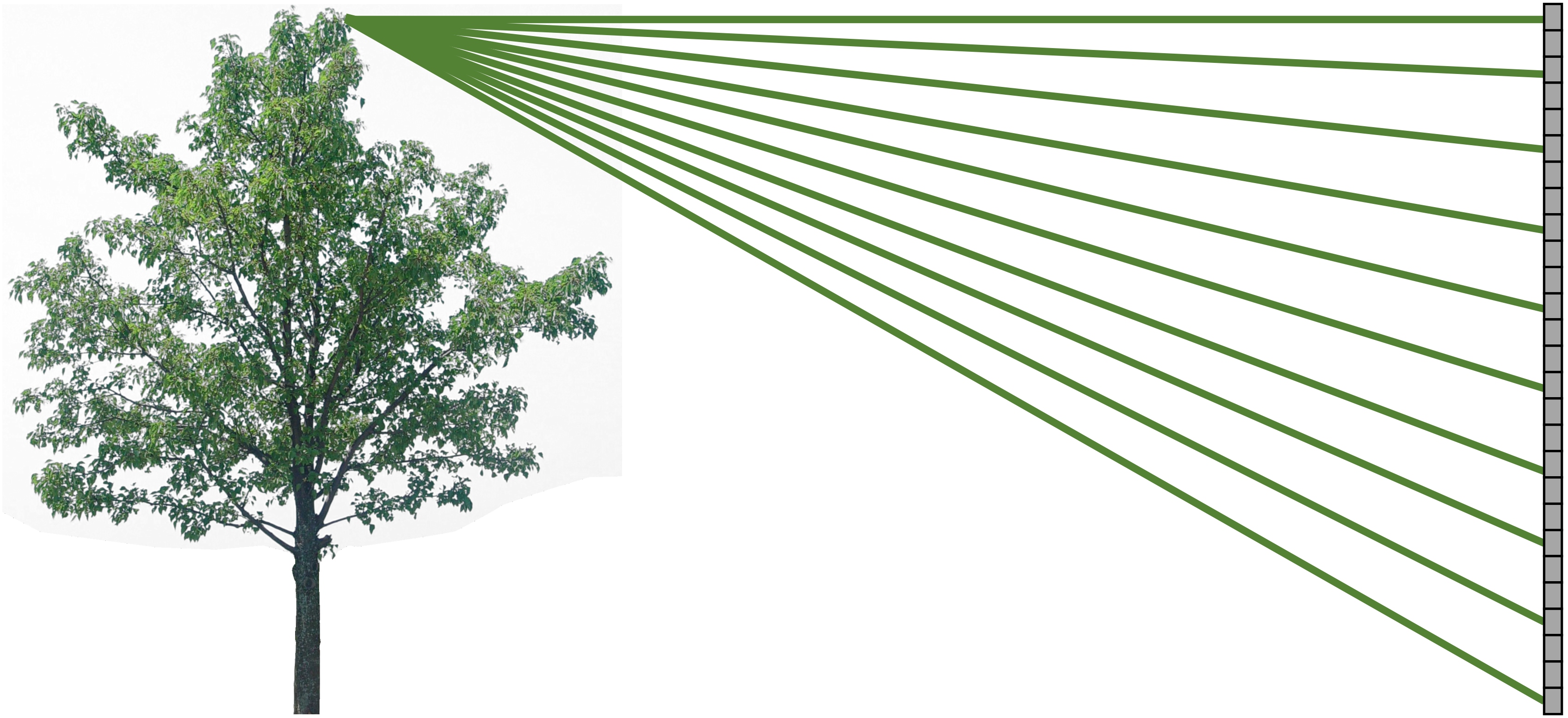
Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Bare-sensor imaging

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

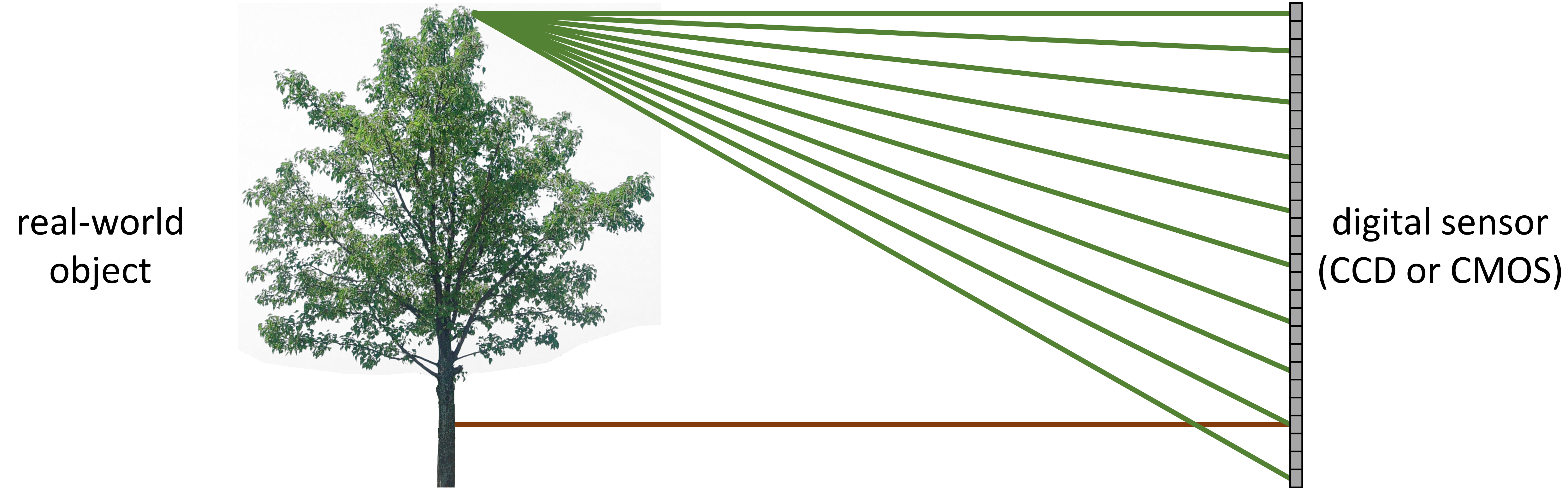
Bare-sensor imaging

real-world object



digital sensor (CCD or CMOS)

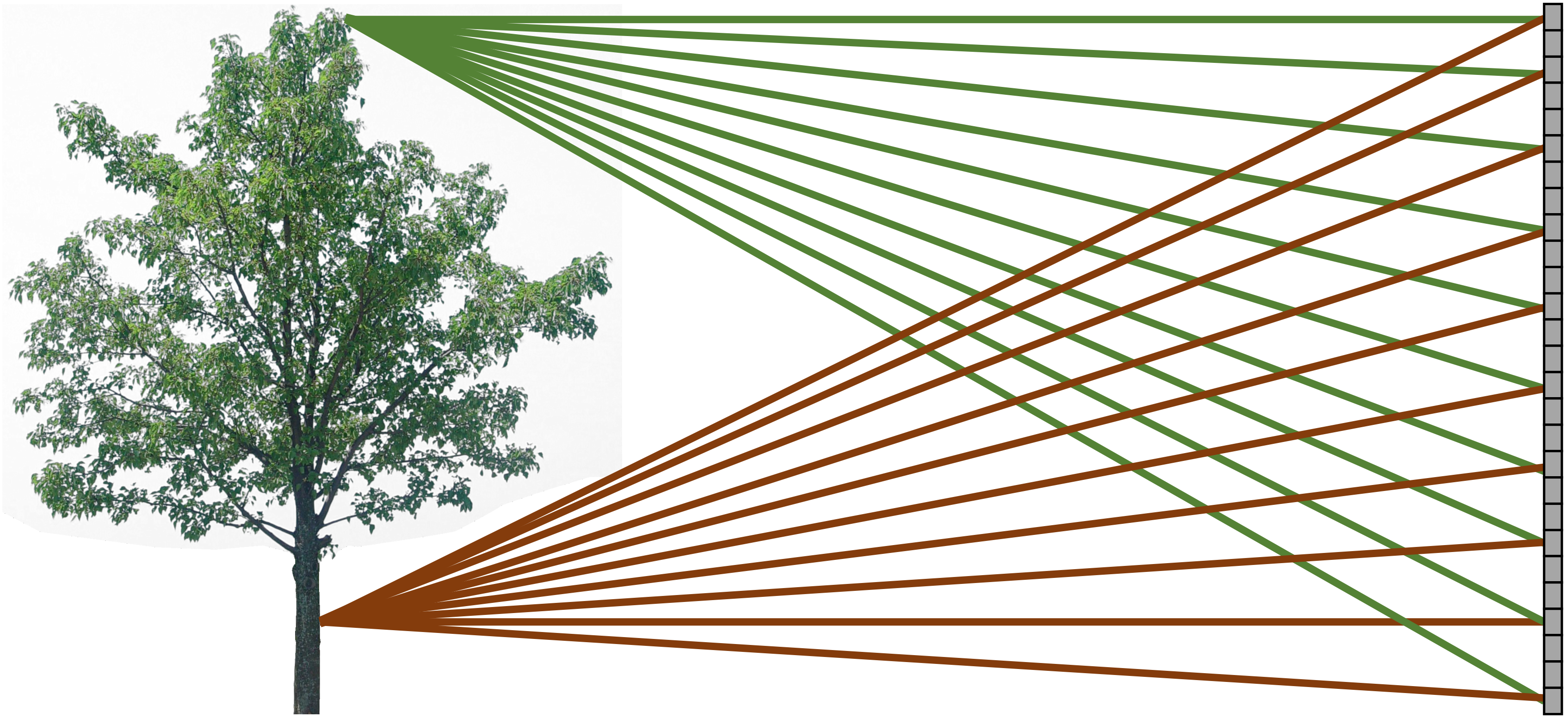
Bare-sensor imaging



Sources: CMU 15-463, Dartmouth College CS89/189.

Bare-sensor imaging

real-world
object



digital sensor
(CCD or CMOS)

All scene points contribute to all sensor pixels

What does the image on
the sensor look like?

Bare-sensor imaging

All scene points contribute to all sensor pixels

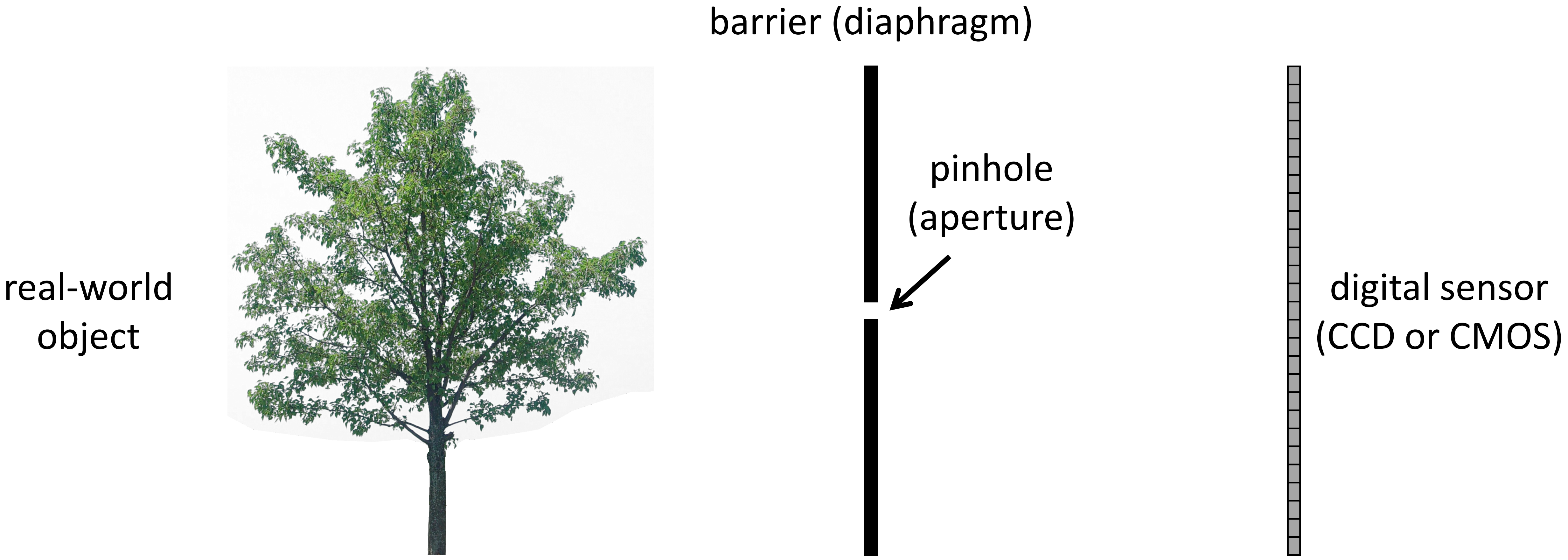
What can we do to make our image look better?

real-world object

digital sensor (CCD or CMOS)

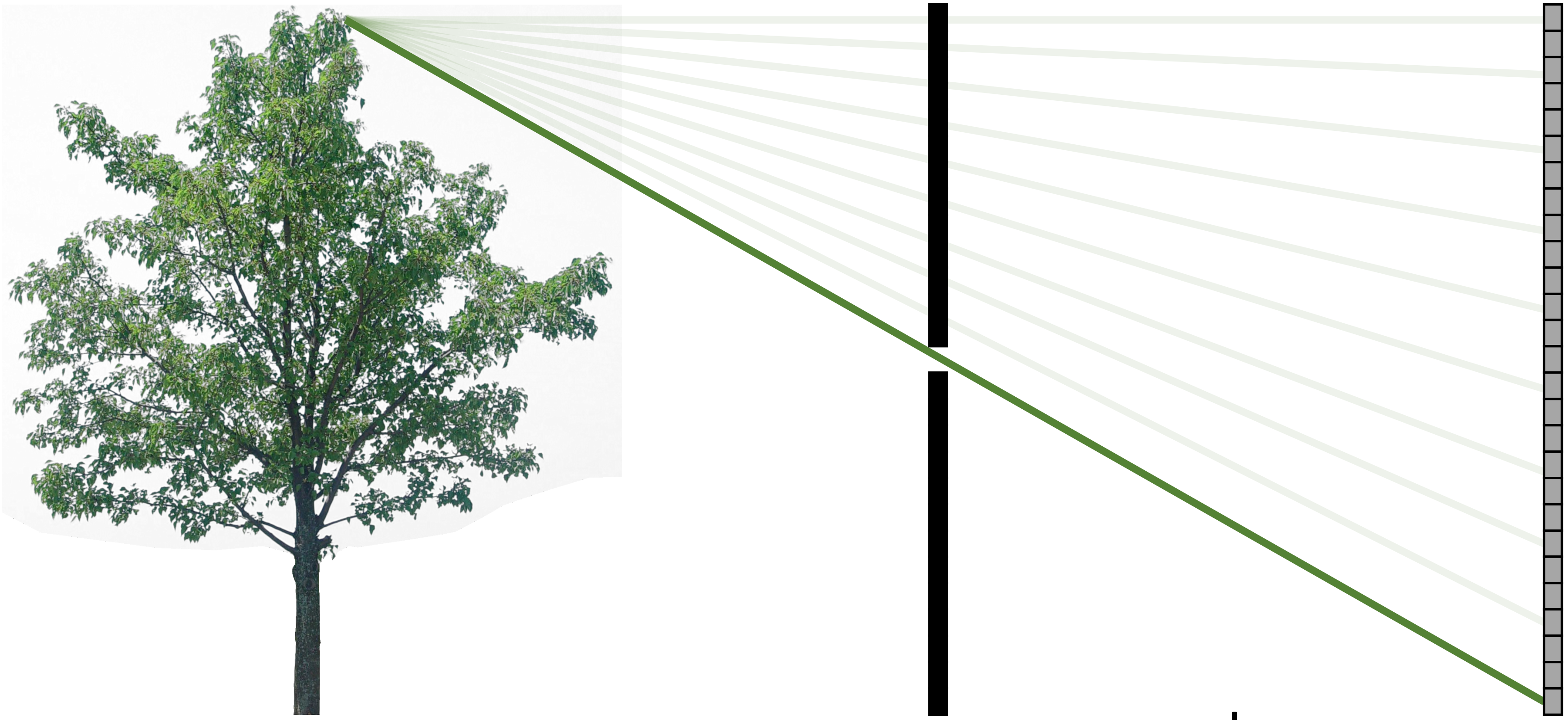
Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Let's add something to this scene



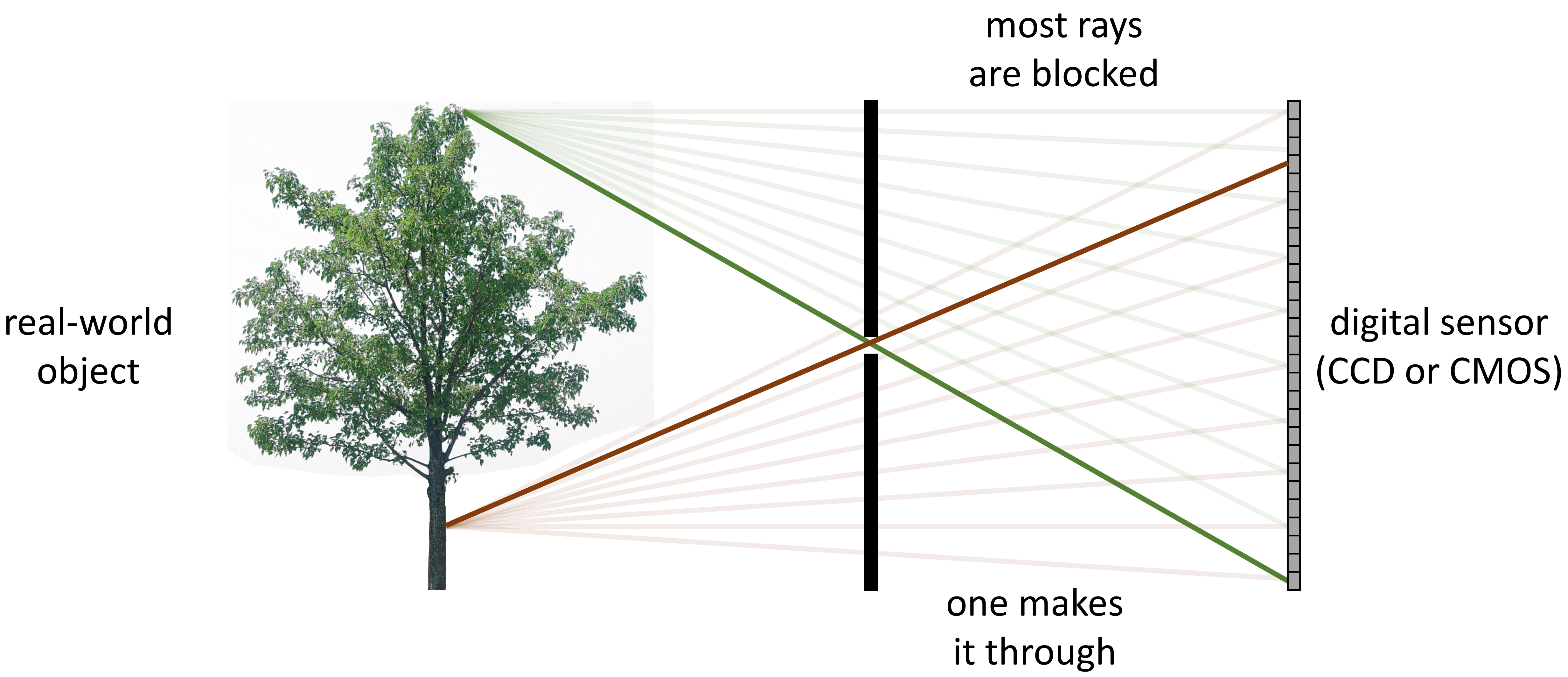
What would an image taken like this look like?

Pinhole imaging



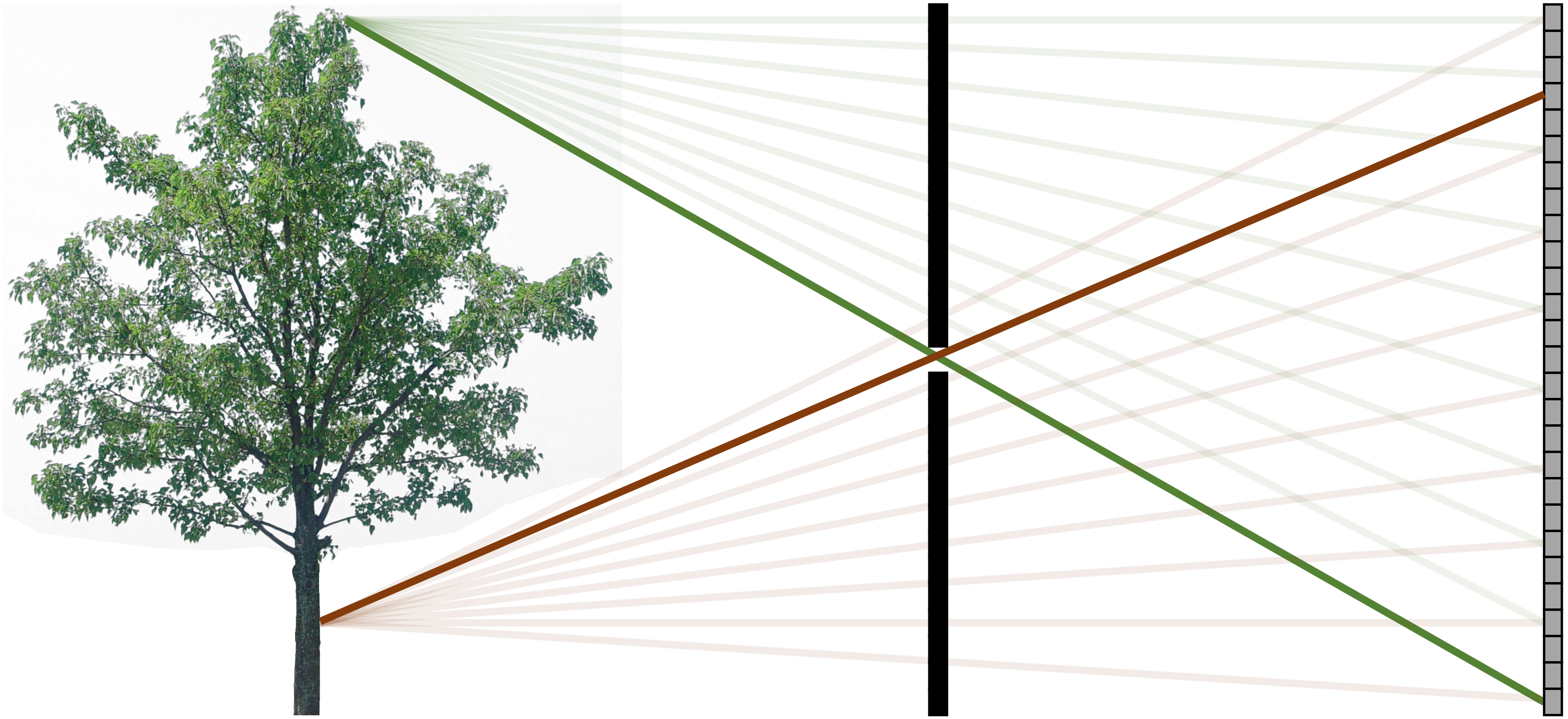
Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Pinhole imaging



Pinhole imaging

real-world
object



digital sensor
(CCD or CMOS)

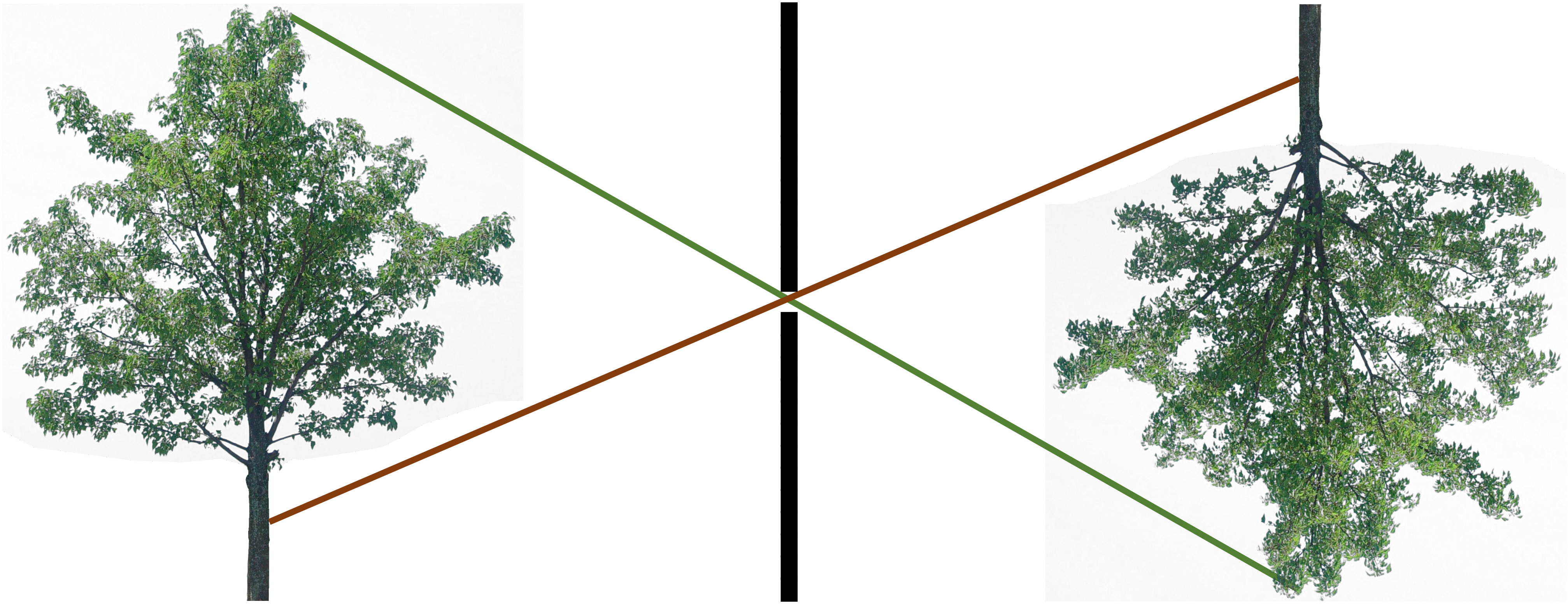
Each scene point contributes to only one sensor pixel

What does the image on the sensor look like?

Sources: CMU 15-463, Dartmouth College CS89/189.

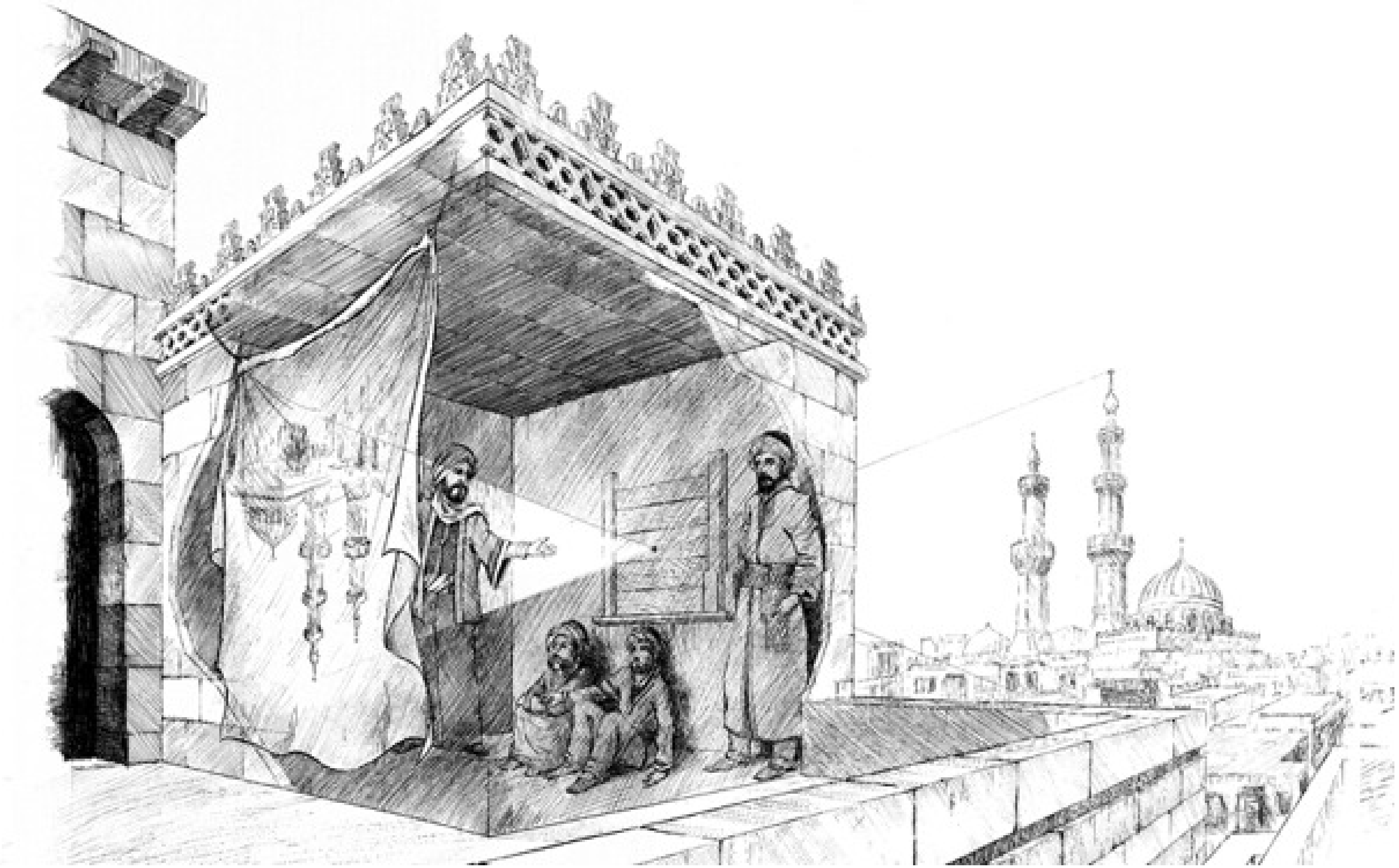
Pinhole imaging

real-world
object



copy of real-world object
(inverted and scaled)

Pinhole camera a.k.a. camera obscura

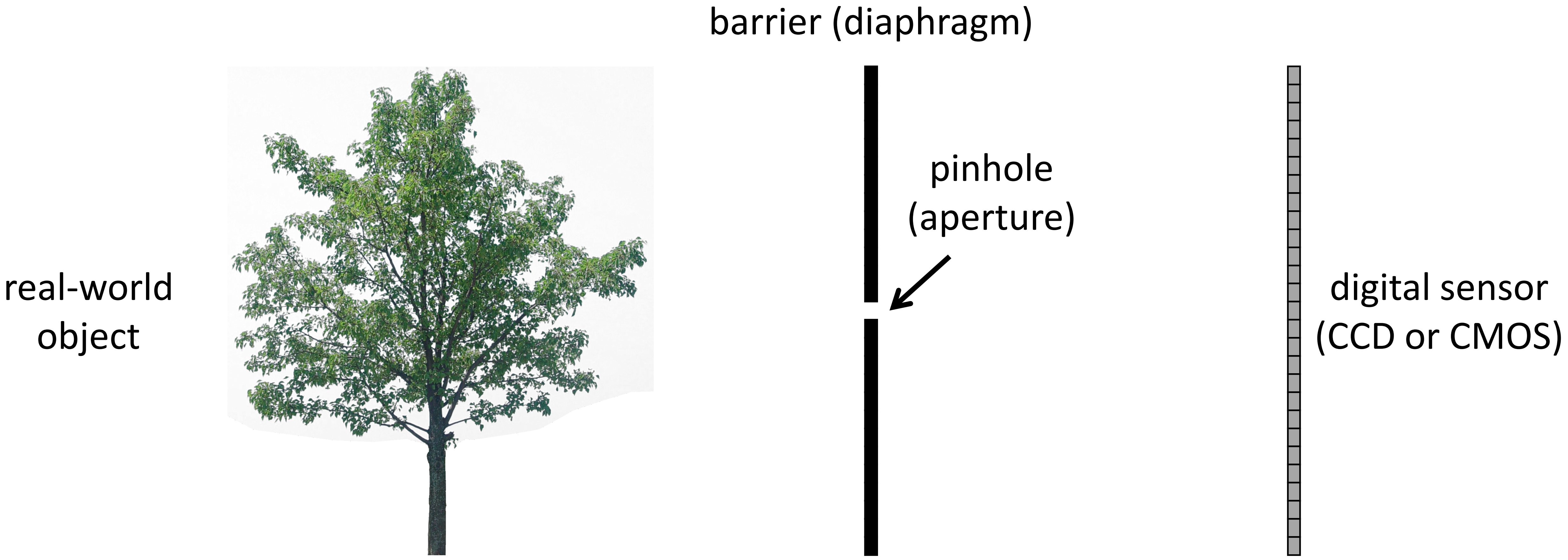


First mention ...

Chinese philosopher Mozi
(470 to 390 BC)

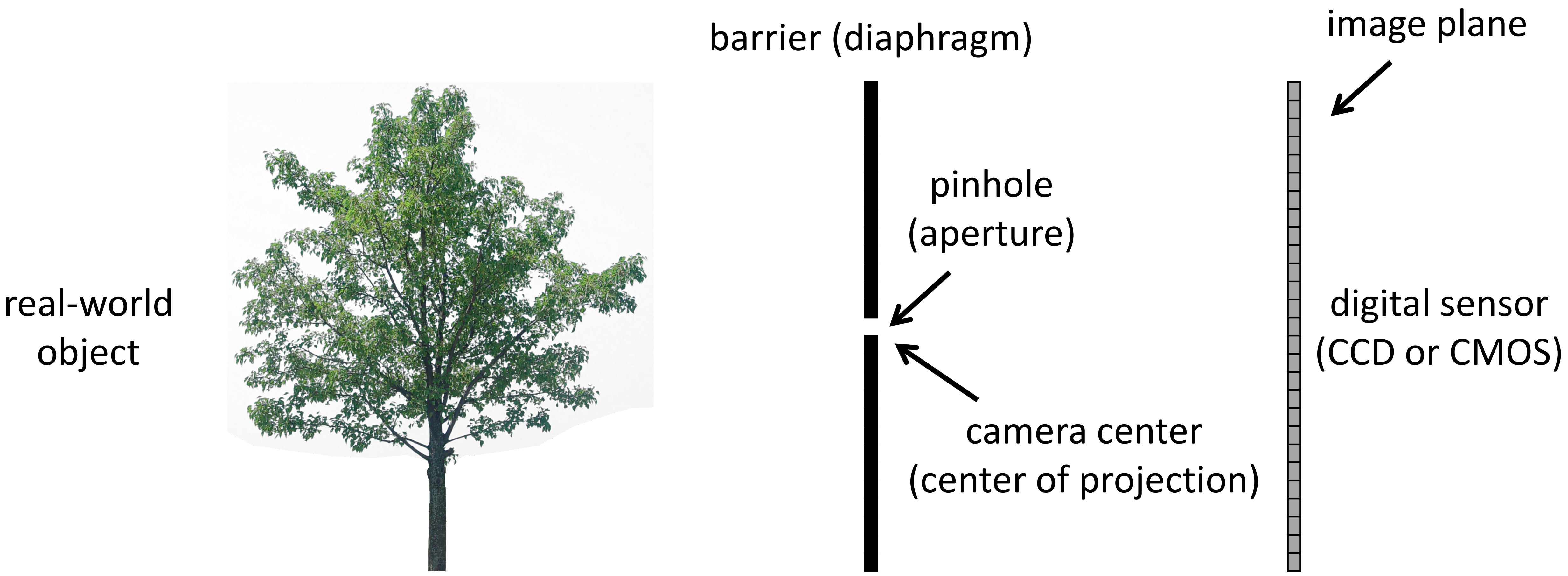
Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Pinhole camera terms



Sources: CMU 15-463, Dartmouth College CS89/189.

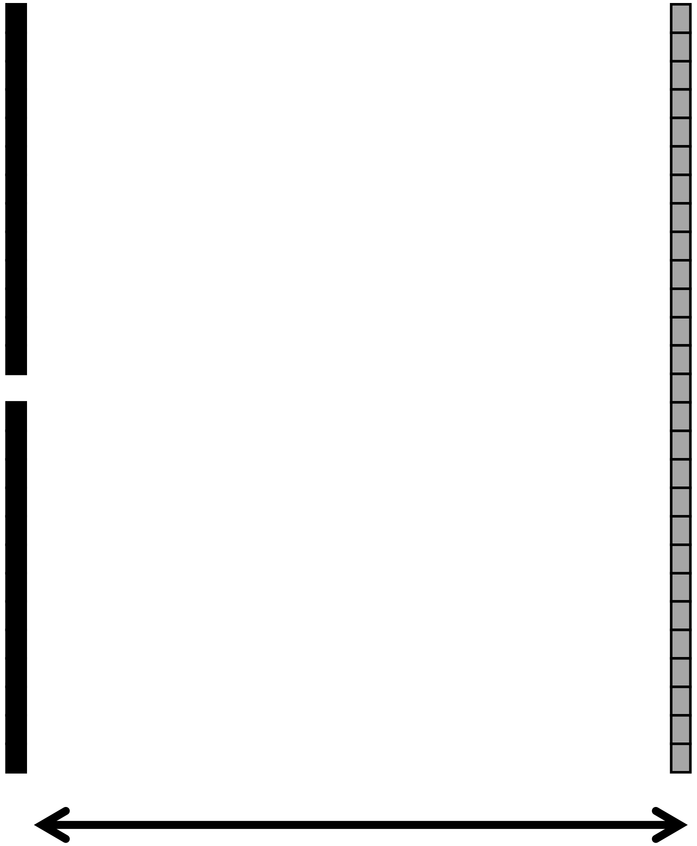
Pinhole camera terms



Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Focal length

real-world object



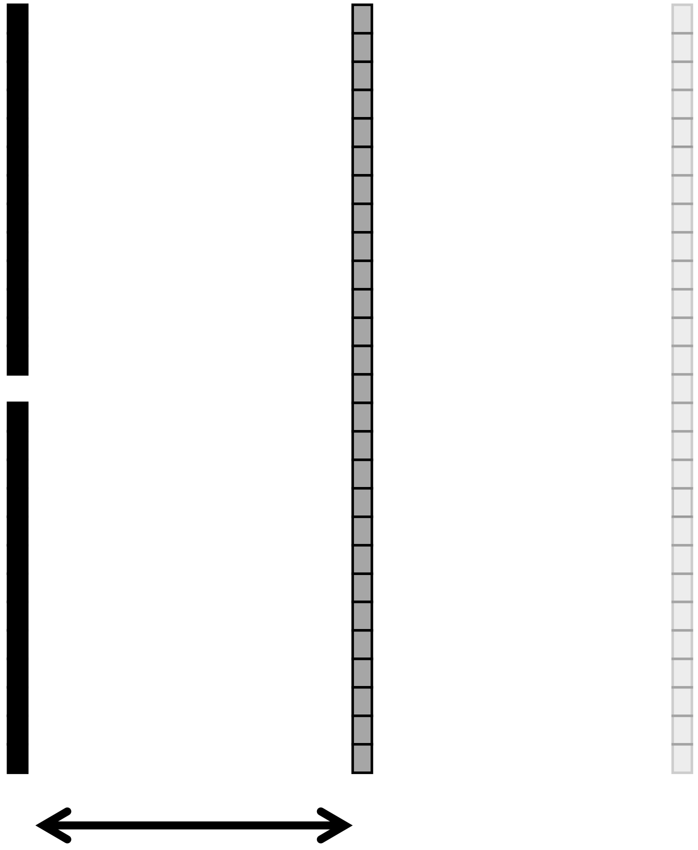
focal length f

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Focal length

What happens as we change the focal length?

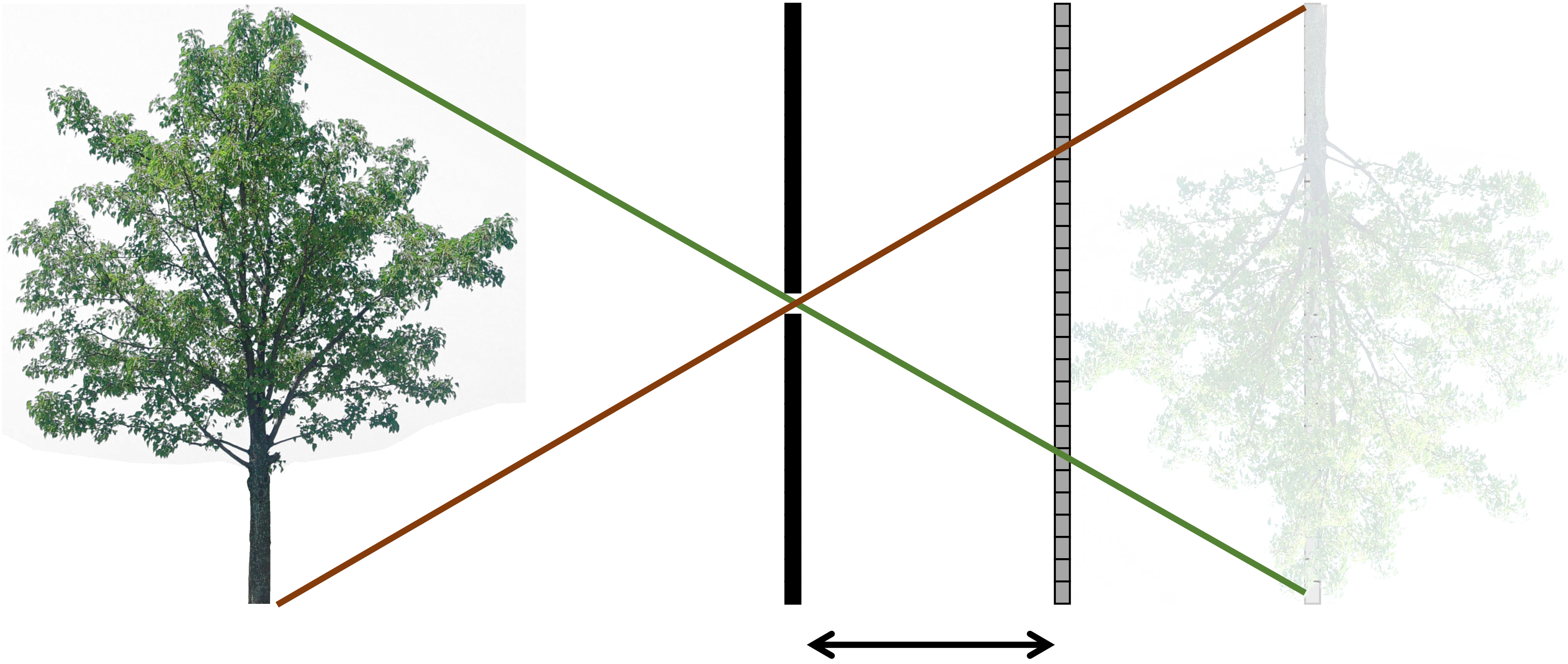
real-world object



Focal length

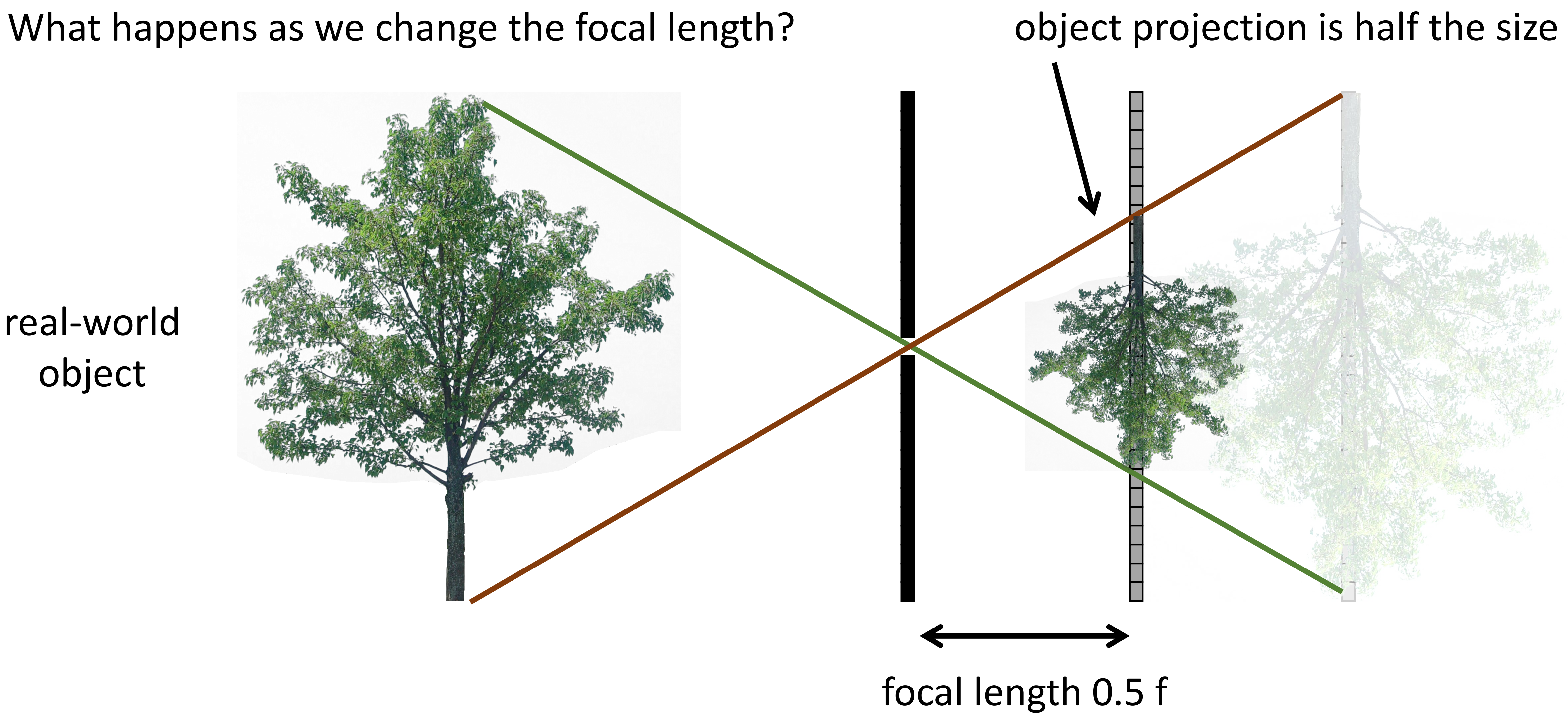
What happens as we change the focal length?

real-world object



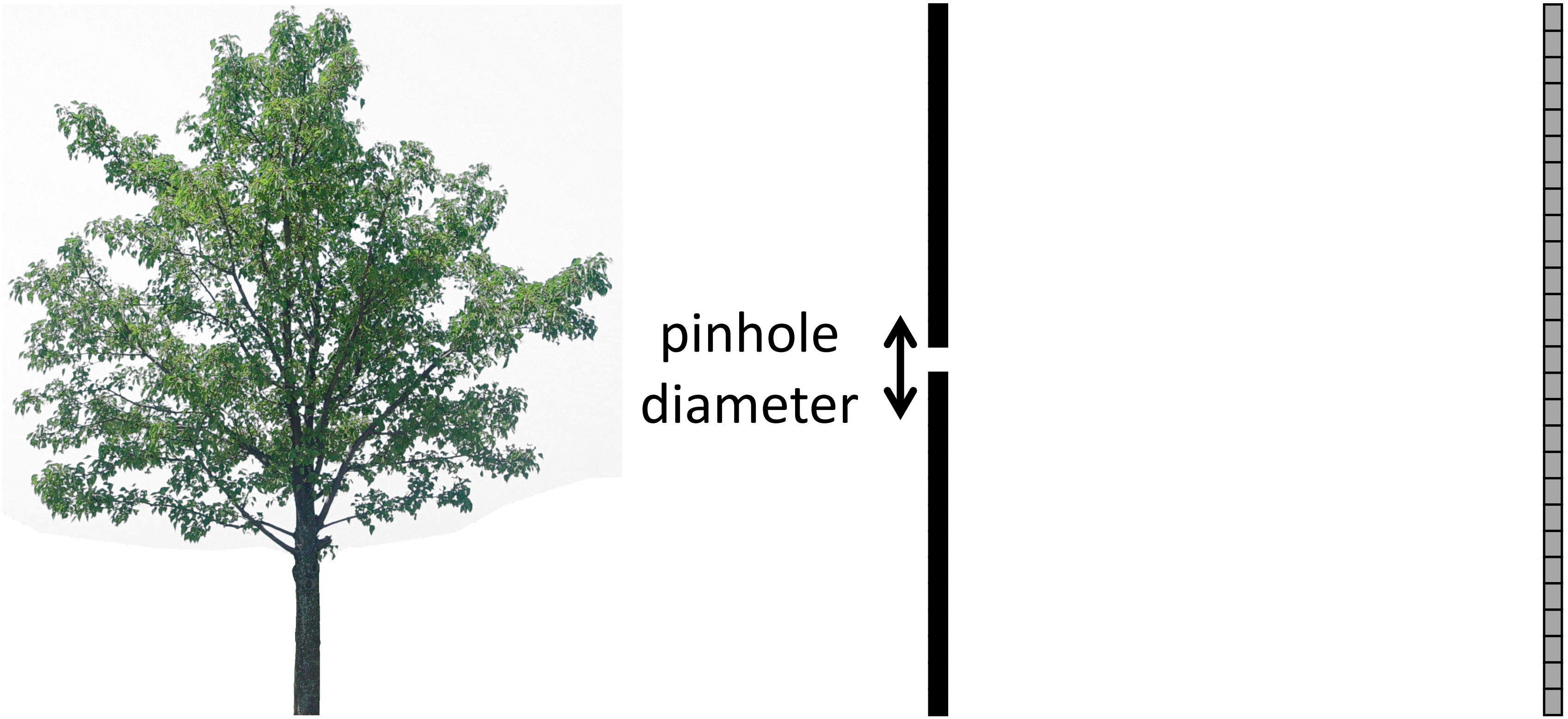
Focal length

What happens as we change the focal length?



Pinhole size

real-world
object

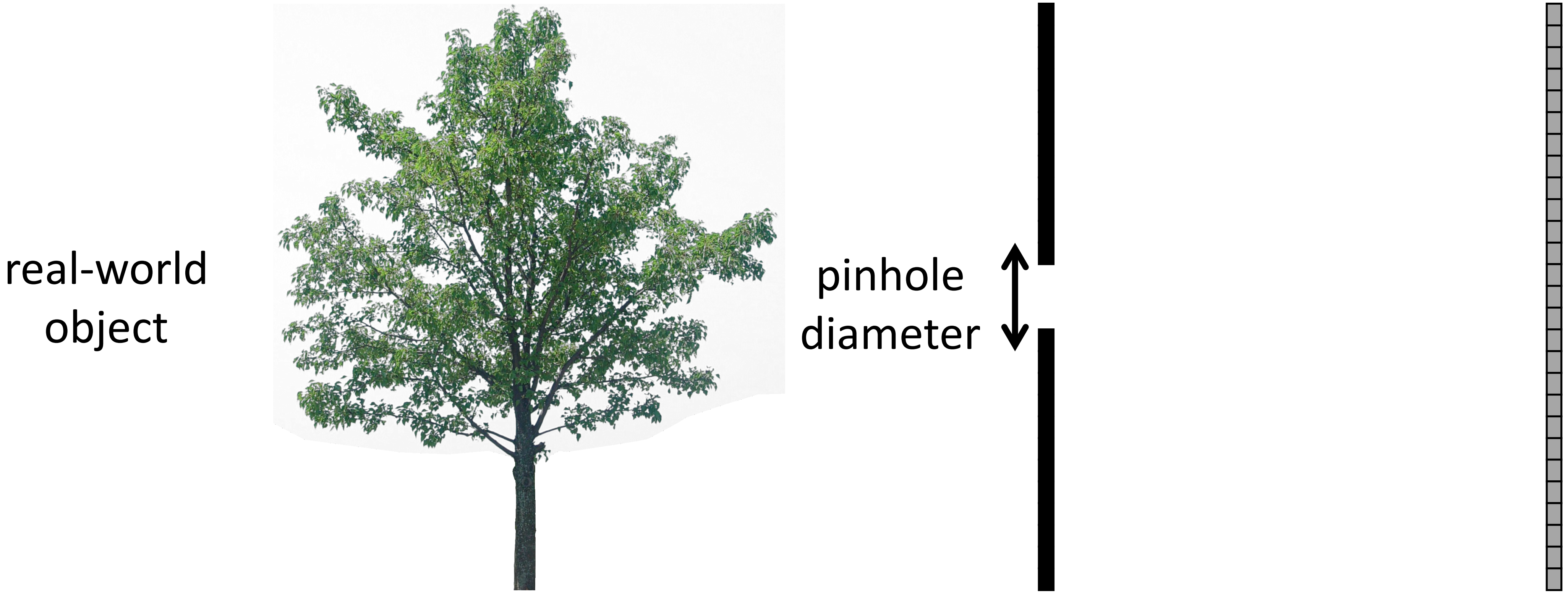


Ideal pinhole has infinitesimally small size

- In practice that is impossible.

Pinhole size

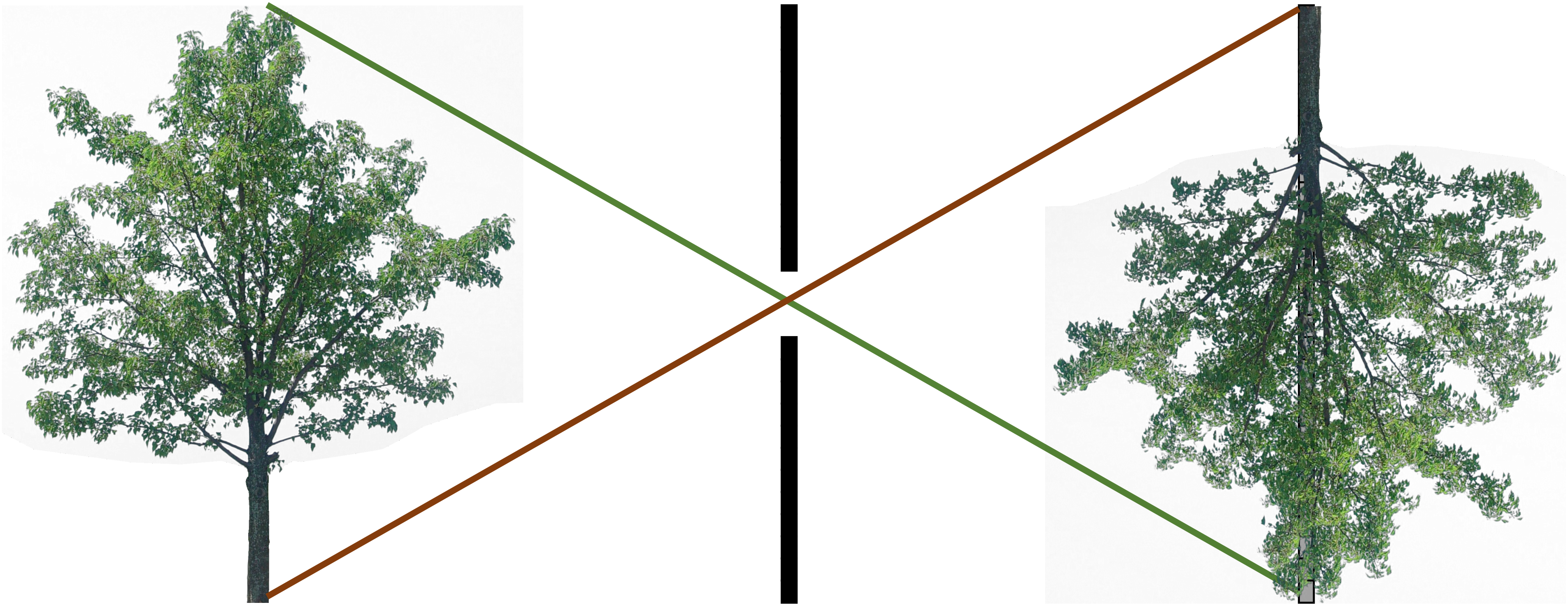
What happens as we change the pinhole diameter?



Pinhole size

What happens as we change the pinhole diameter?

real-world
object

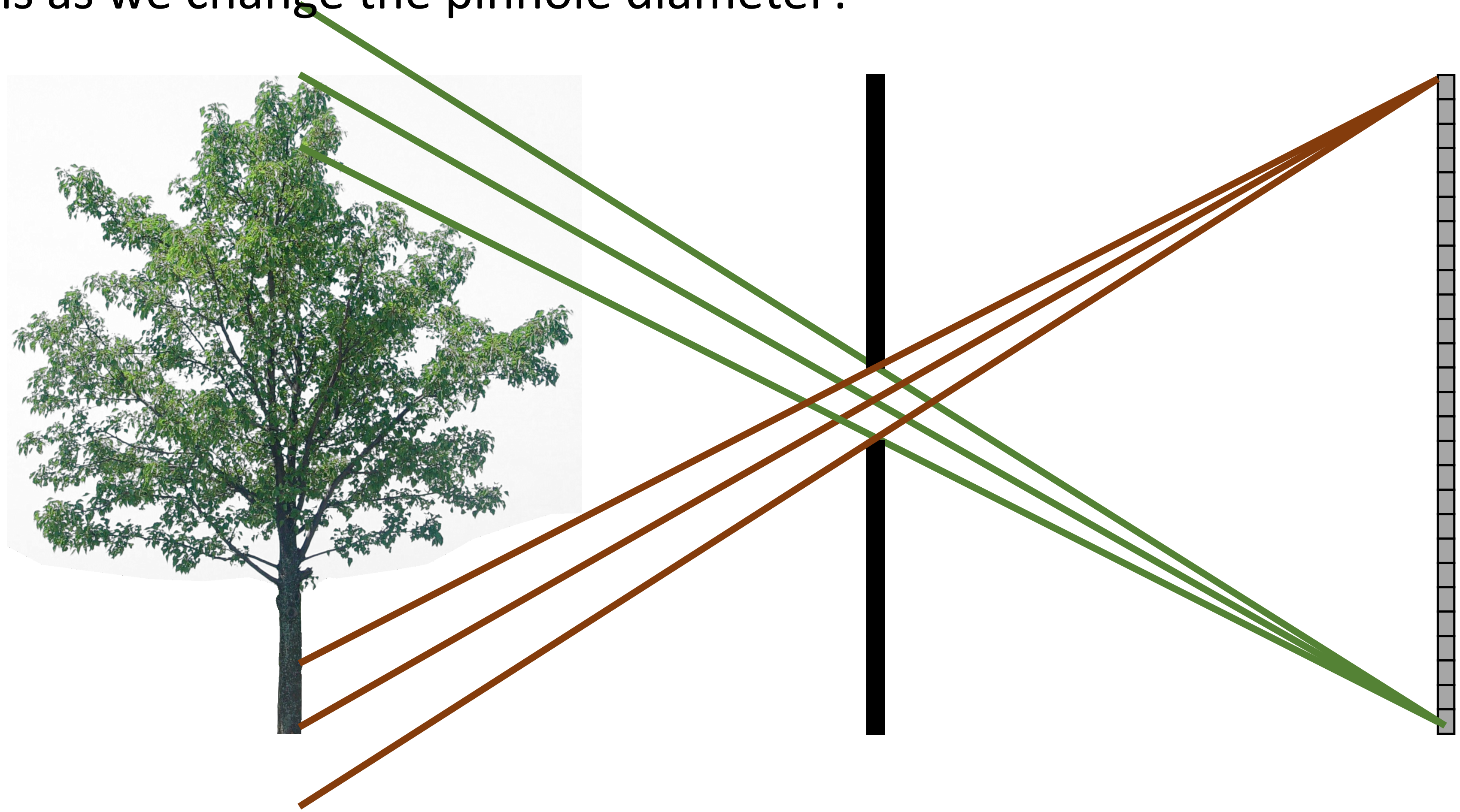


Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Pinhole size

What happens as we change the pinhole diameter?

real-world
object



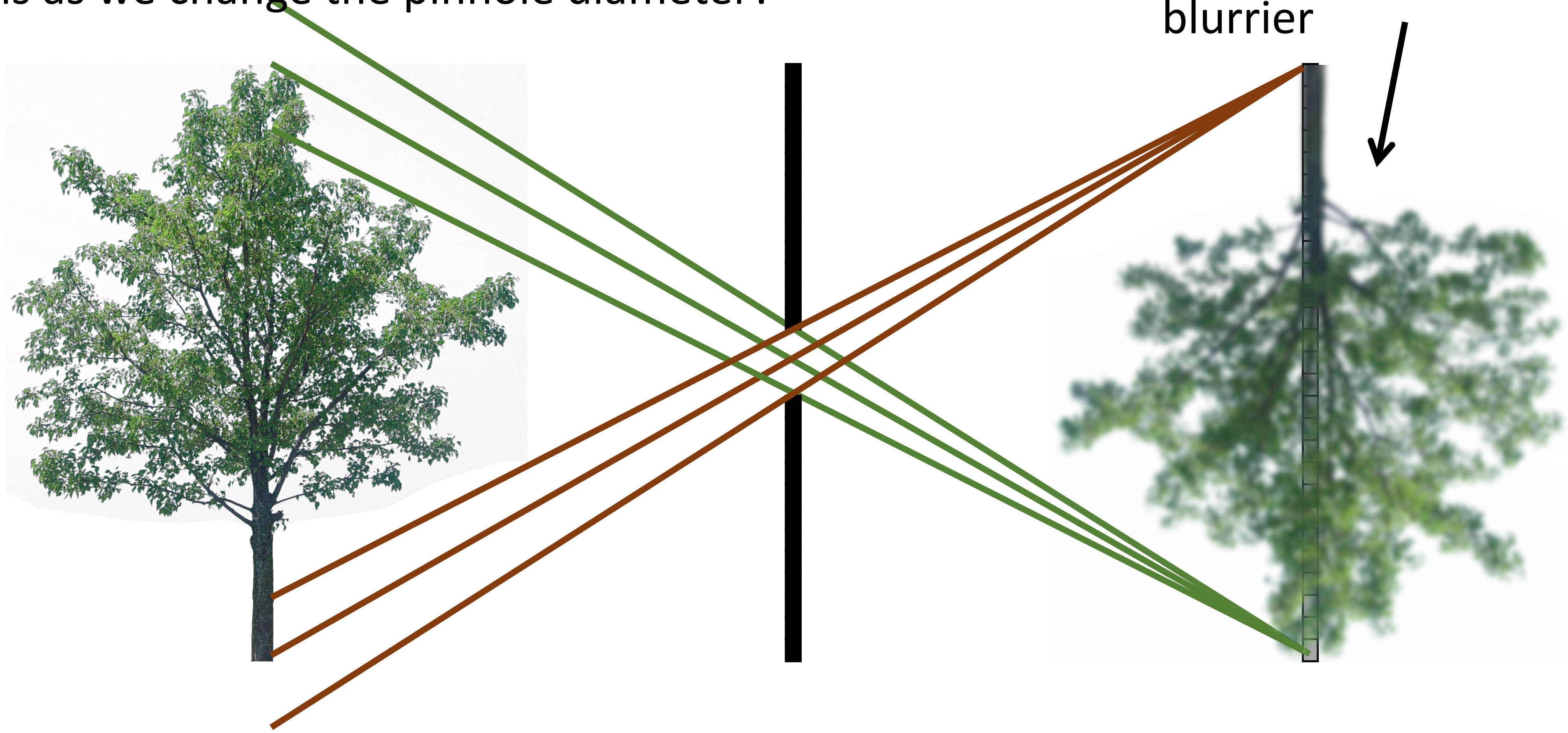
Sources: CMU 15-463, Dartmouth College CS89/189.

Pinhole size

What happens as we change the pinhole diameter?

object projection becomes blurrier

real-world object

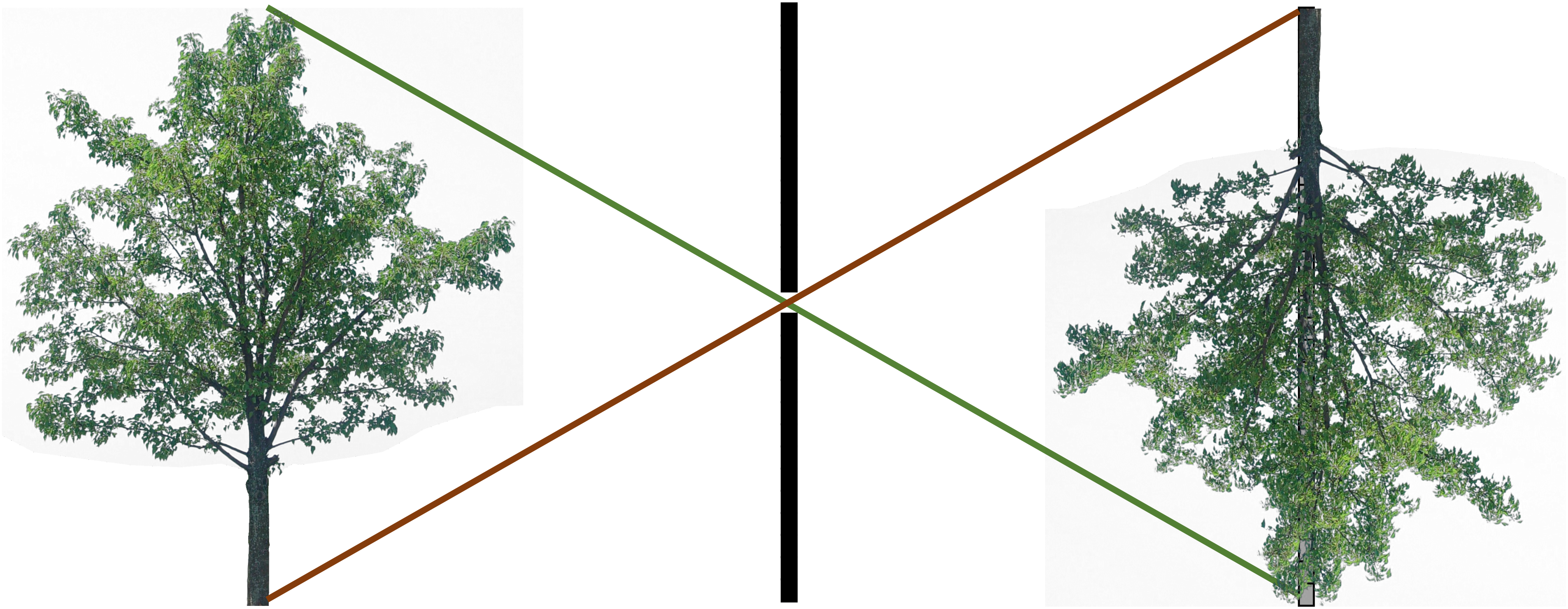


Sources: CMU 15-463, Dartmouth College CS89/189.

Pinhole size

What happens as we change the pinhole diameter?

real-world
object

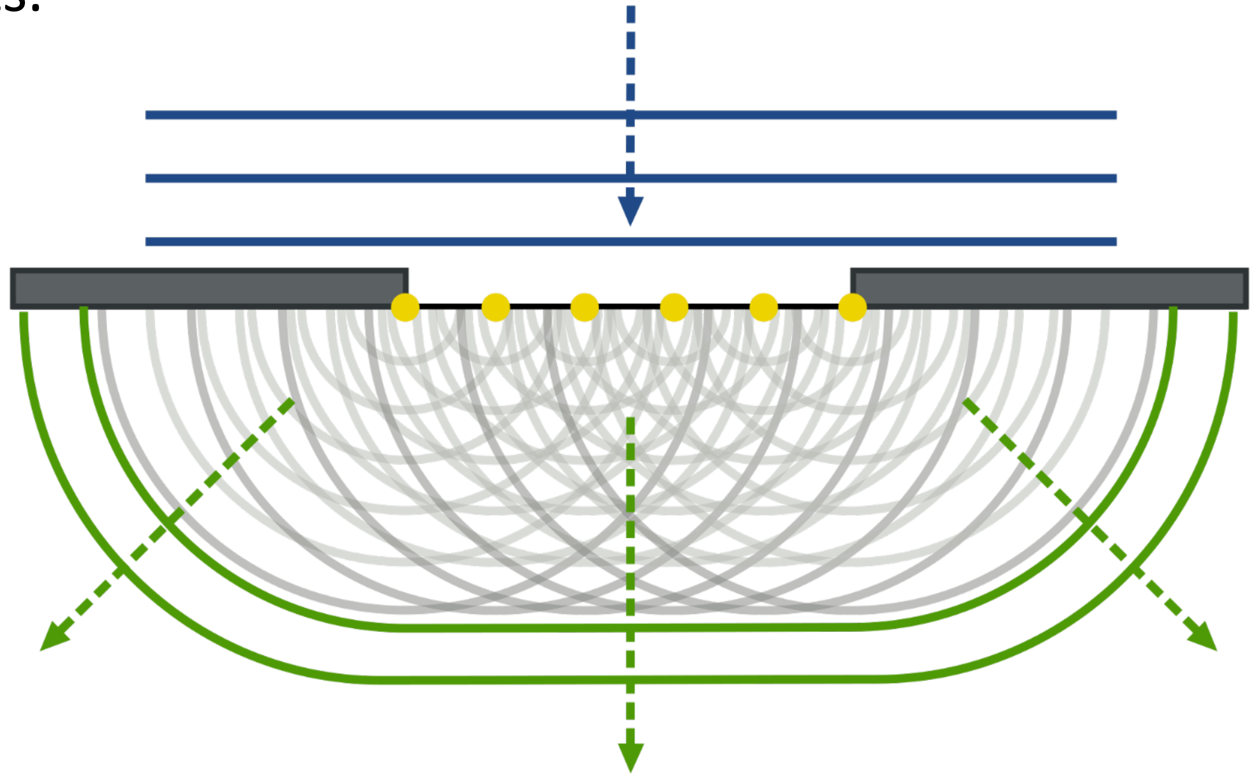


Sources: CMU 15-463, Dartmouth College CS89/189.

Pinhole size – Diffraction limit

Huygens-Fresnel Principle: Every point on a wavefront is itself the source of spherical wavelets, and the sum of these wavelets forms the wavefront at a later time.

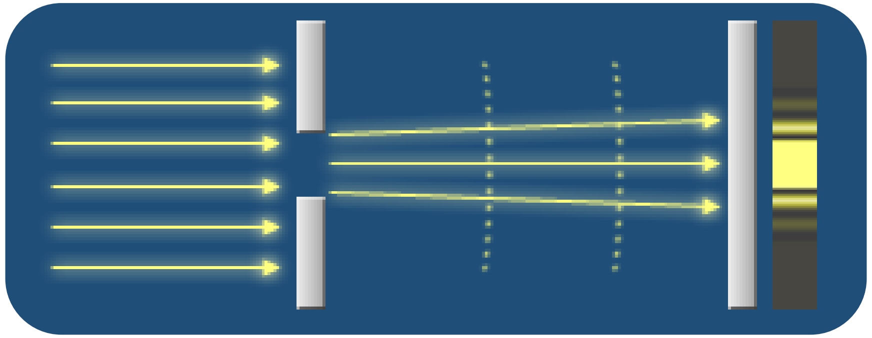
When applied to light waves encountering a slit, each point within the slit acts as a source of new spherical wavelets.

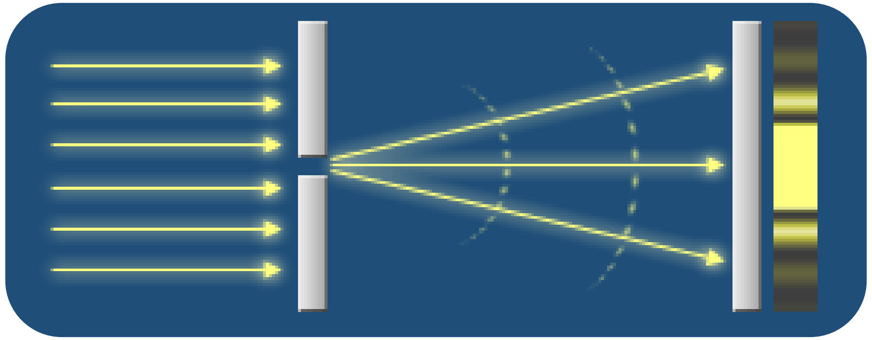


Pinhole size – Diffraction limit

Diffraction Limit: Point at which the size of the pinhole causes the image quality to degrade due to diffraction. Determined y the **wavelength** of light and size of the **aperture**.

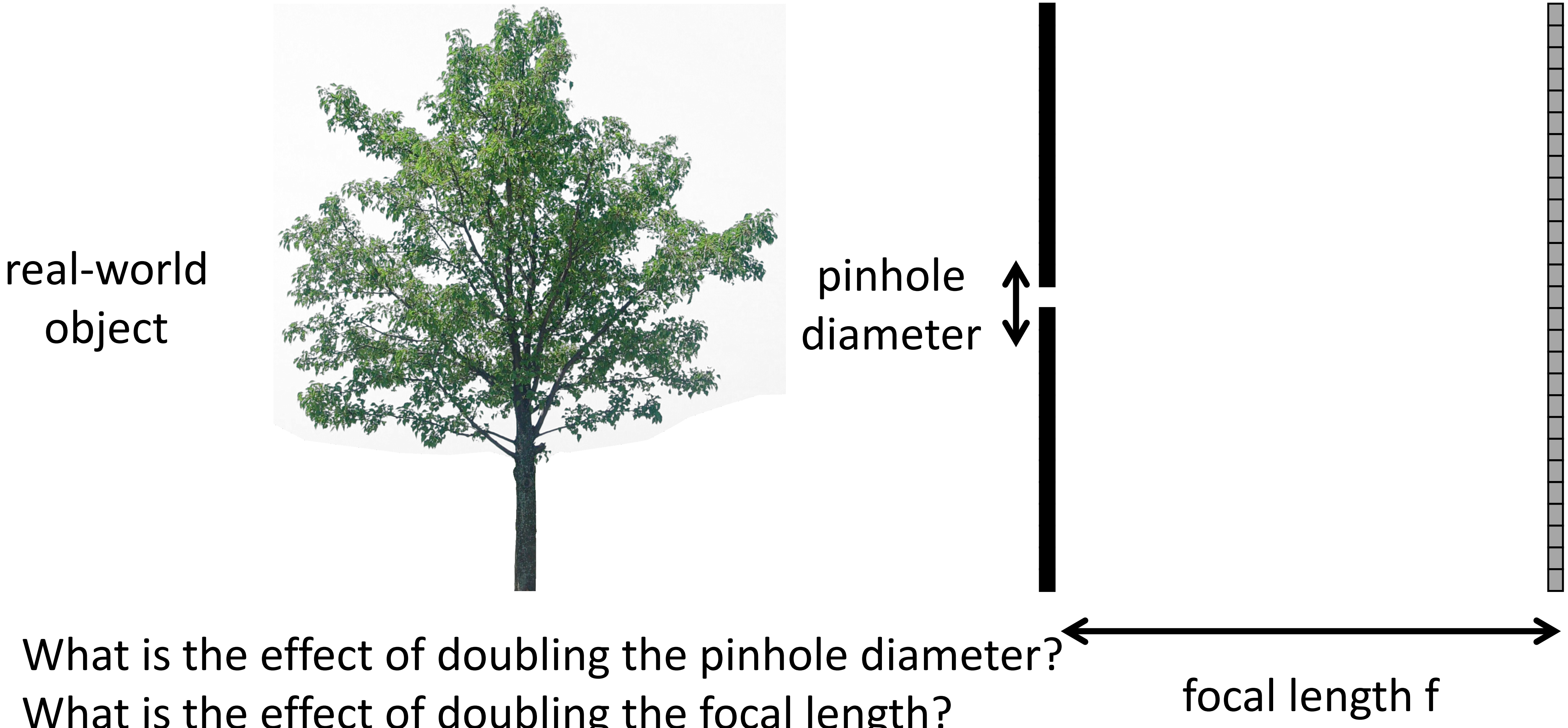
Fundamental limit to the resolution (or sharpness) of an imaging system given its aperture size.





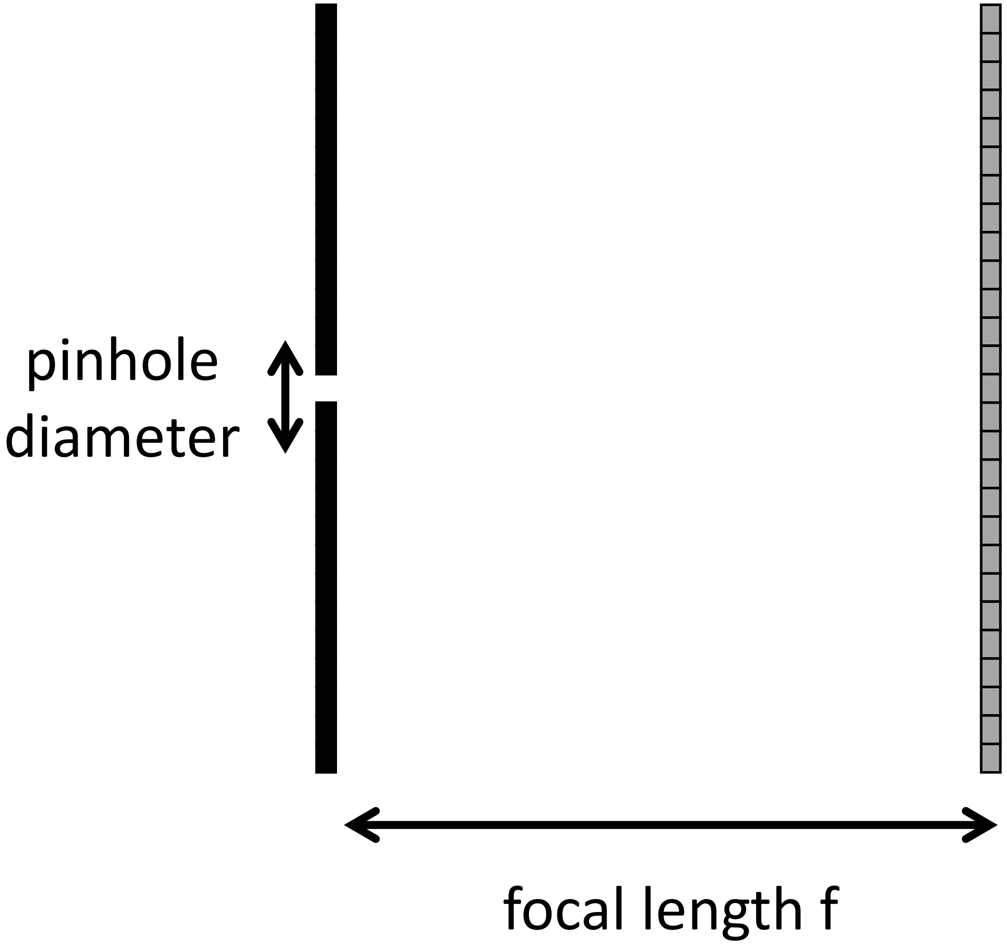
Sources: CMU 15-463, Dartmouth College CS89/189.

What about light efficiency?



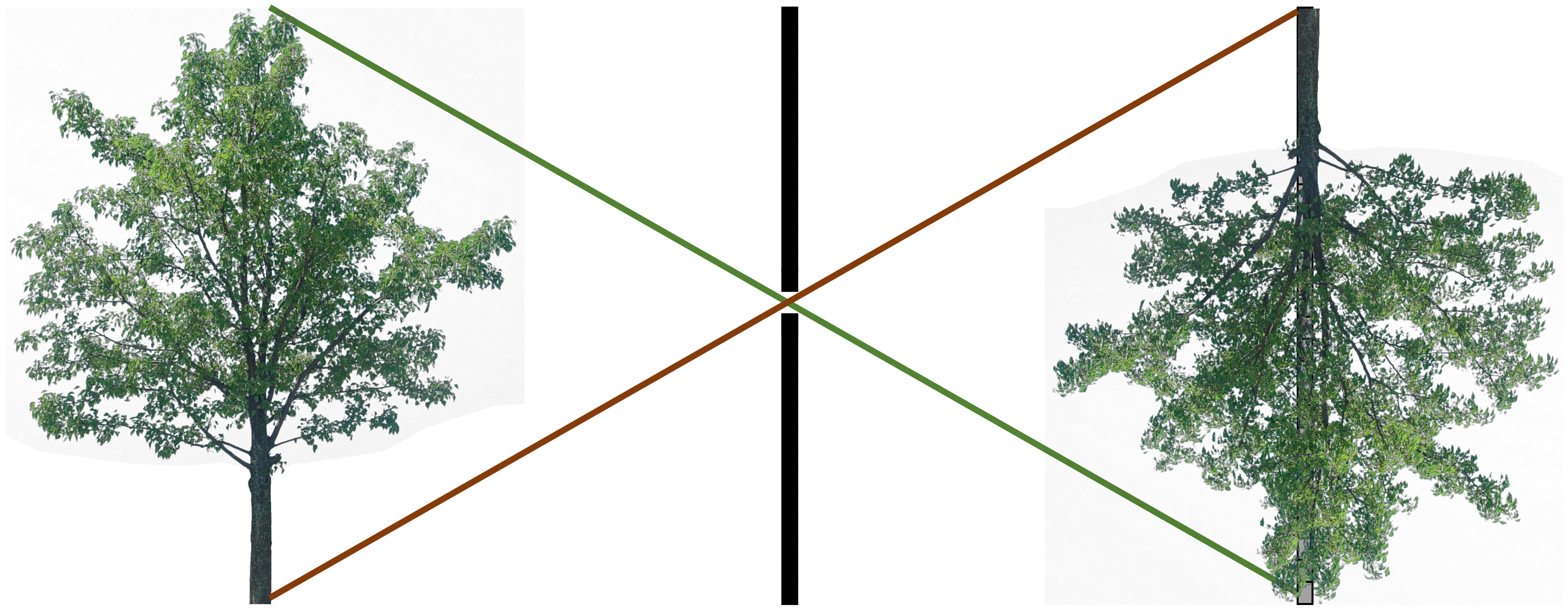
- What is the effect of doubling the pinhole diameter?
- What is the effect of doubling the focal length?

What about light efficiency?



- 2x pinhole diameter → 4x light
- 2x focal length → ¼x light

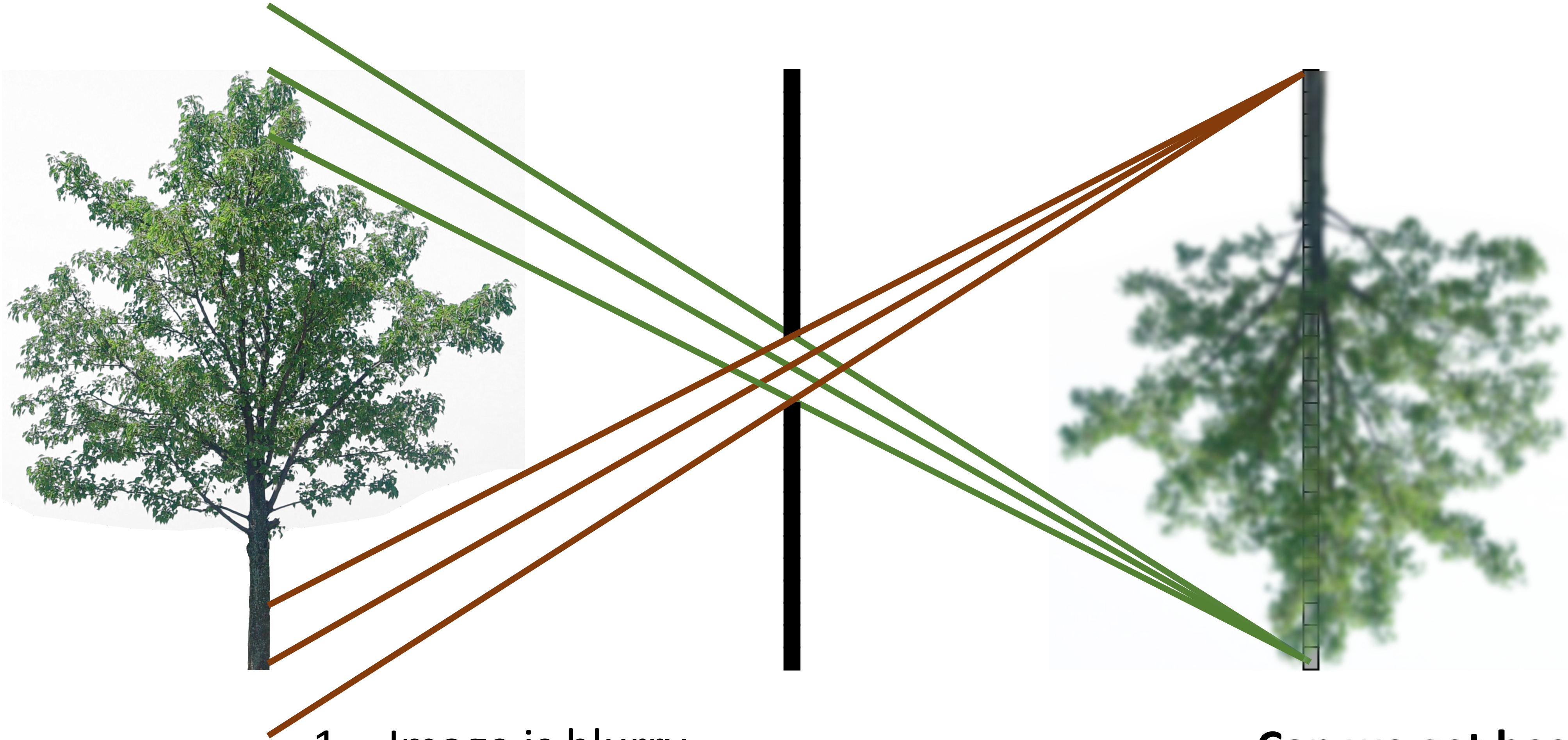
Pinhole camera trade-off



Small (ideal) pinhole:
1. Image is sharp.
2. Signal-to-noise ratio is low.

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Pinhole camera trade-off



Large pinhole:

1. Image is blurry.
2. Signal-to-noise ratio is high.

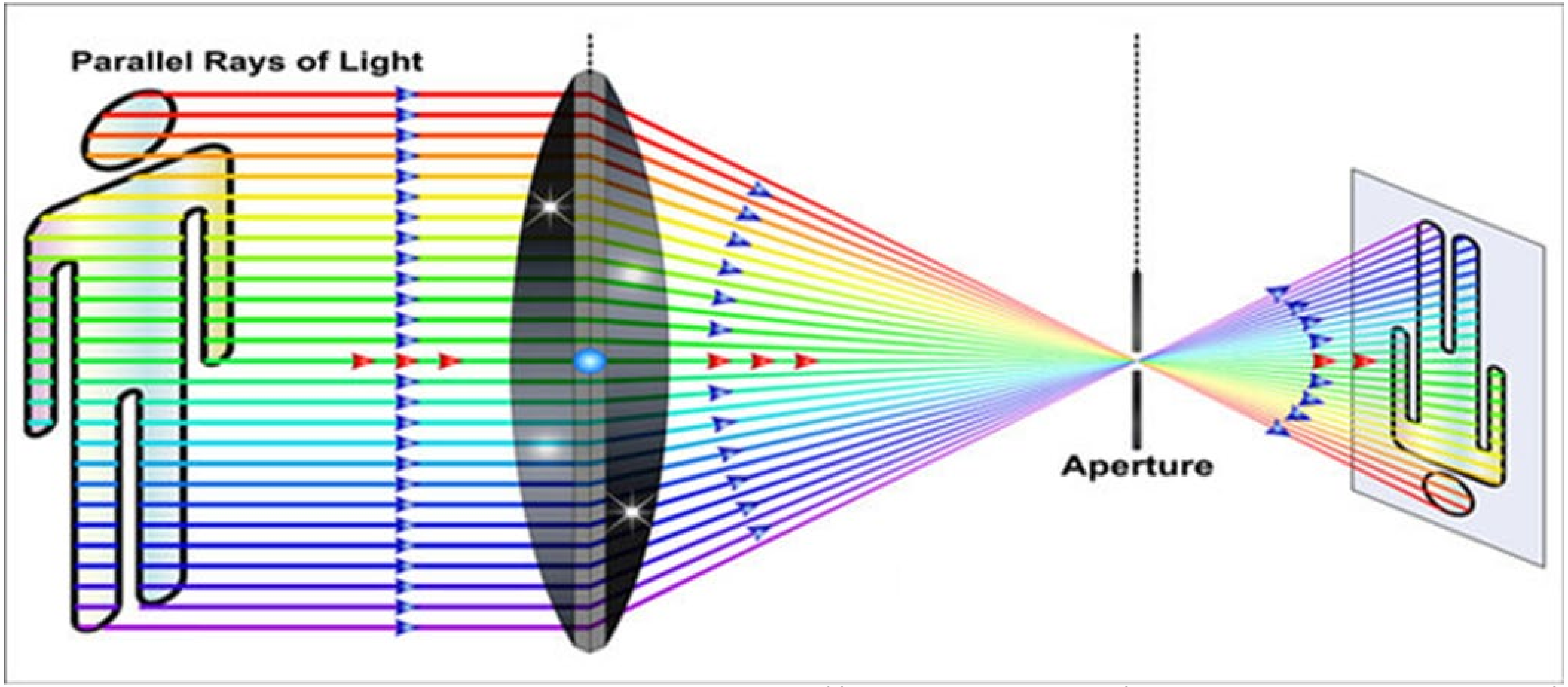
Can we get best of both worlds?

Almost, by using lenses

- Larger apertures
  ➔ Gather more light
- But need to be **focused**

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Almost, by using lenses



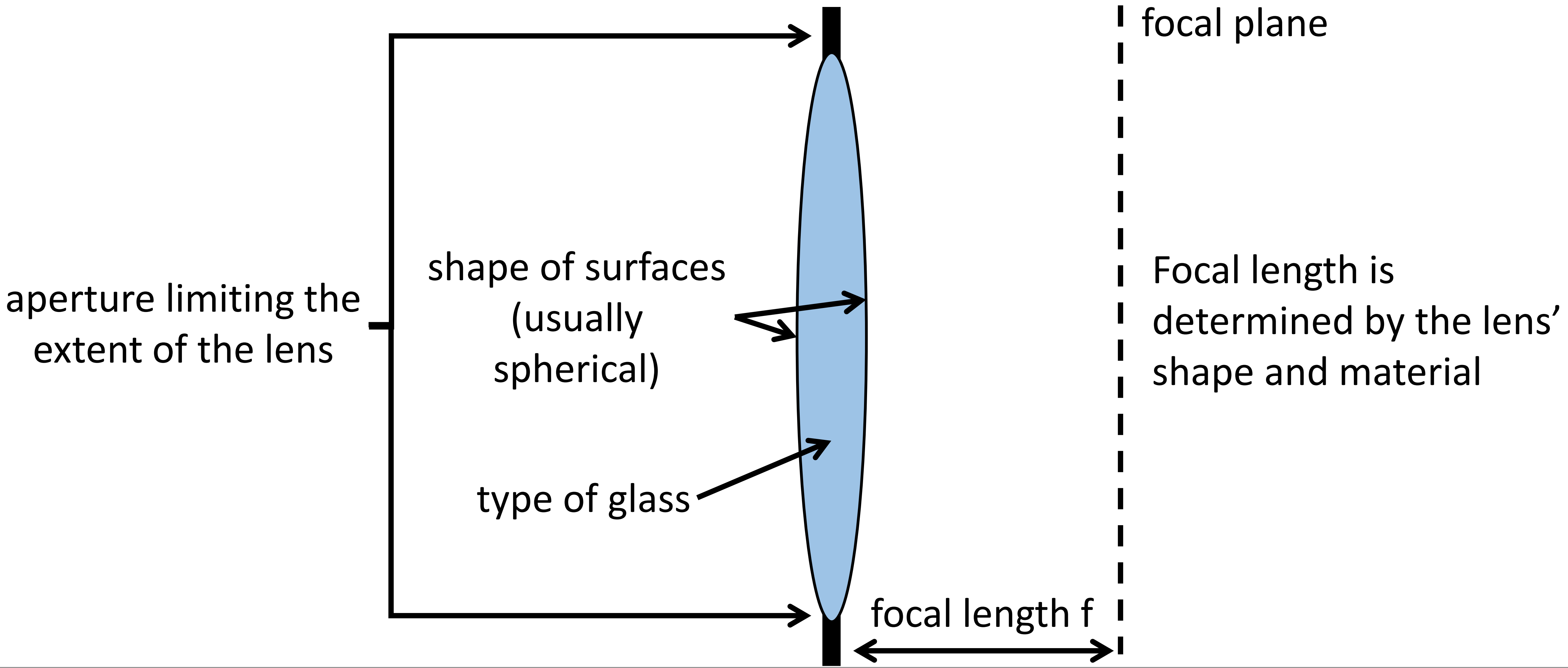
https://gadgetpursuit.com/how-do-camera-lenses-work/

Lenses map “bundles” of rays from points on the scene to the sensor.

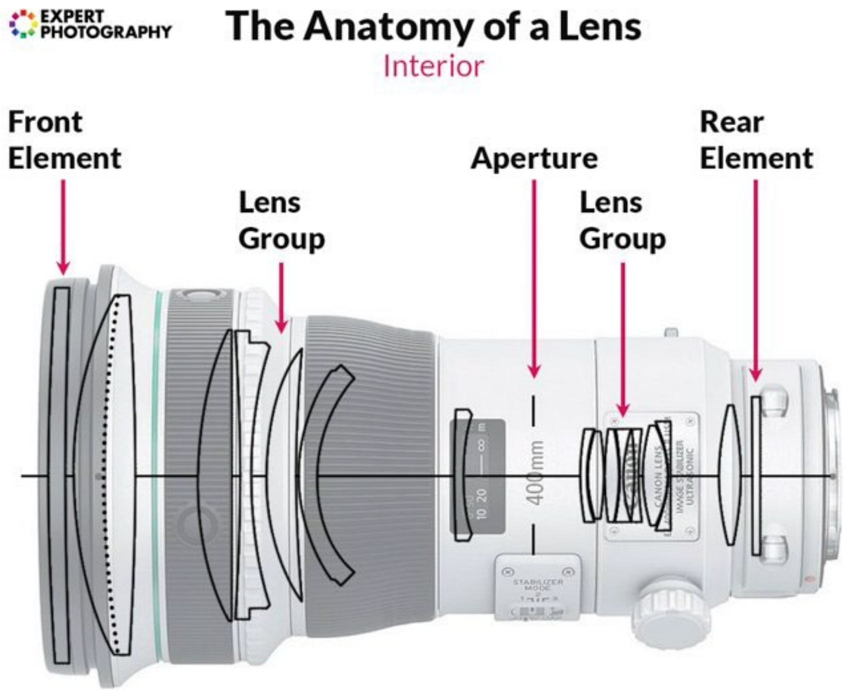
How does this mapping work exactly?

What is a lens?

A piece of glass manufactured to have a specific shape.



The lens on your camera



Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Aperture size

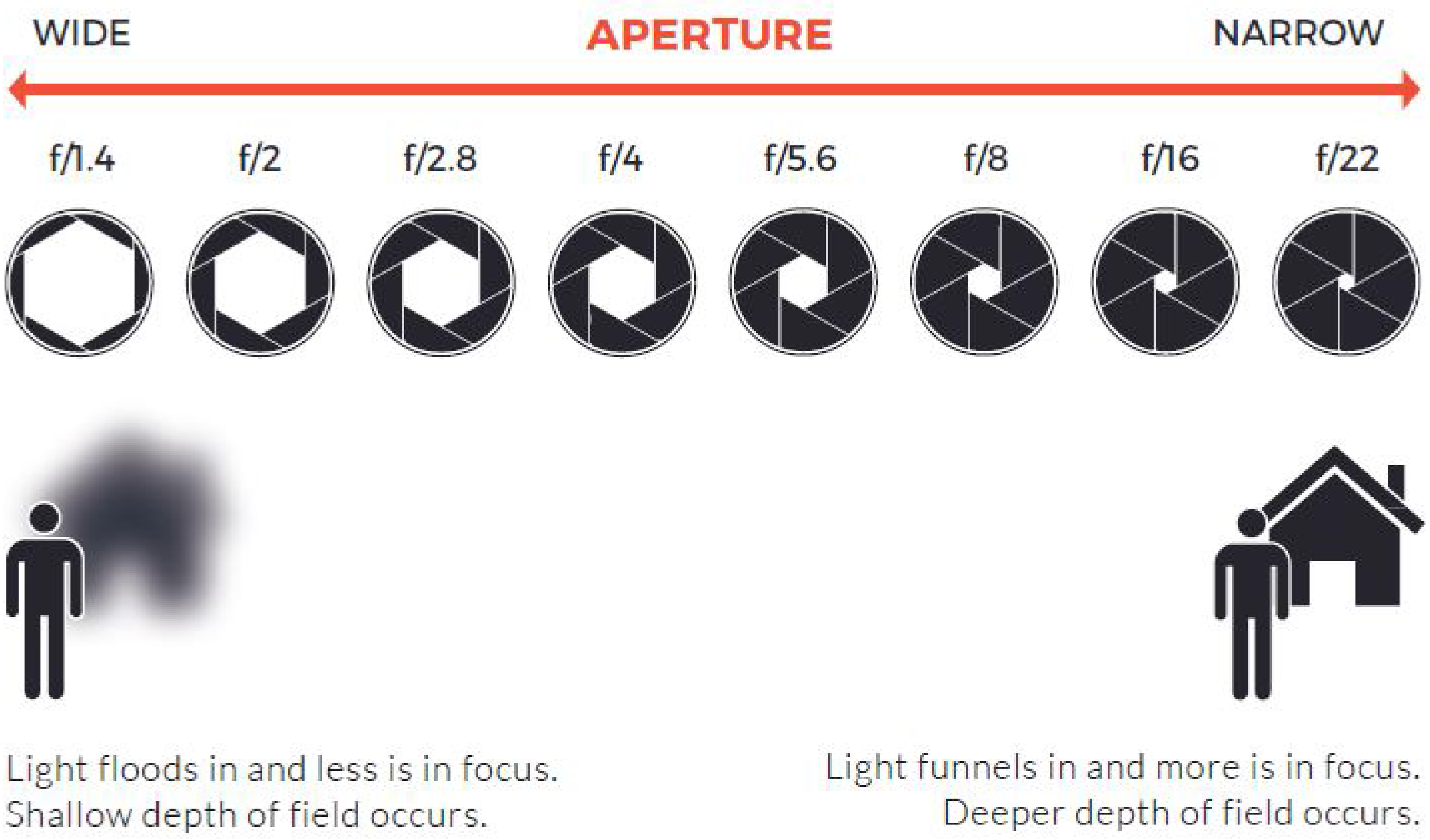
Most lenses have apertures of variable size.

- The size of the aperture is expressed as the “f-number”: The larger this number, the smaller the aperture.

f / 1.4 | f / 2.8 | f / 4 | f / 8 | f / 16

You can see the aperture by removing the lens and looking inside it.

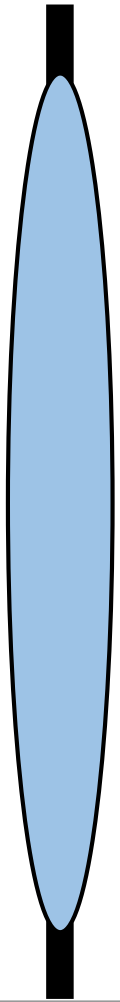
Aperture size



https://www.city-academy.com/news/what-is-aperture-in-photography/

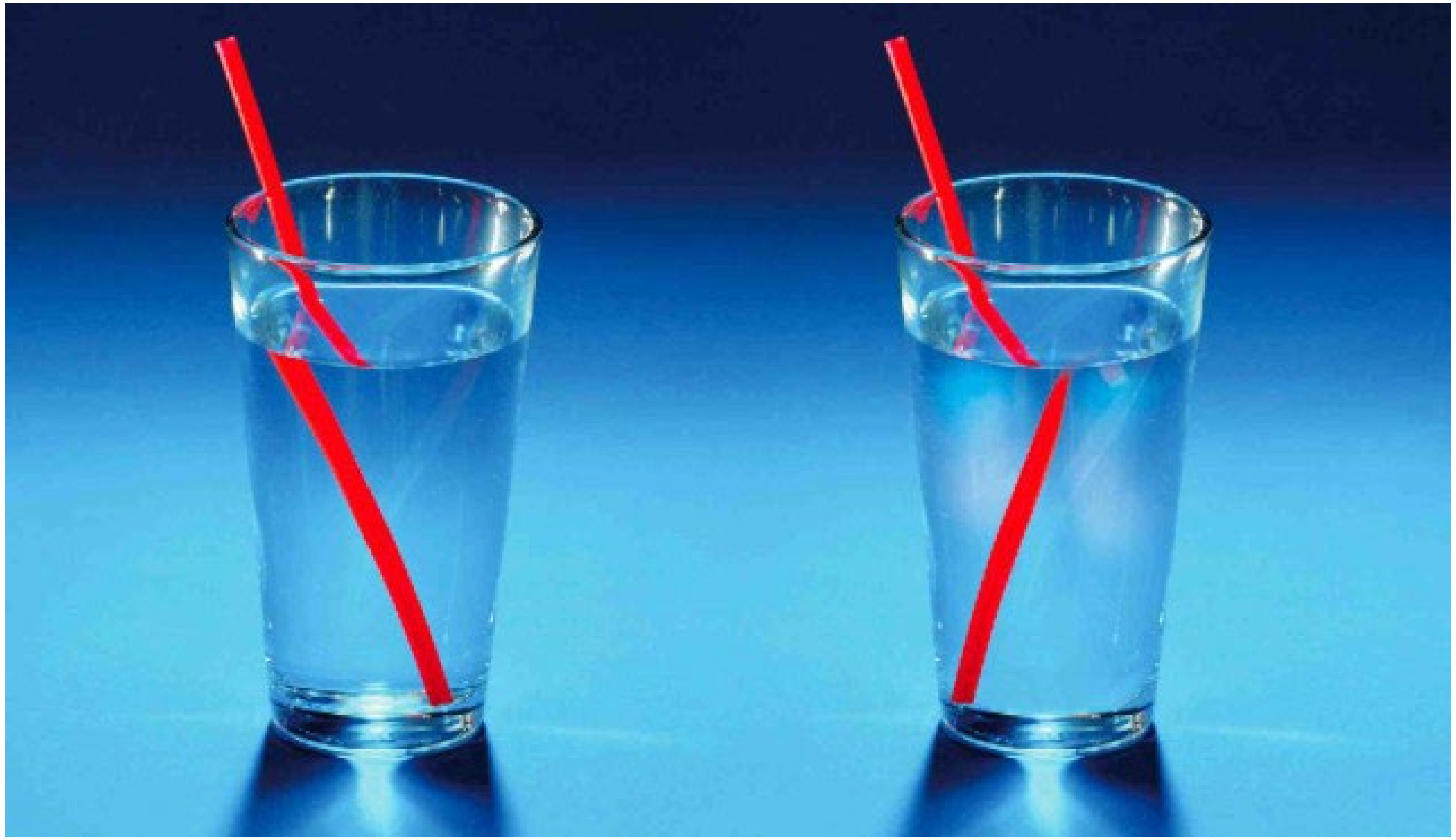
How does a lens work?

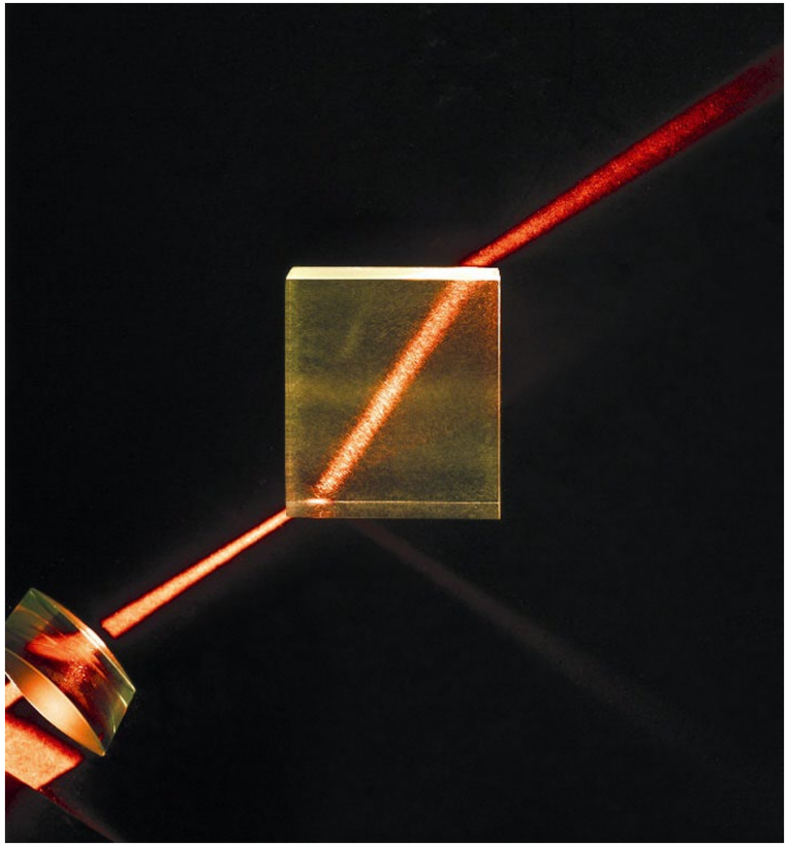
Lenses are design so that their refraction makes light rays bend in a very specific way.



Refraction

Refraction is the bending of rays of light when they move from one material to another

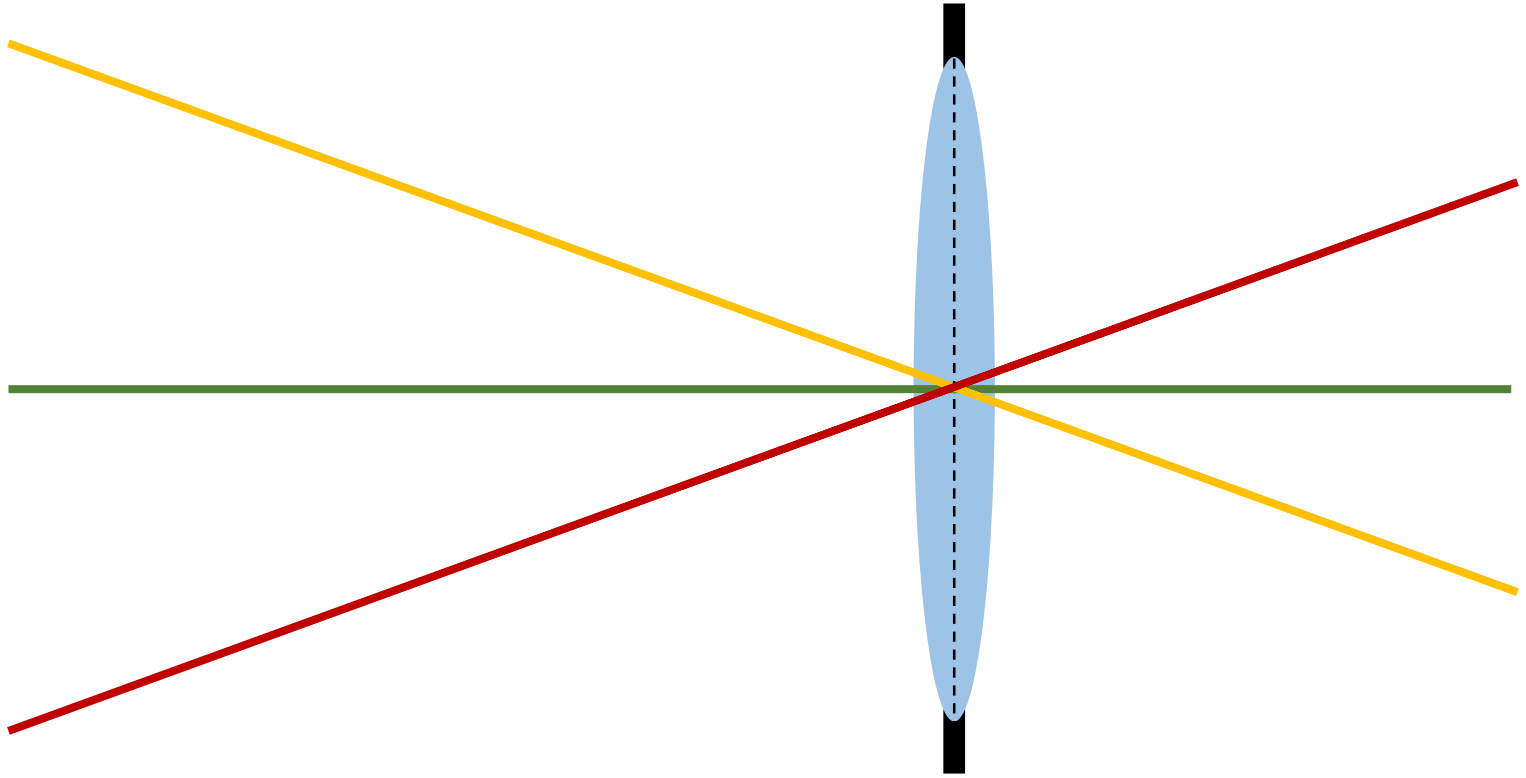




Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Thin lens model

Simplification of geometric optics for well-designed lenses.



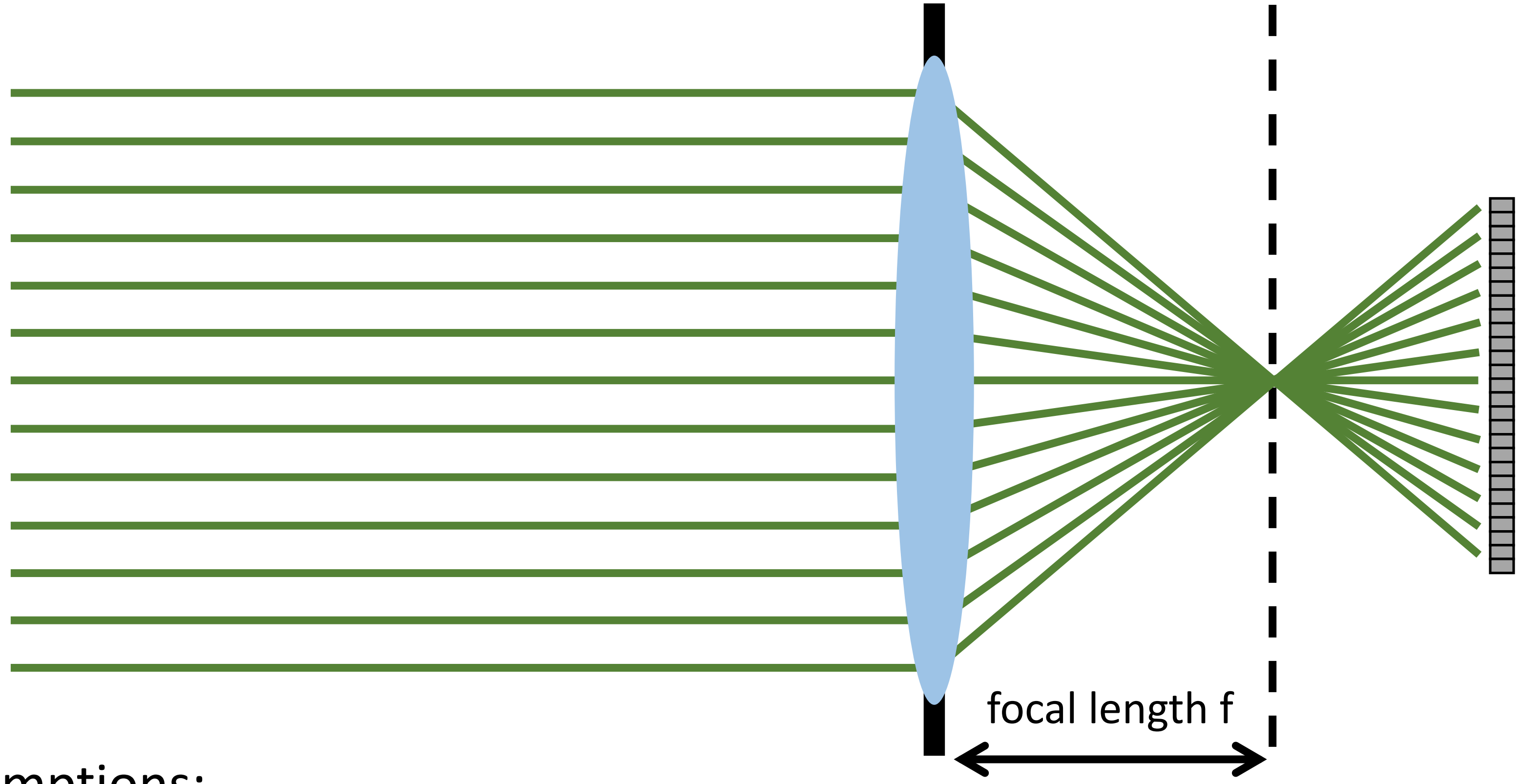
Two assumptions:

1. Rays passing through lens center are unaffected.

Sources: CMU 15-463, Dartmouth College CS89/189.

Thin lens model

Simplification of geometric optics for well-designed lenses.



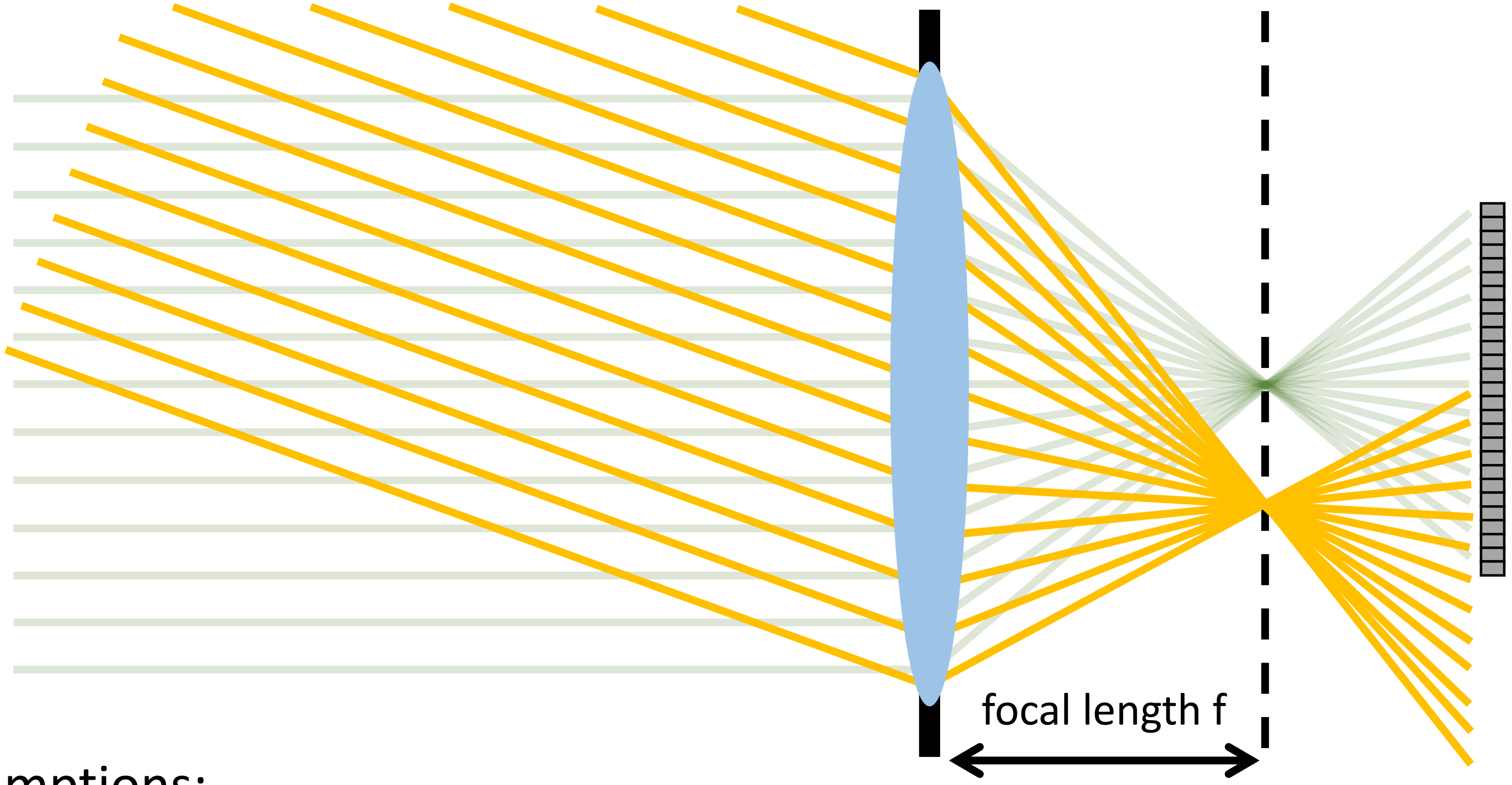
Two assumptions:

1. Rays passing through lens center are unaffected.
2. Parallel rays converge to a single point located on focal plane.

Sources: [CMU 15-463], [Dartmouth College CS89/189].

Thin lens model

Simplification of geometric optics for well-designed lenses.



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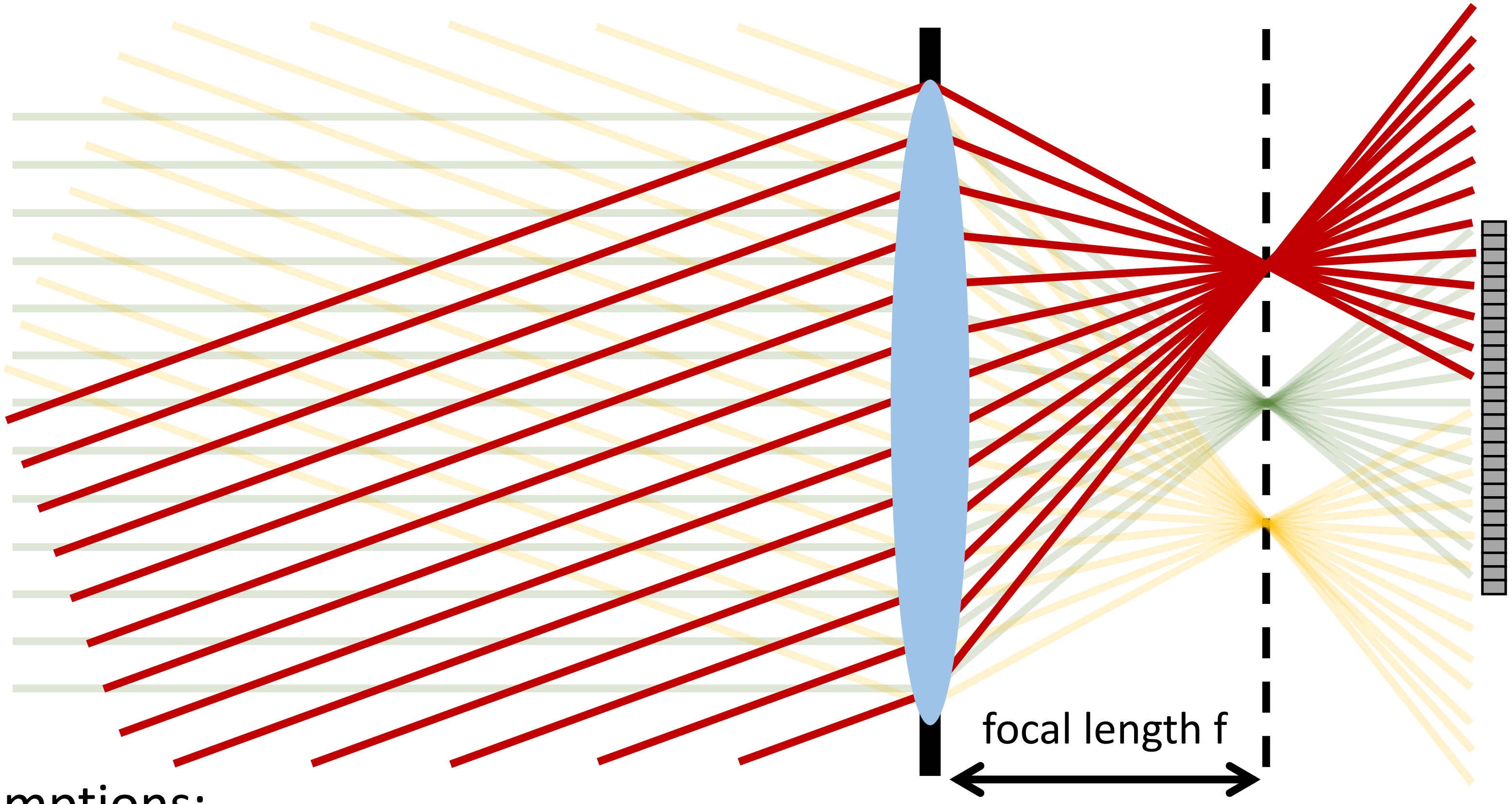
Thin lens model

Simplification of geometric optics for well-designed lenses.

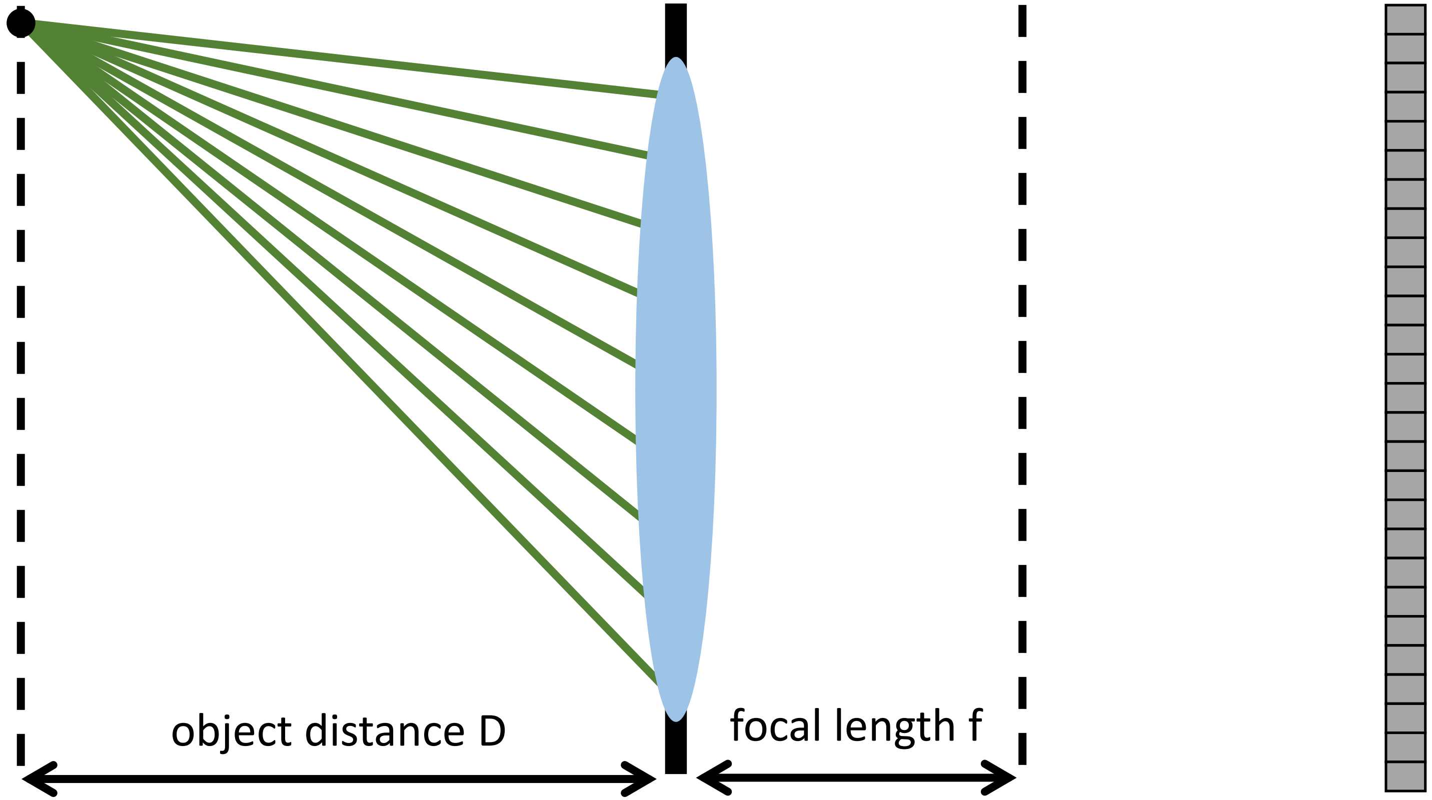


Two assumptions:

1. Rays passing through lens center are unaffected.
2. Parallel rays converge to a single point located on focal plane.

Tracing rays through a thin lens

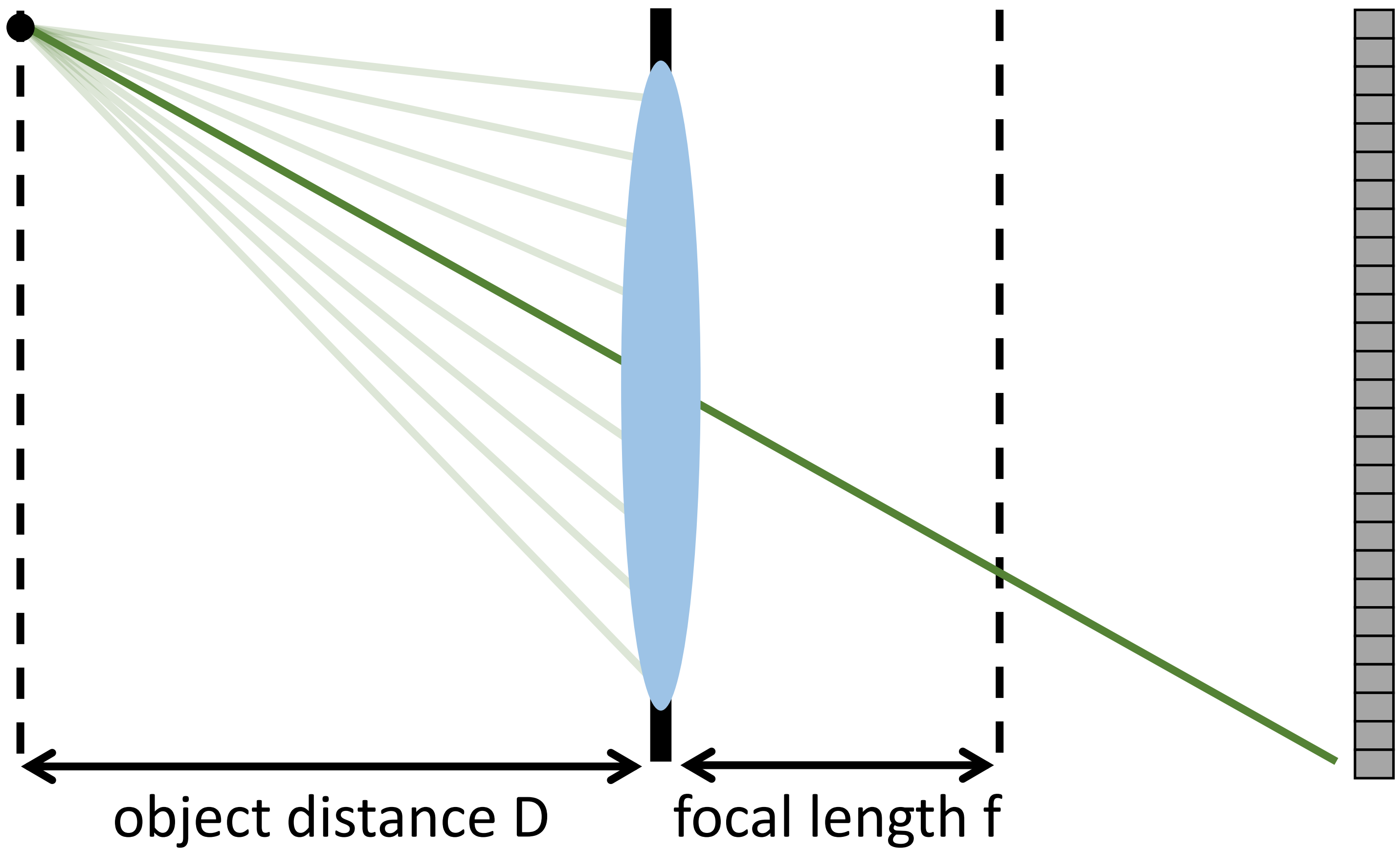
Consider an object emitting a bundle of rays. How do they propagate through the lens?



Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?

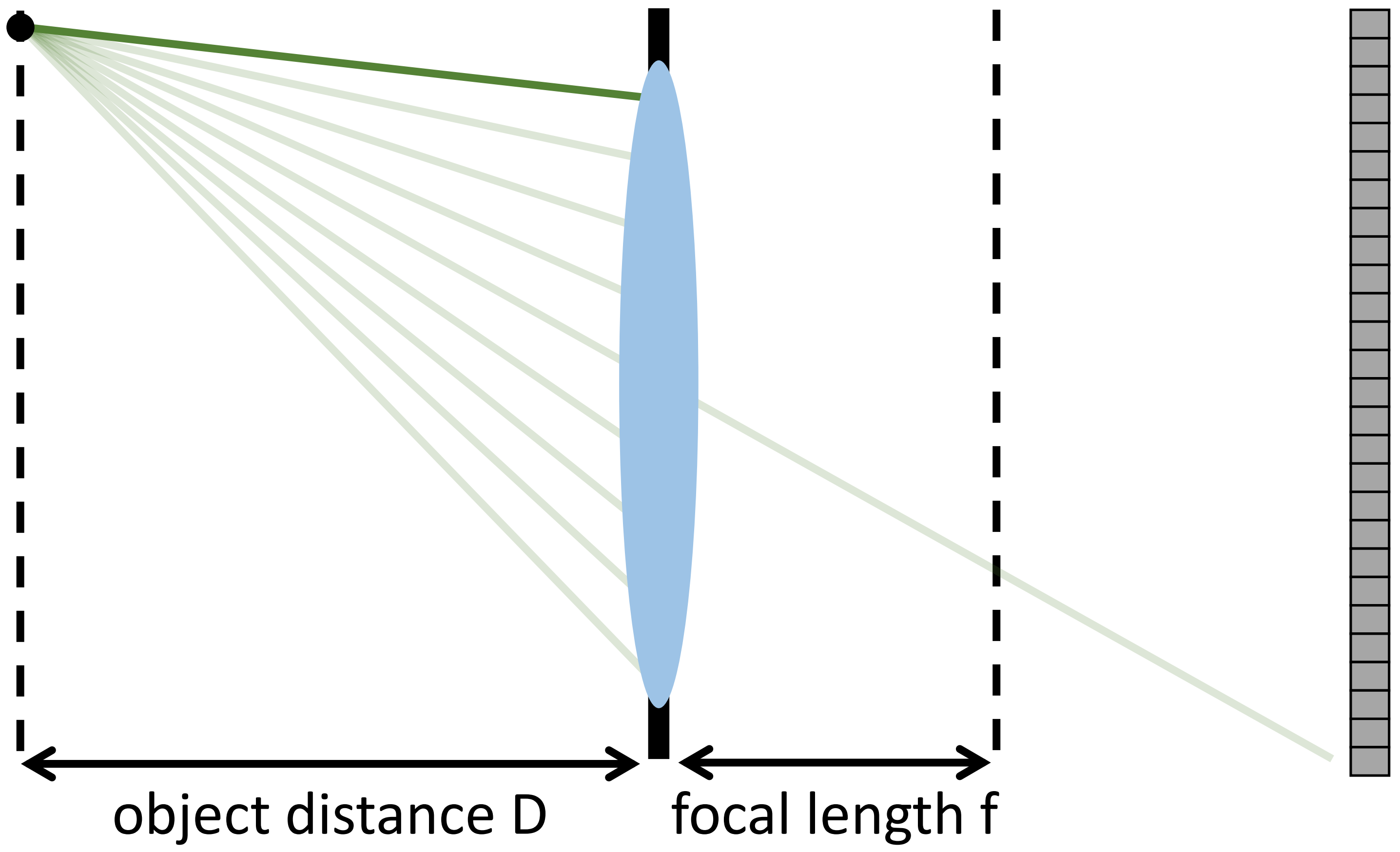
1. Trace rays through lens center.



Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?

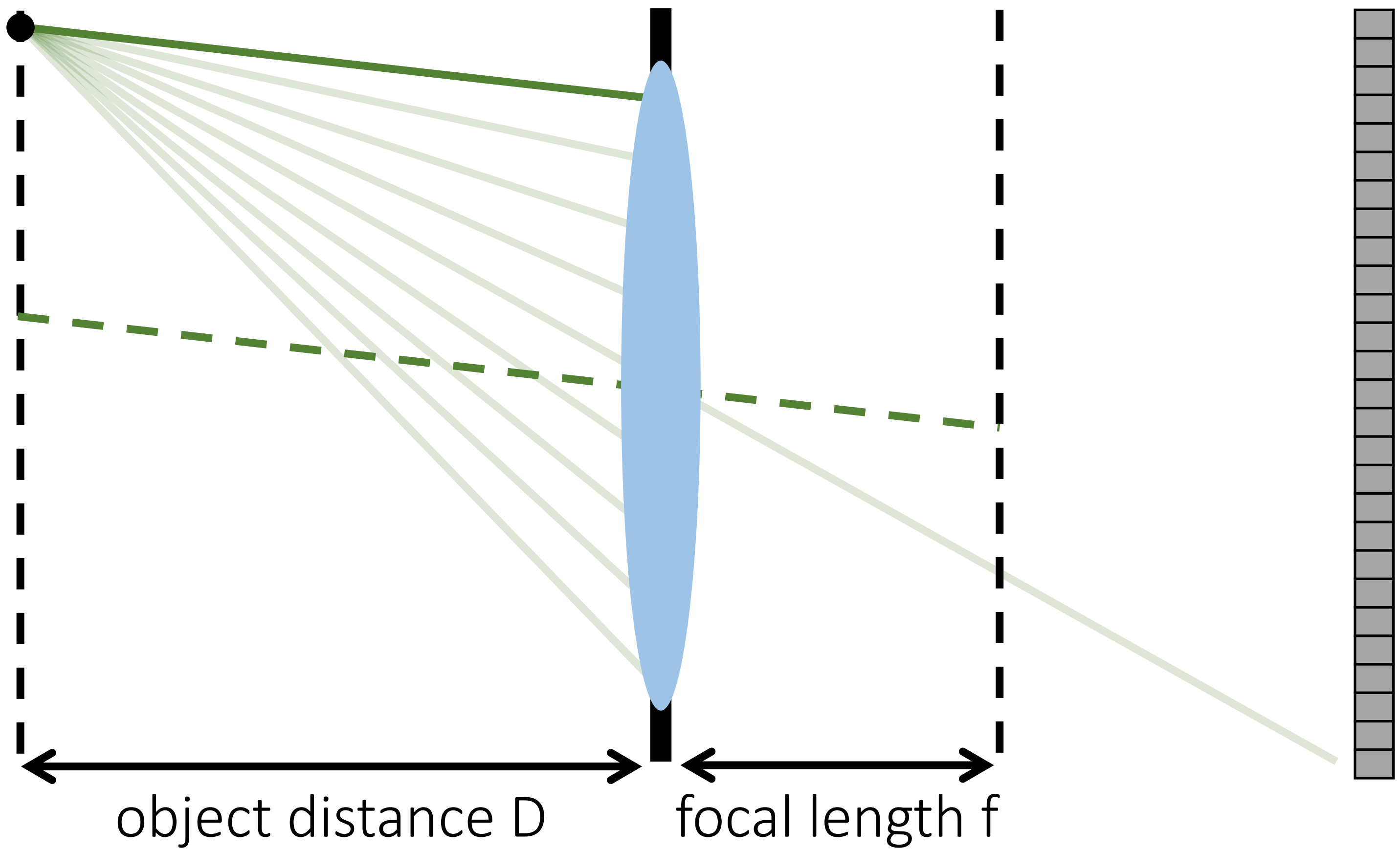
1. Trace rays through lens center.
2. For all other rays:



Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?

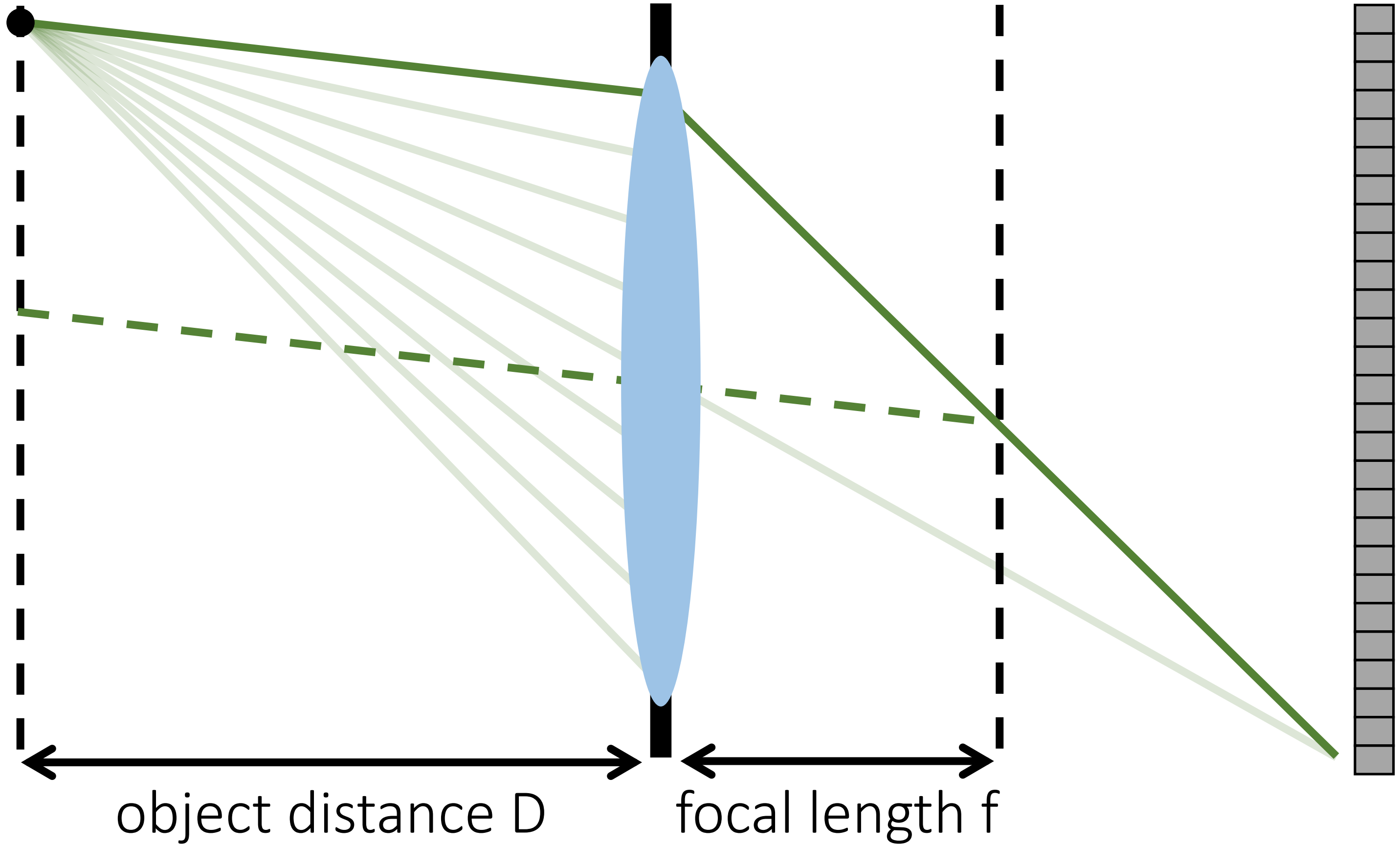
1. Trace rays through lens center.
2. For all other rays:
   a. Trace their parallel through lens center.



Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?

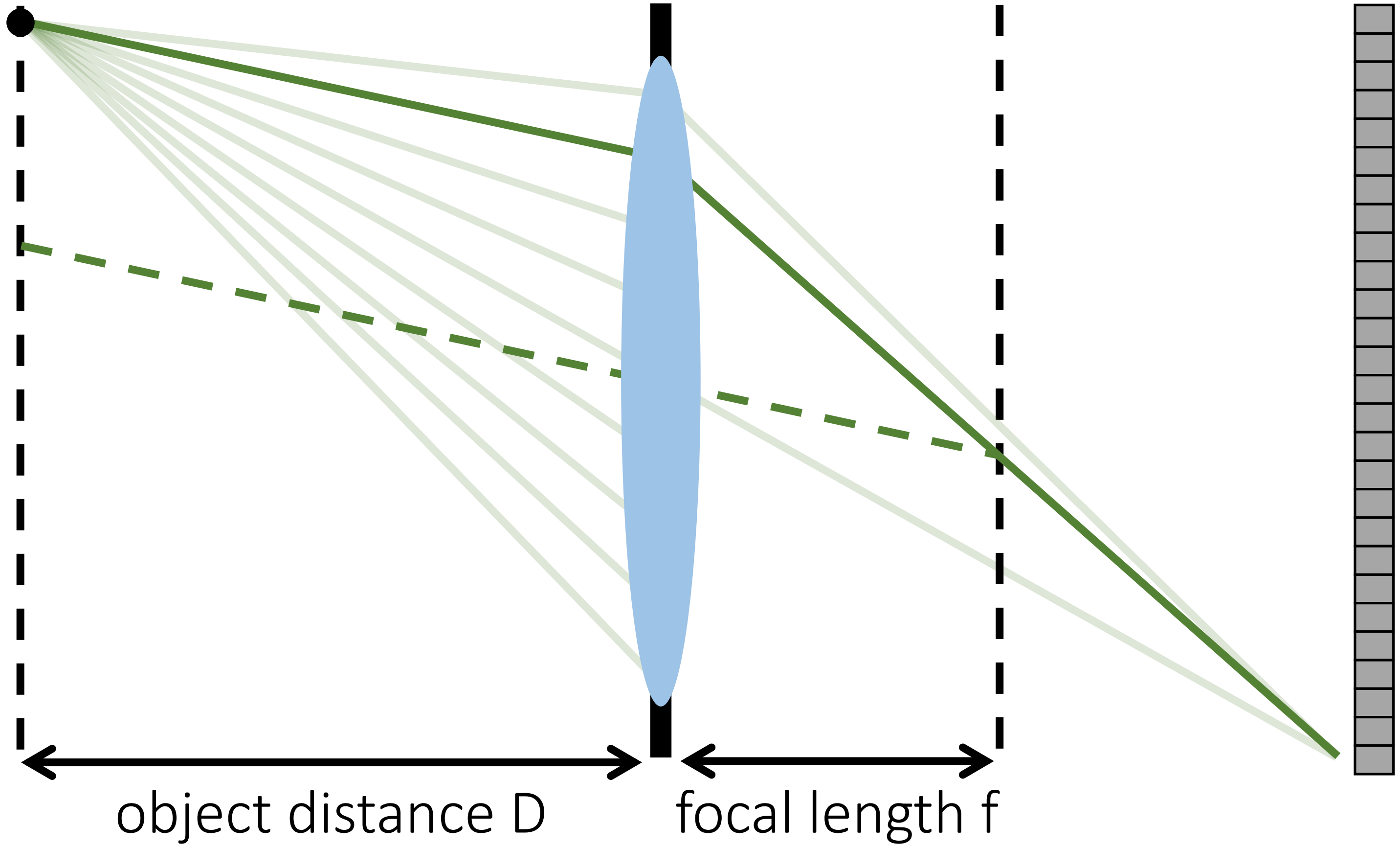
1. Trace rays through lens center.
2. For all other rays:
    a. Trace their parallel through lens center.
    b. Connect on focal plane.



Tracing rays through a thin lens

Consider an object emitting a bundle of rays. How do they propagate through the lens?

1. Trace rays through lens center.
2. For all other rays:
   a. Trace their parallel through lens center.
   b. Connect on focal plane.



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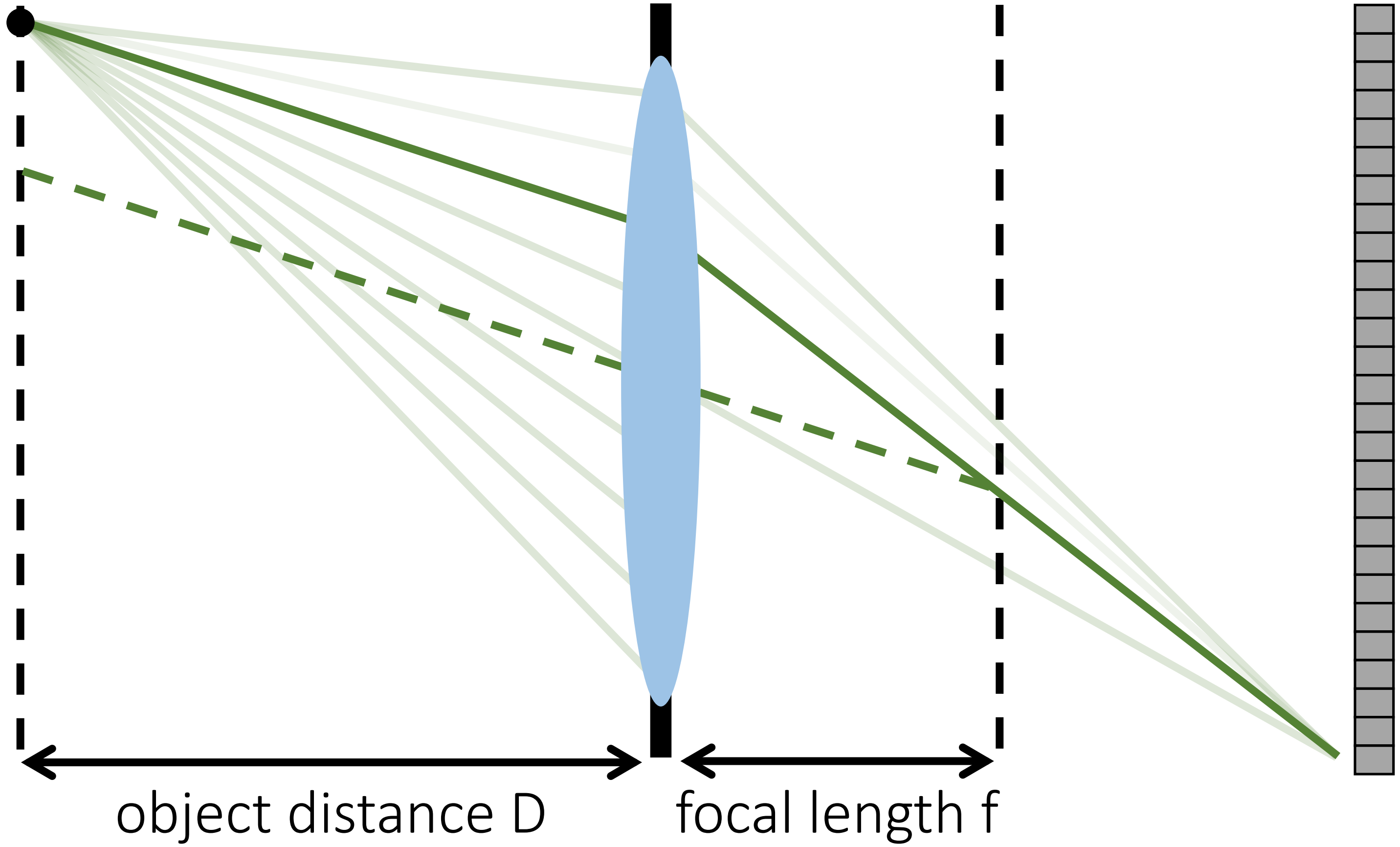

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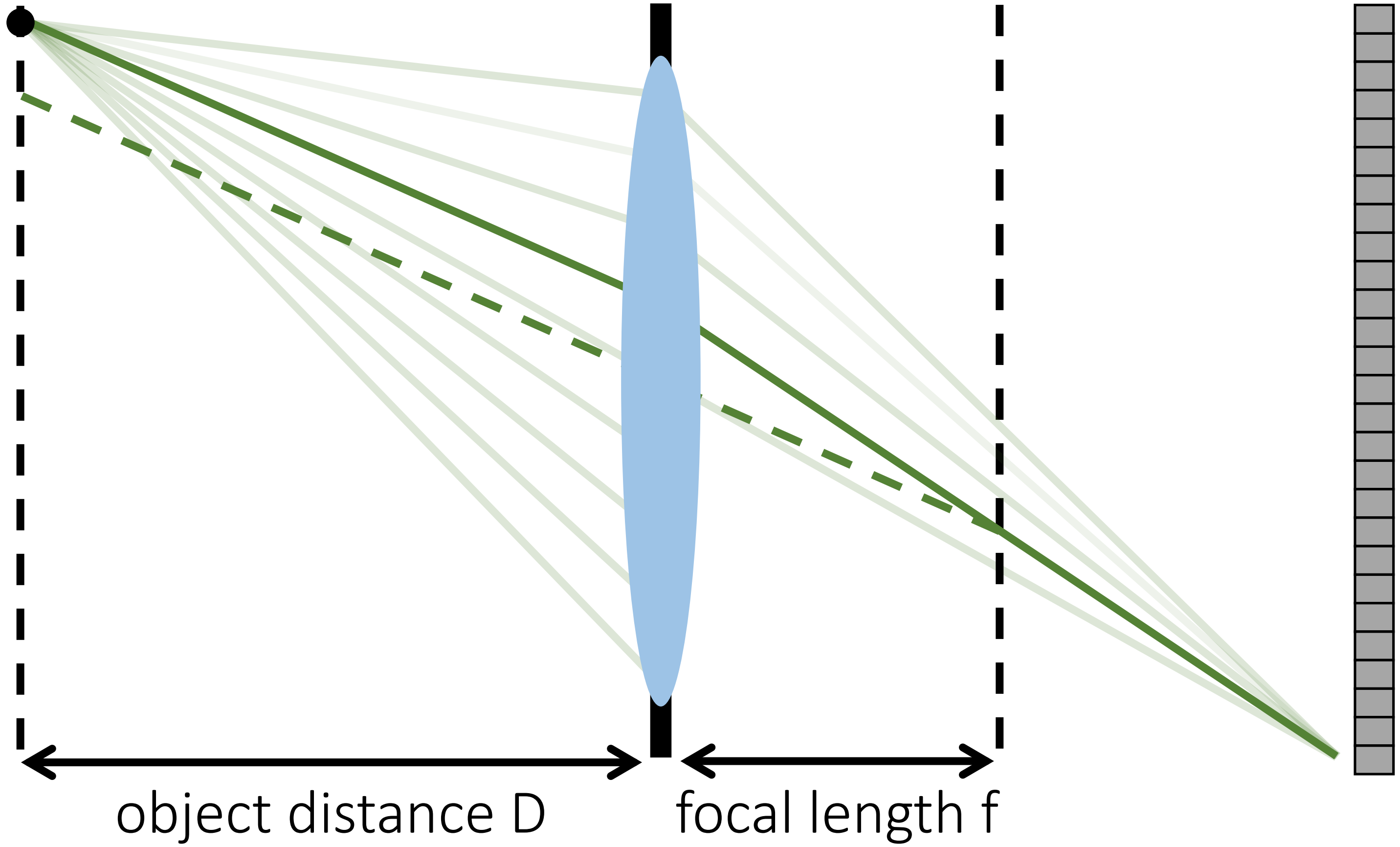

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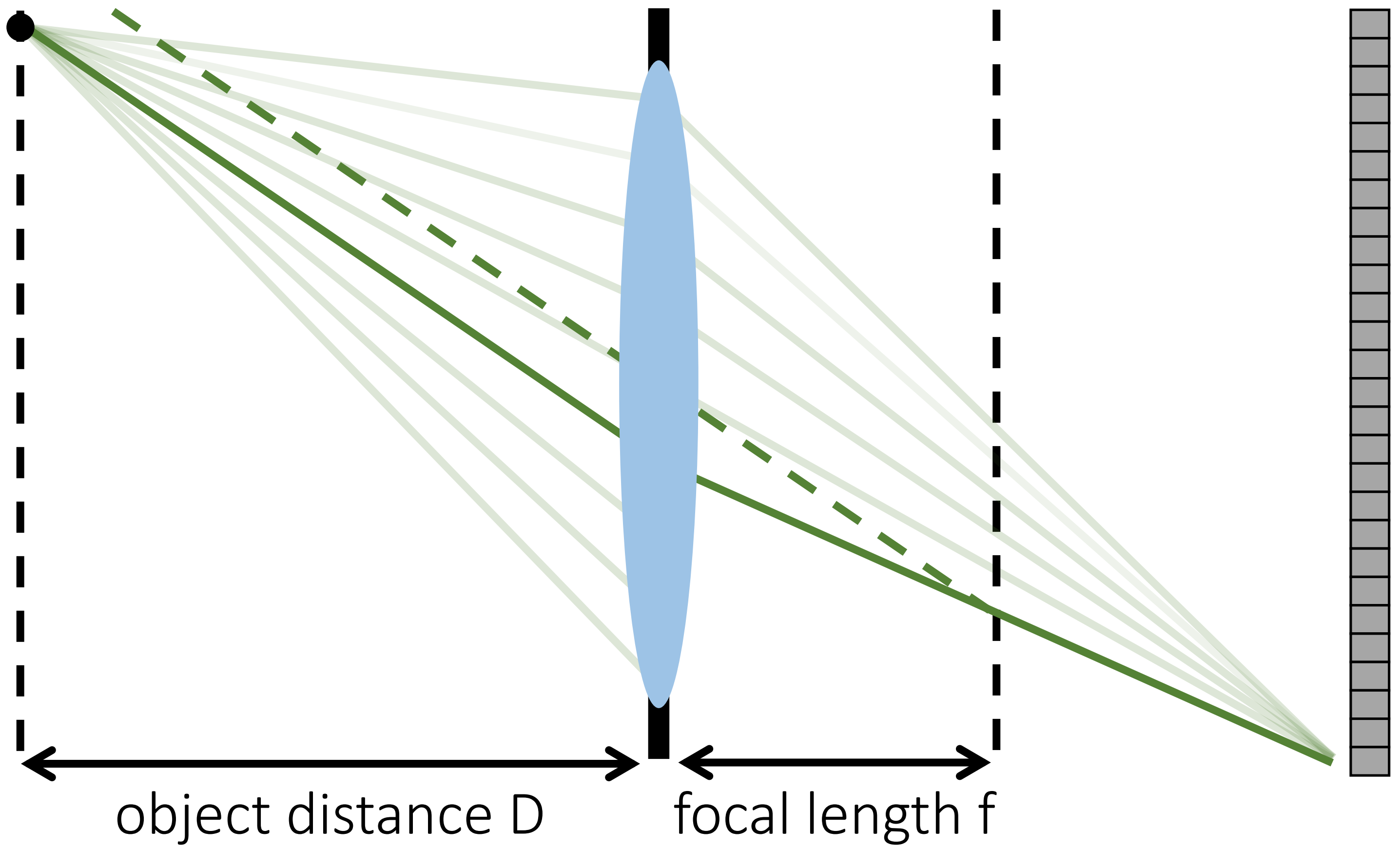

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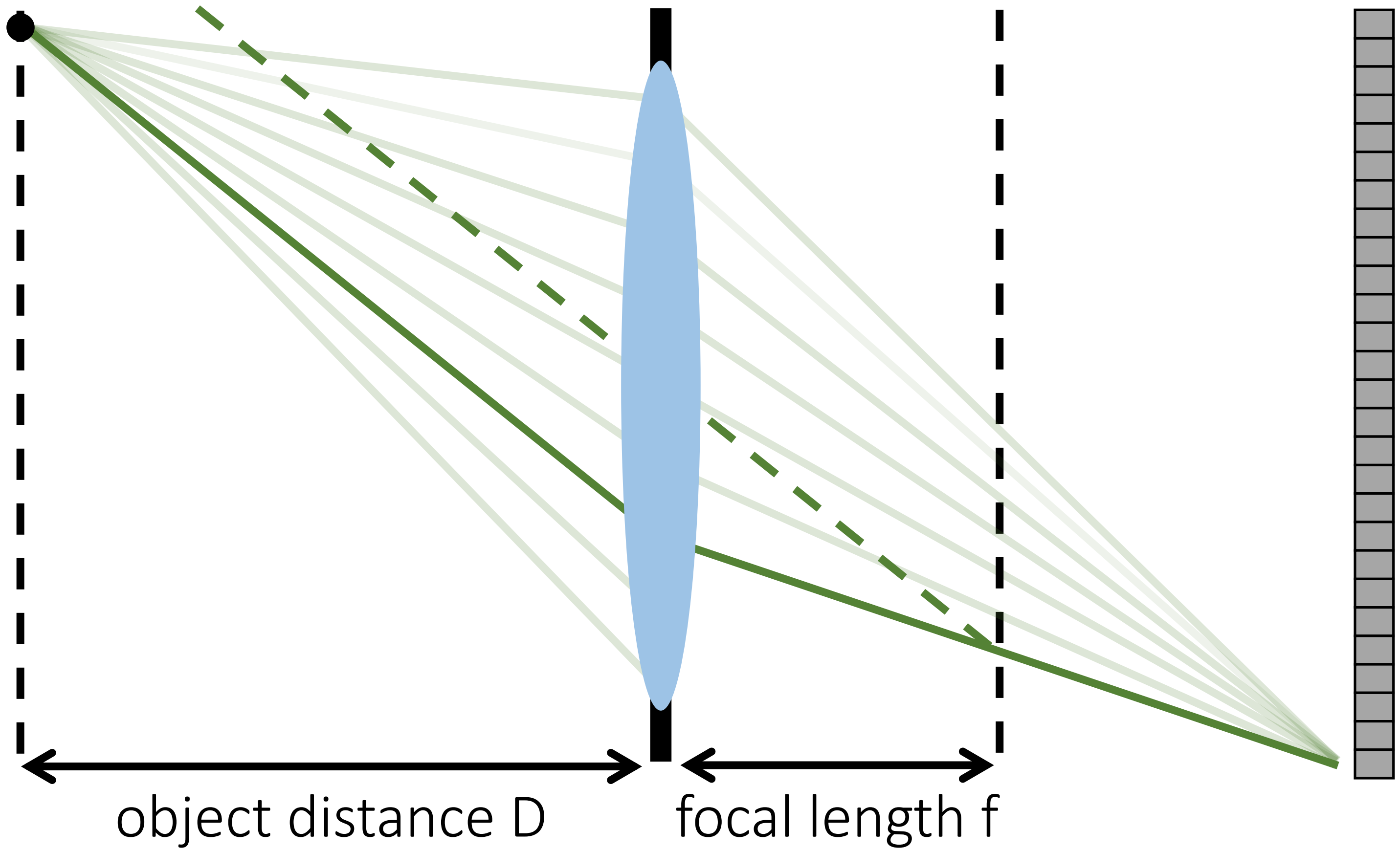

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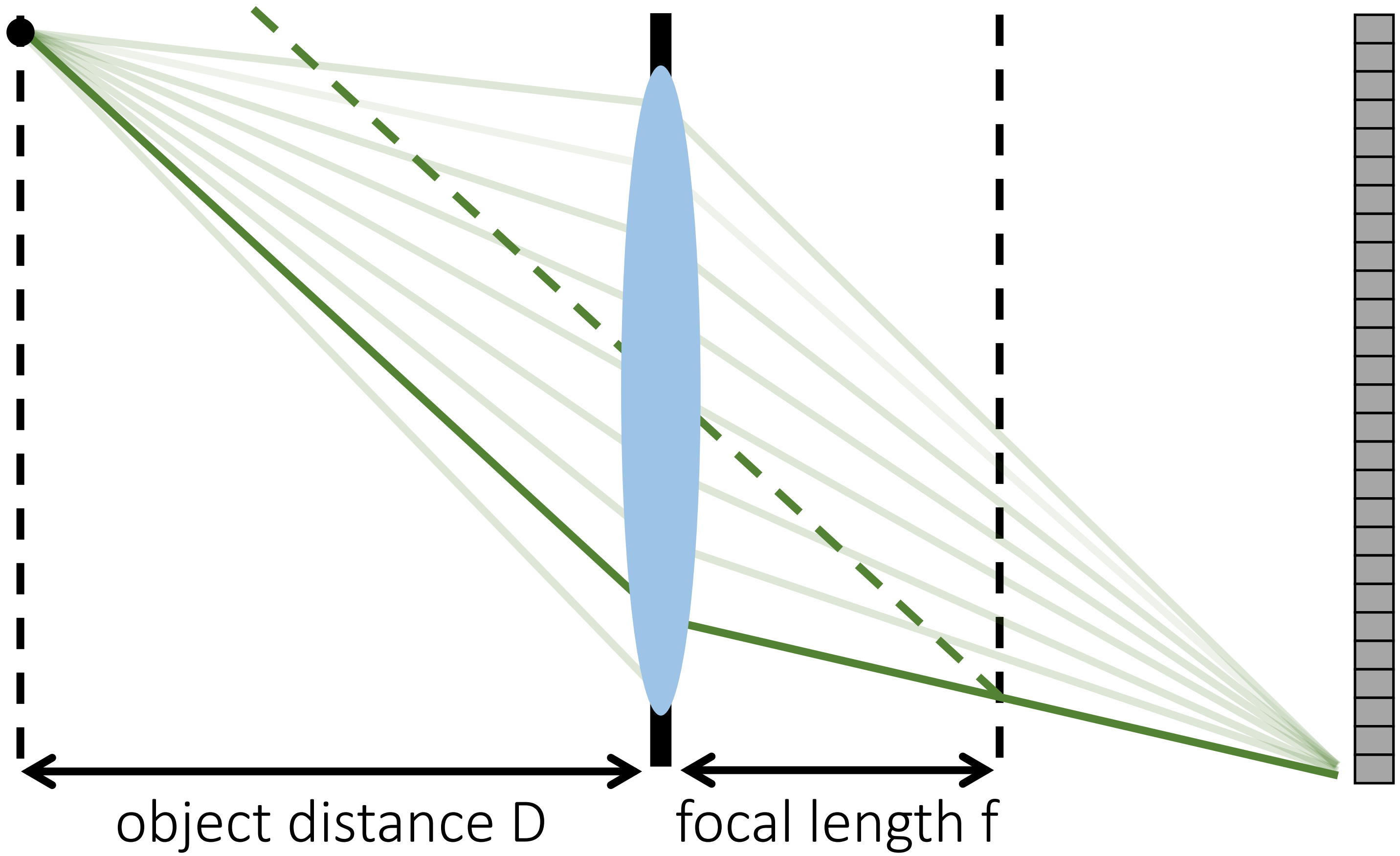

Tracing rays through a thin lens

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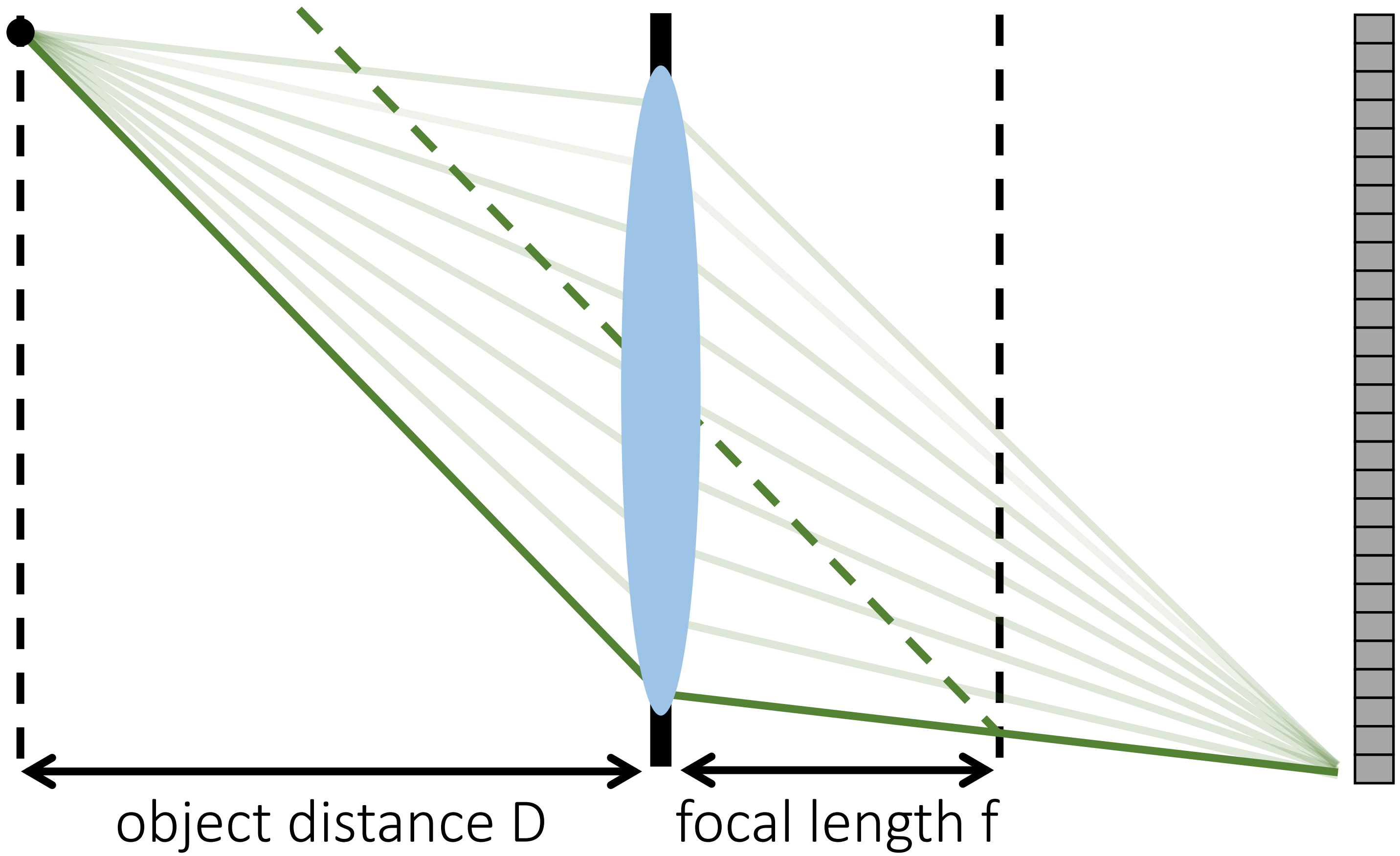

Tracing rays through a thin lens

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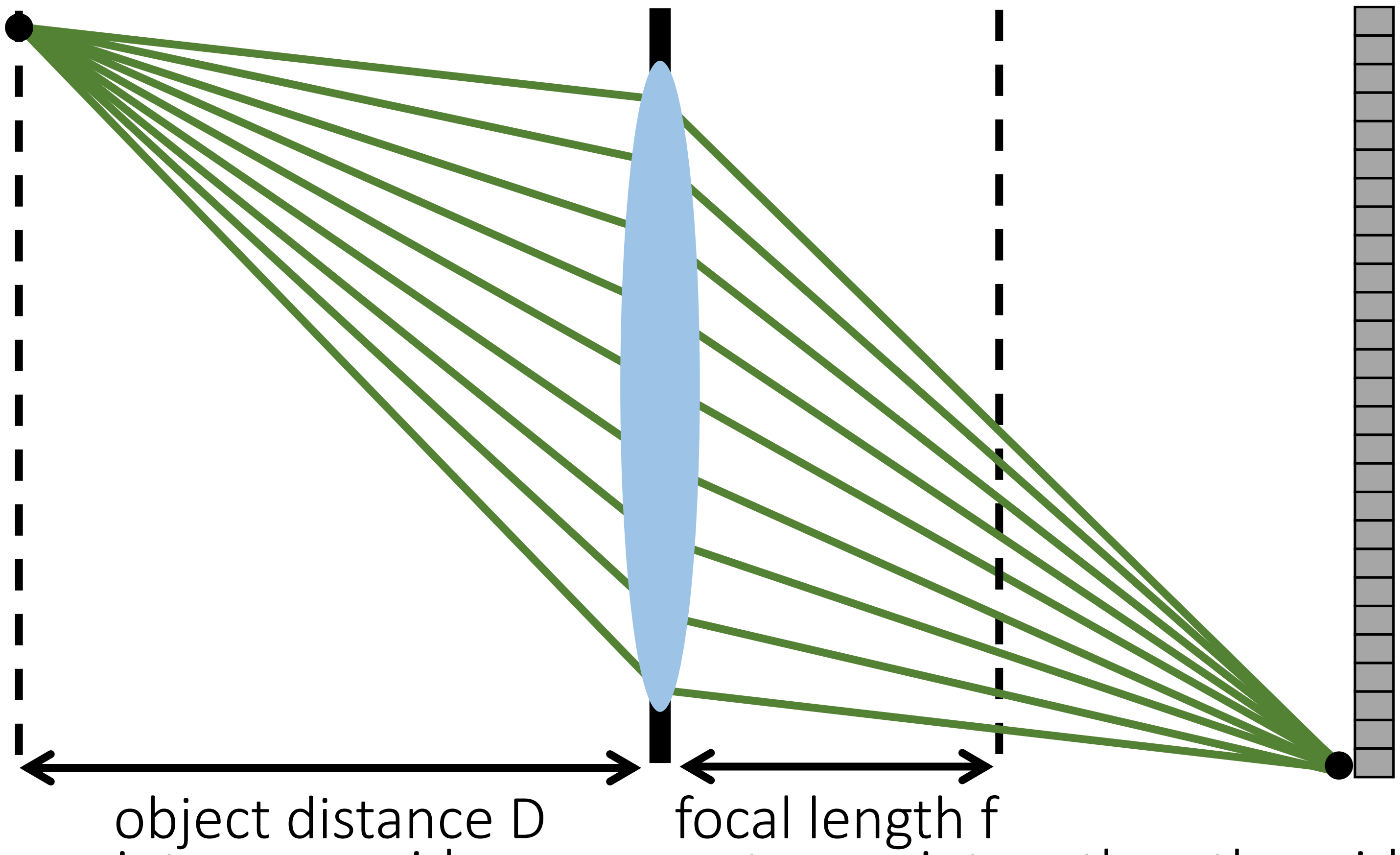
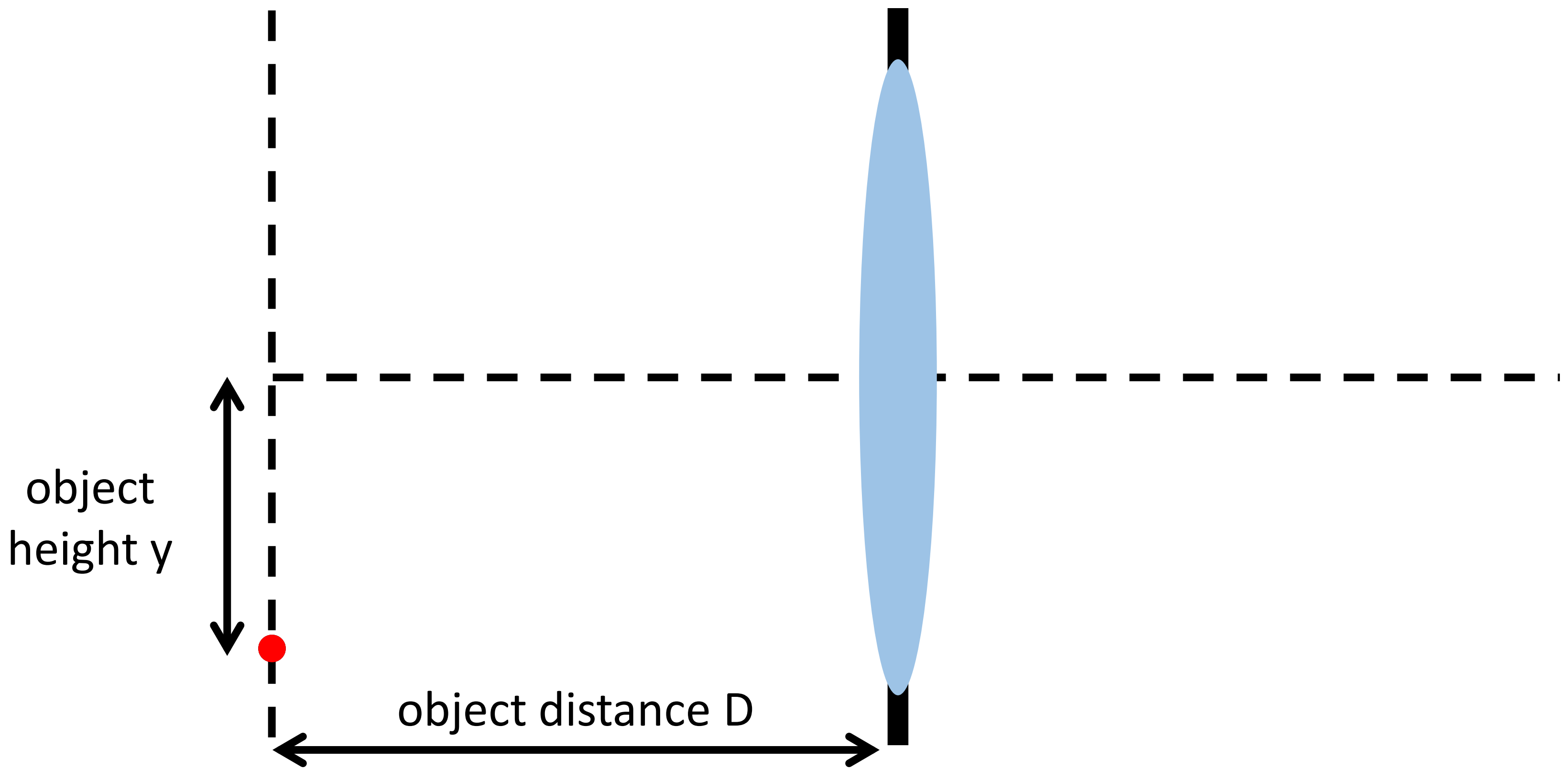


Focusing property:

1. Rays emitted from a point on one side converge to a point on the other side.

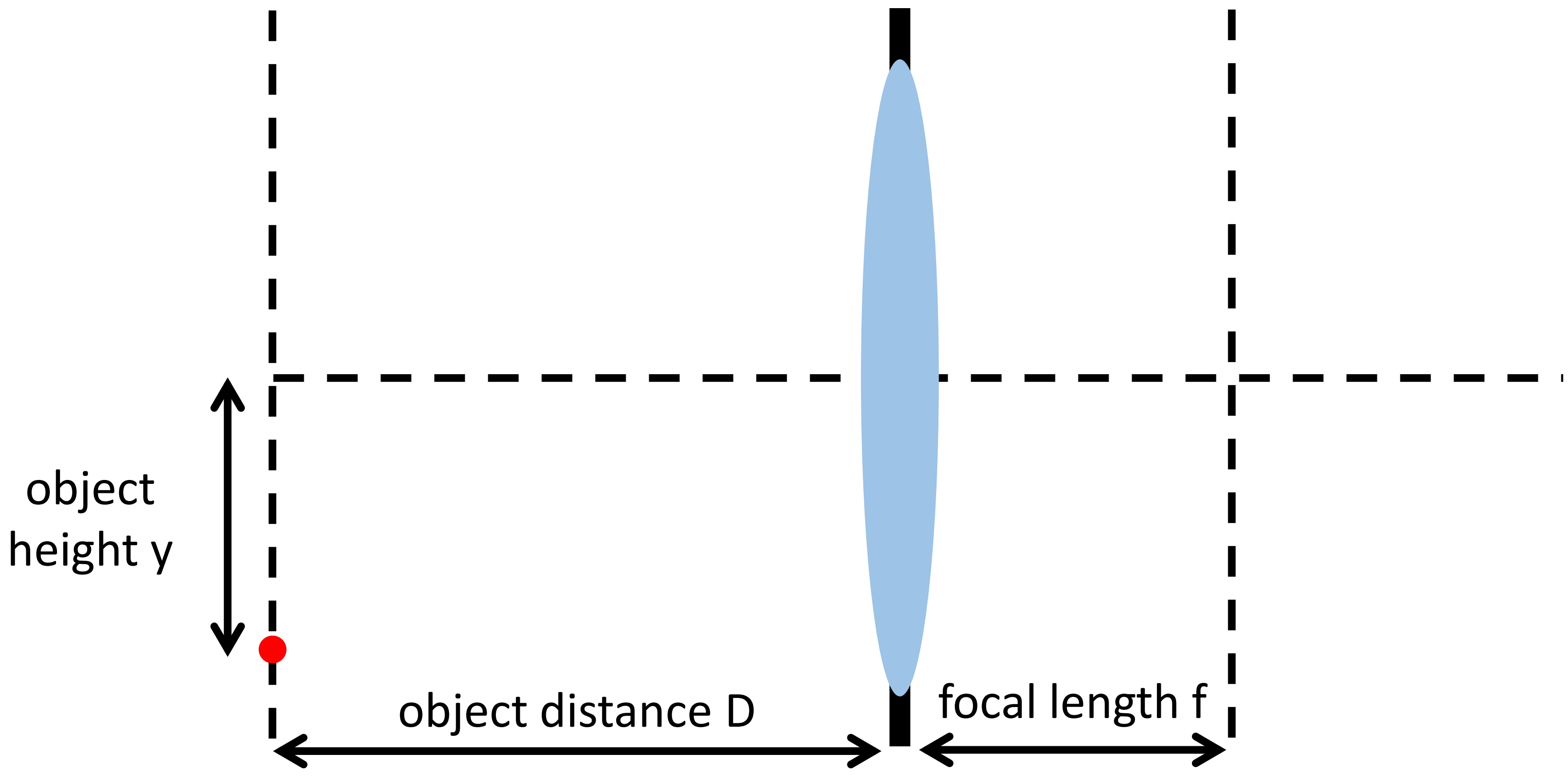
Gaussian (thin) lens formula

How can we relate scene-space (D, y) and image space (D’, y’) quantities?



Gaussian (thin) lens formula

How can we relate scene-space (D, y) and image space (D', y') quantities?



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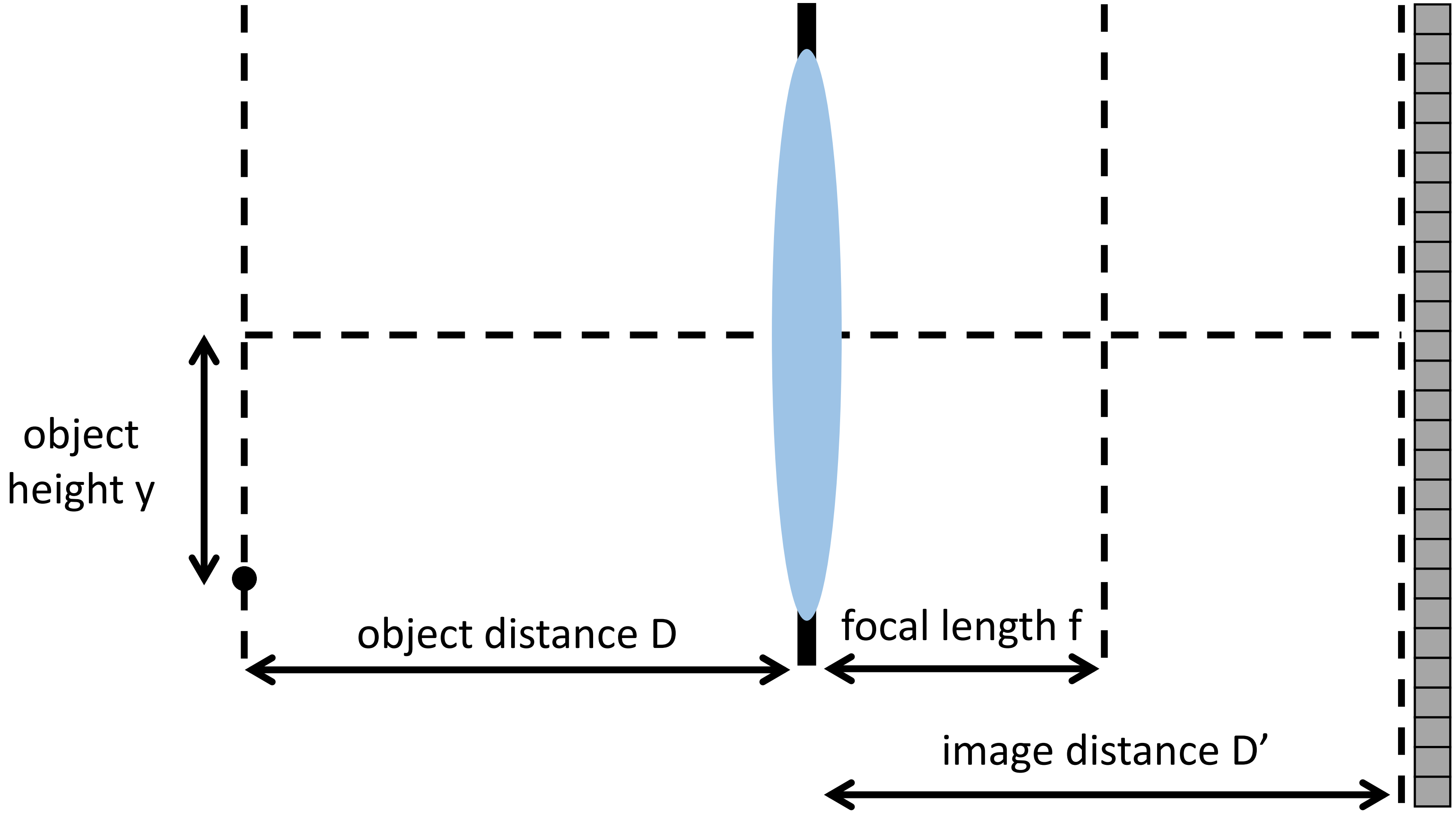

Gaussian (thin) lens formula

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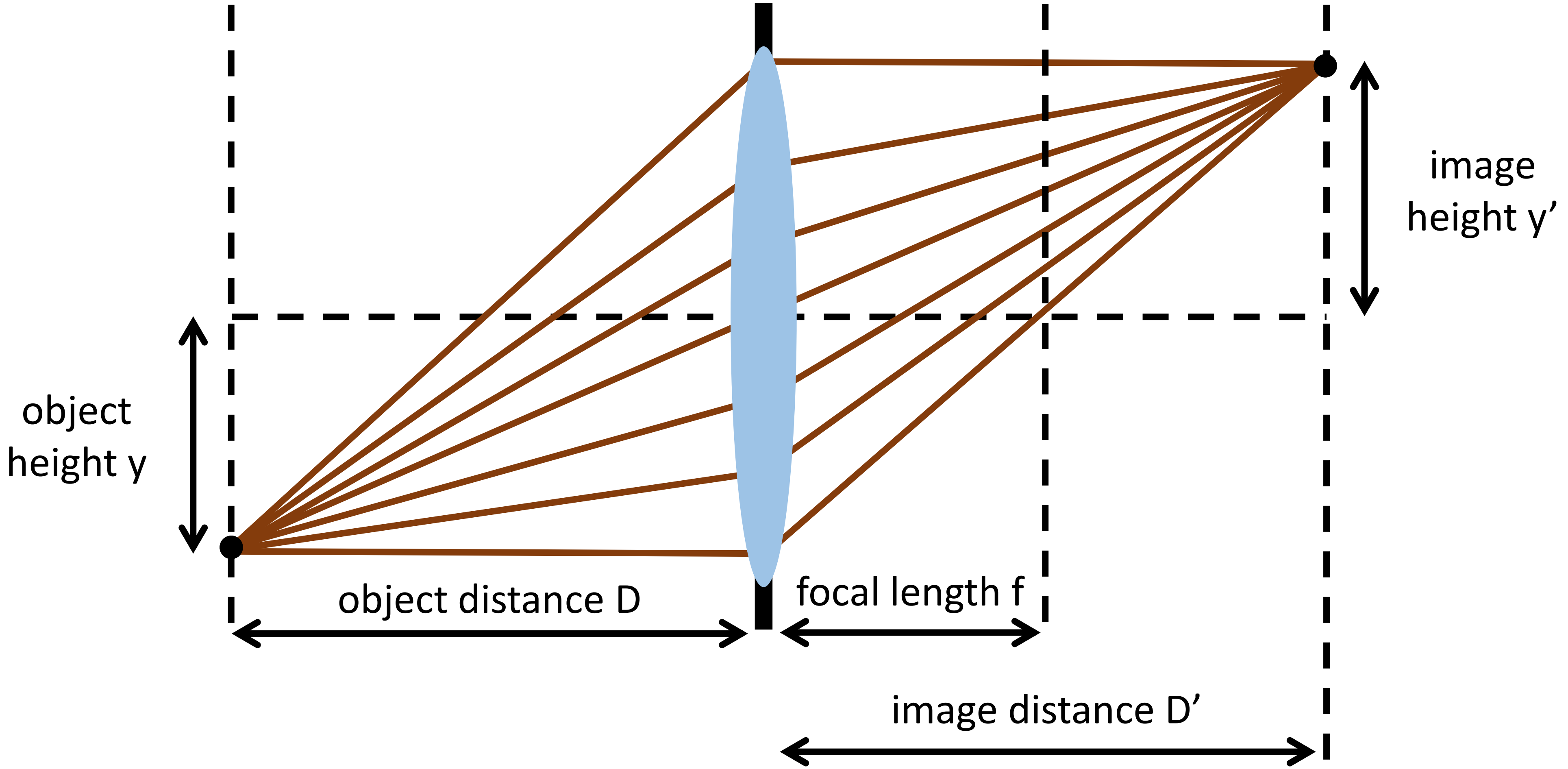
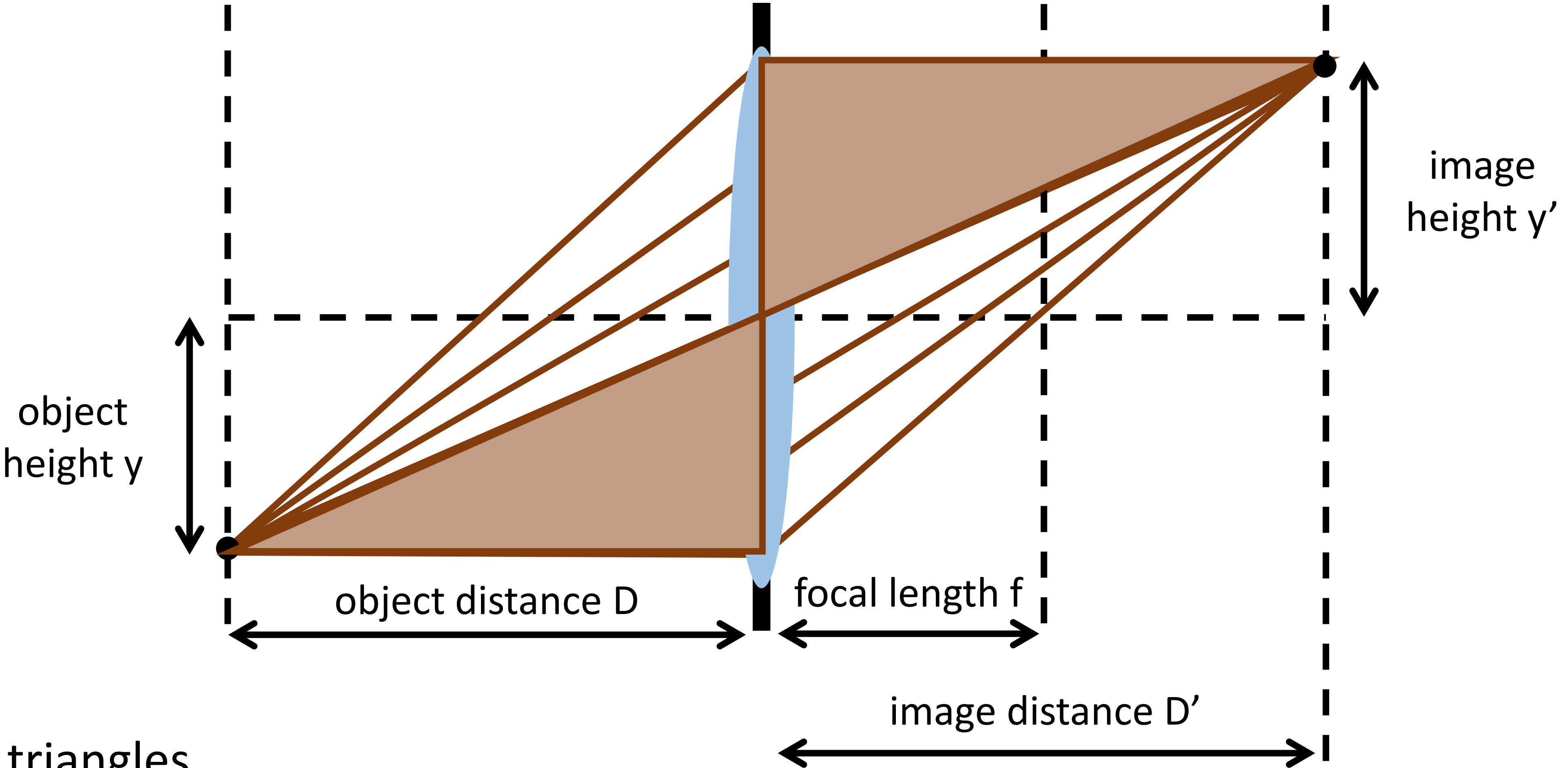


Gaussian (thin) lens formula

How can we relate scene-space (D, y) and image space (D’, y’) quantities?

$$\frac{y}{y'} = ?$$

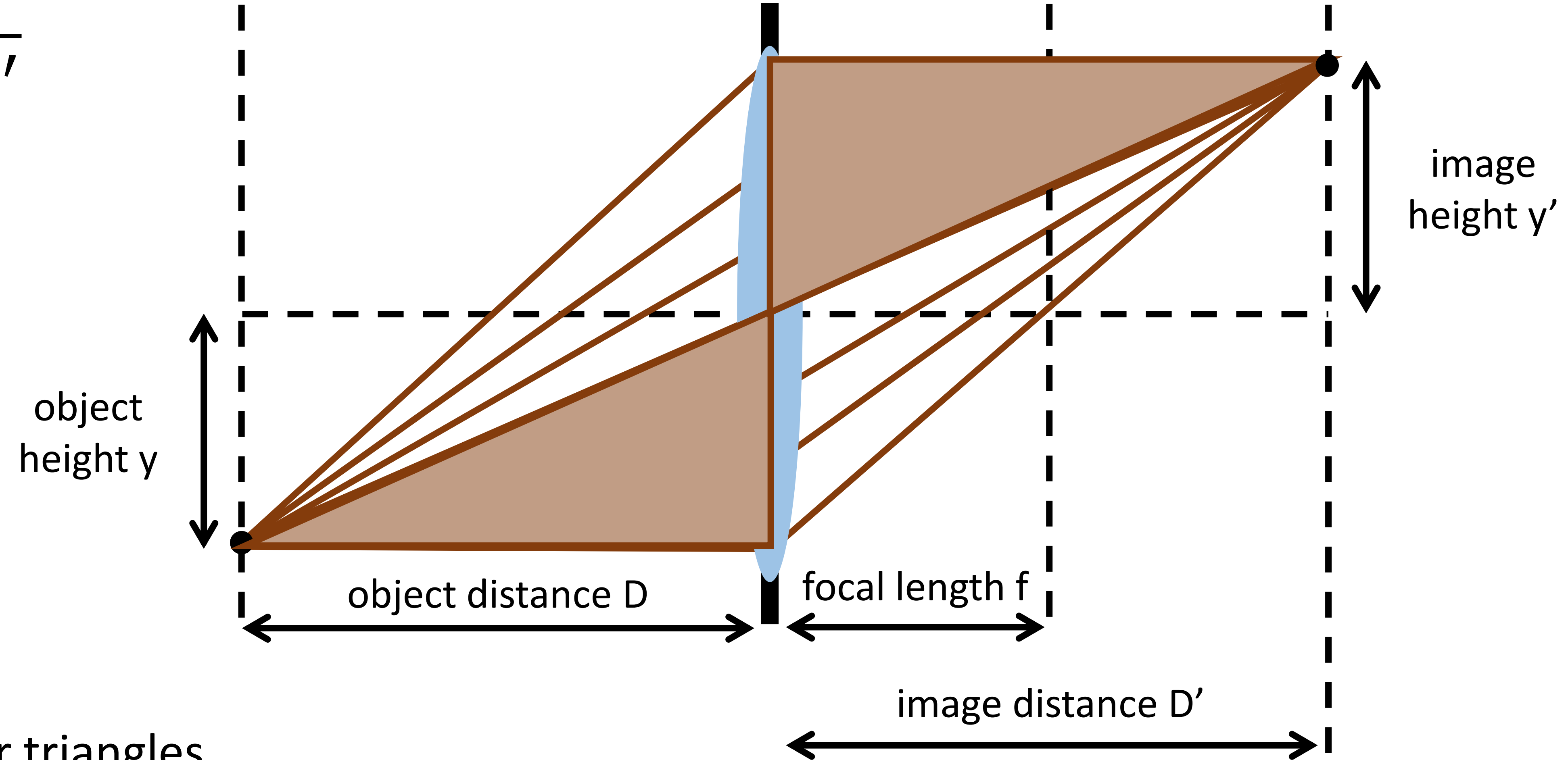


Use similar triangles

Gaussian (thin) lens formula

How can we relate scene-space (D, y) and image space (D’, y’) quantities?

$$\frac{y}{y'} = \frac{D}{D'}$$



Use similar triangles

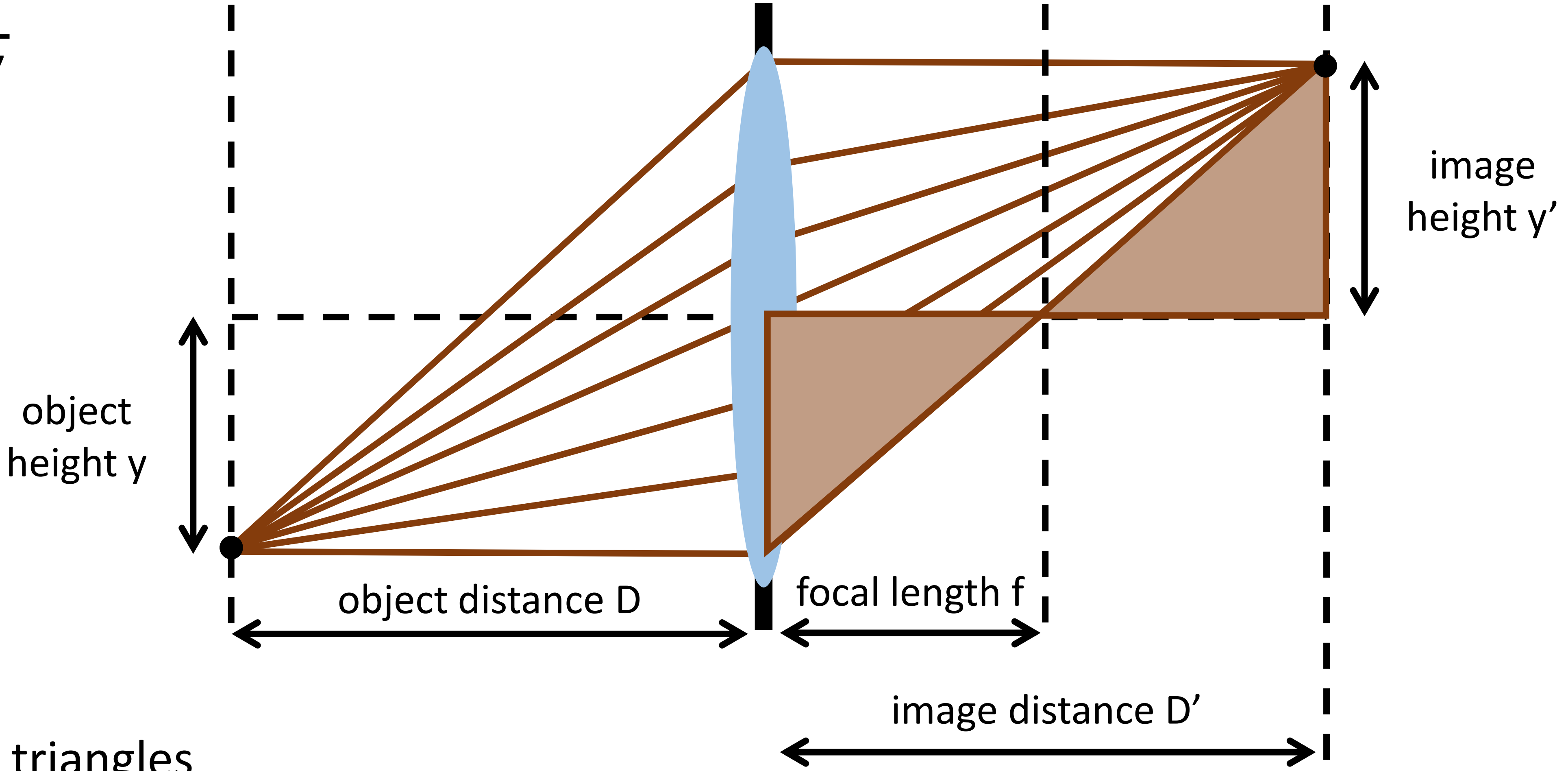
Sources: CMU 15-463, Dartmouth College CS89/189.

Gaussian (thin) lens formula

How can we relate scene-space (D, y) and image space (D', y') quantities?

$$\frac{y}{y'} = \frac{D}{D'}$$

$$\frac{y}{y'} = ?$$



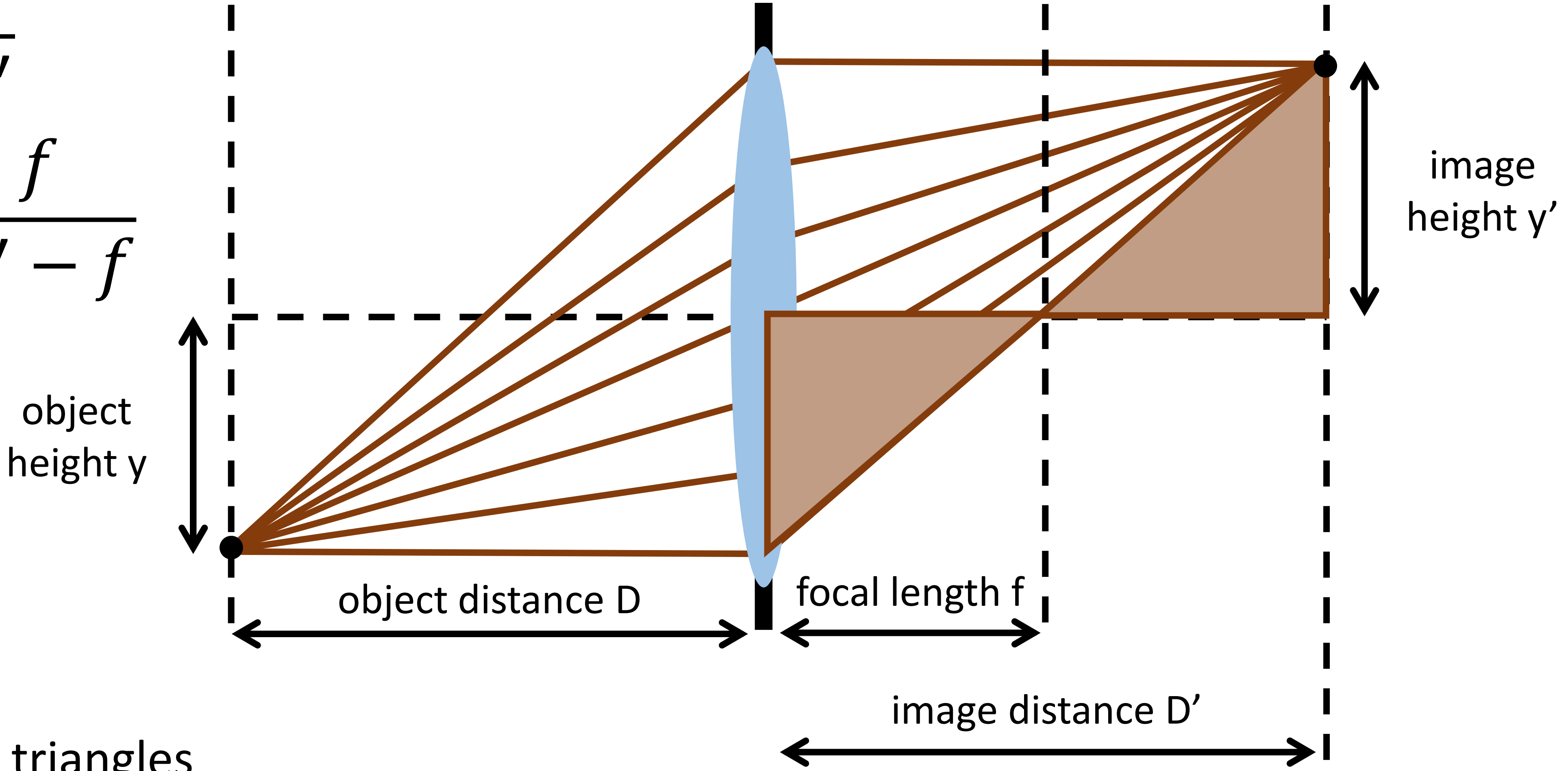
Use similar triangles

Gaussian (thin) lens formula

How can we relate scene-space (D, y) and image space (D', y') quantities?

$$\frac{y}{y'} = \frac{D}{D'}$$

$$\frac{y}{y'} = \frac{f}{D' - f}$$



Use similar triangles

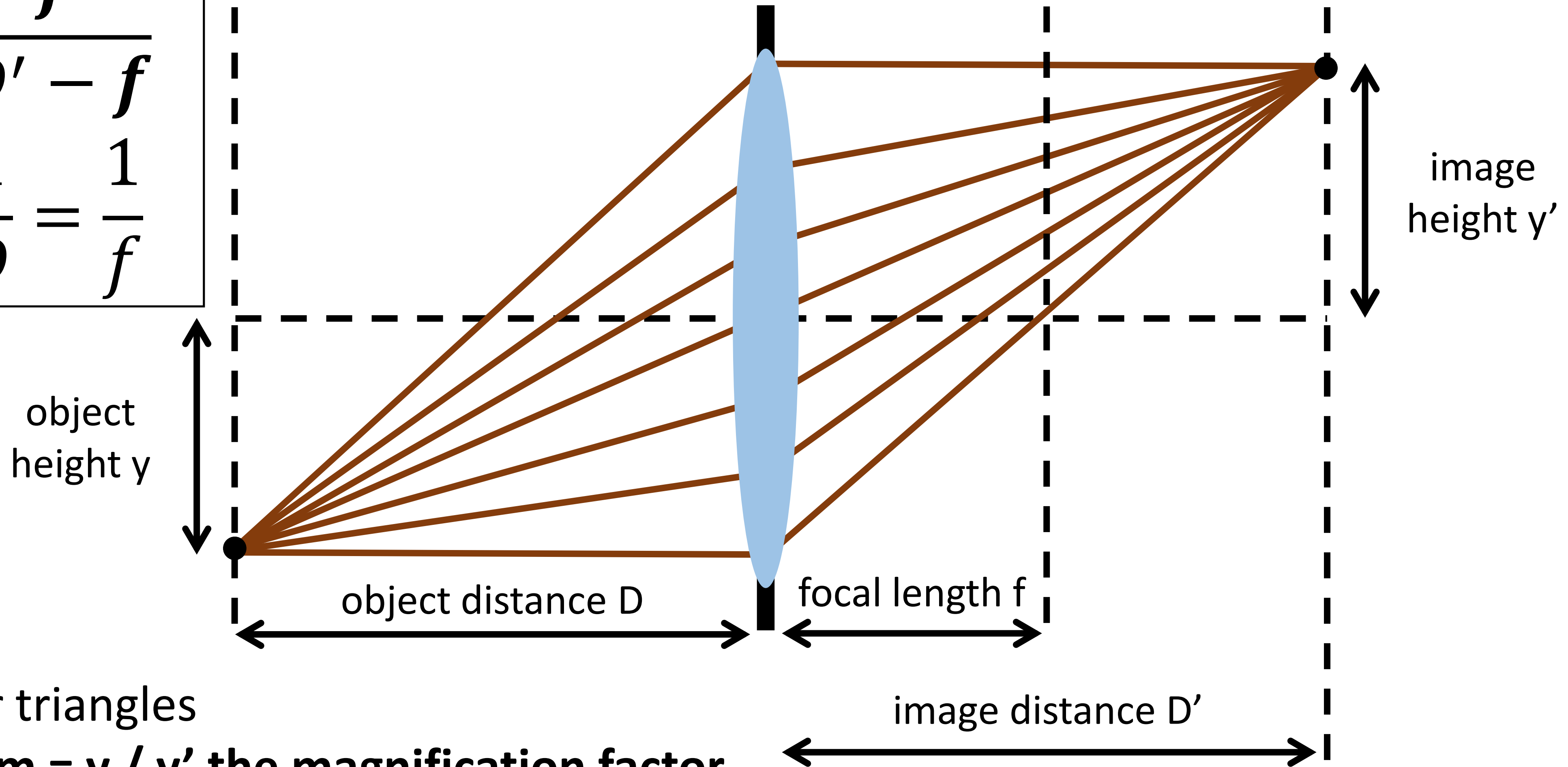
Gaussian (thin) lens formula

How can we relate scene-space (D, y) and image space (D’, y’) quantities?

$$m = \frac{f}{D' - f}$$

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

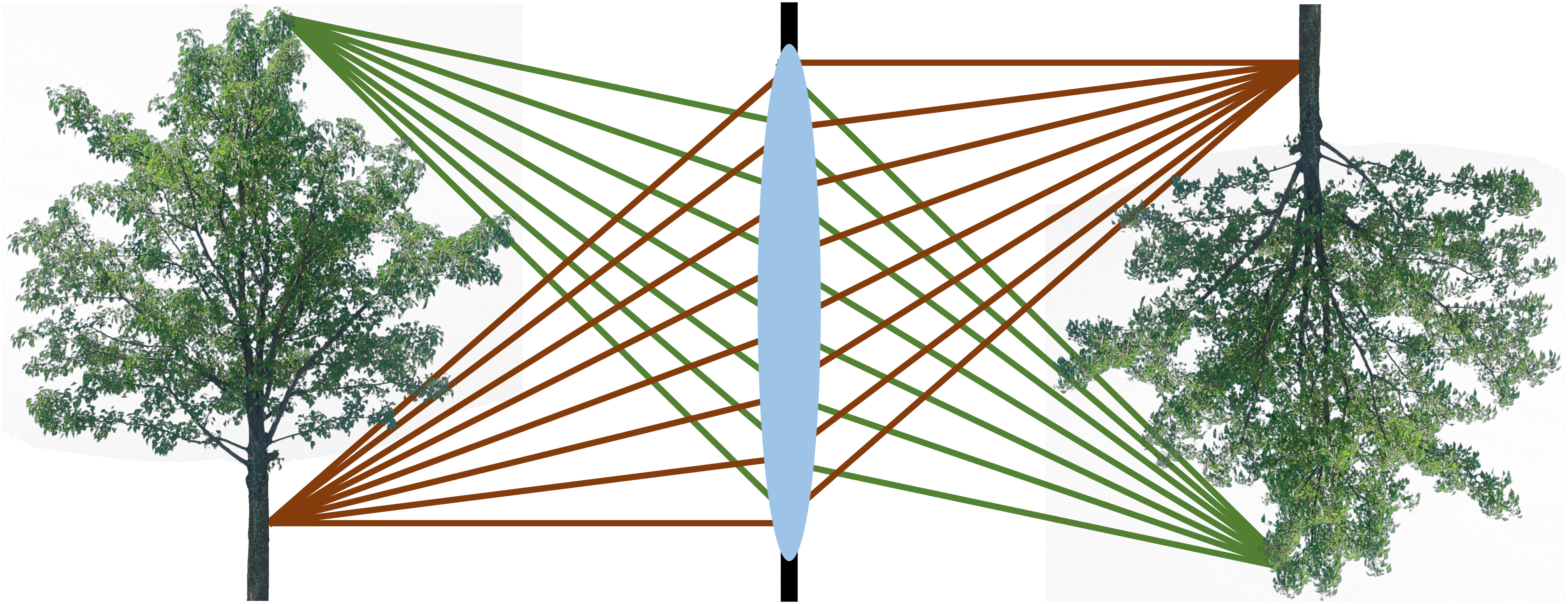
proof



Use similar triangles

- We call **m = y / y’ the magnification factor**

Problems?



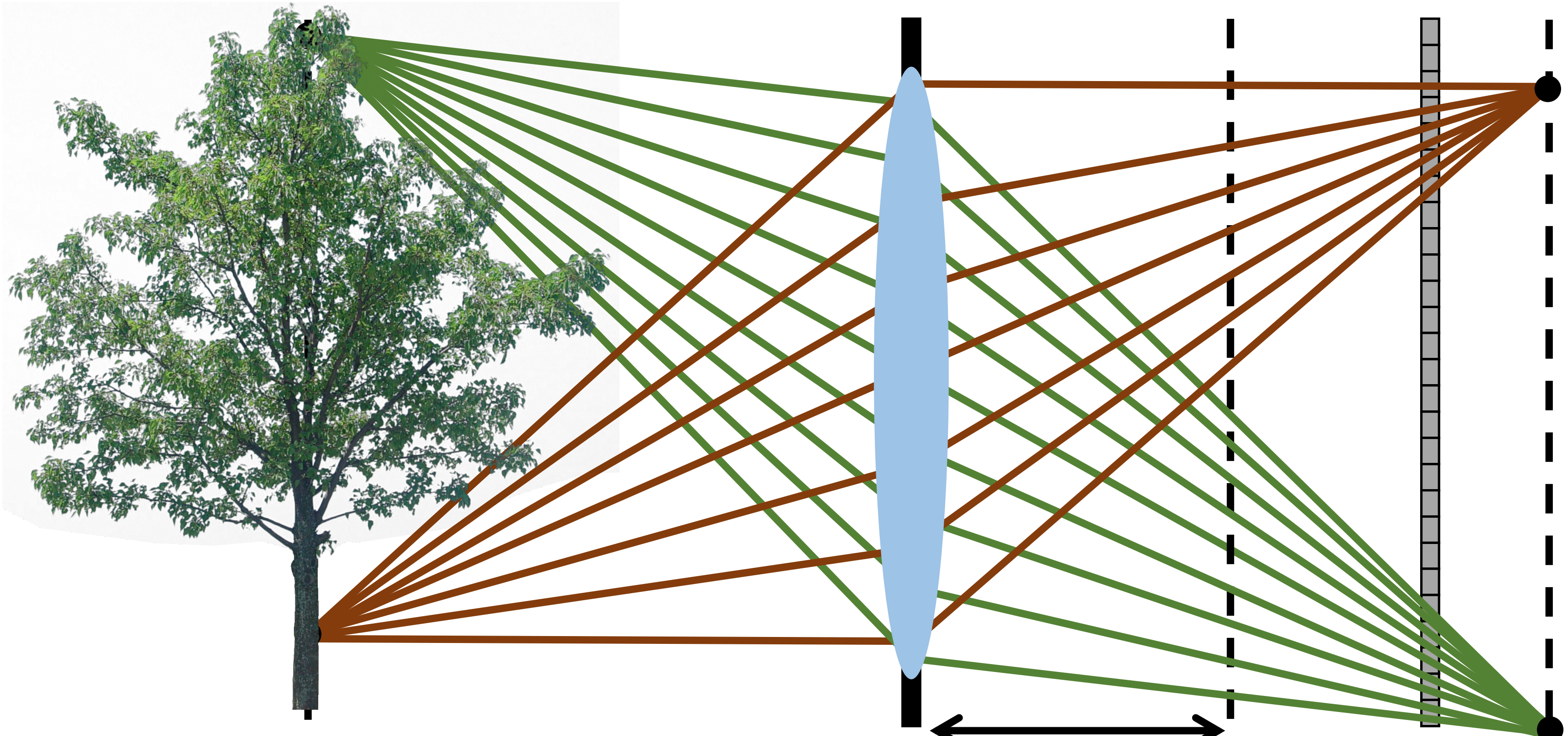
Do we lose anything by using a lens?

Sources: CMU 15-463, Dartmouth College CS89/189.

Defocus

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

What happens if we don't place the sensor at the focus distance?



Defocus

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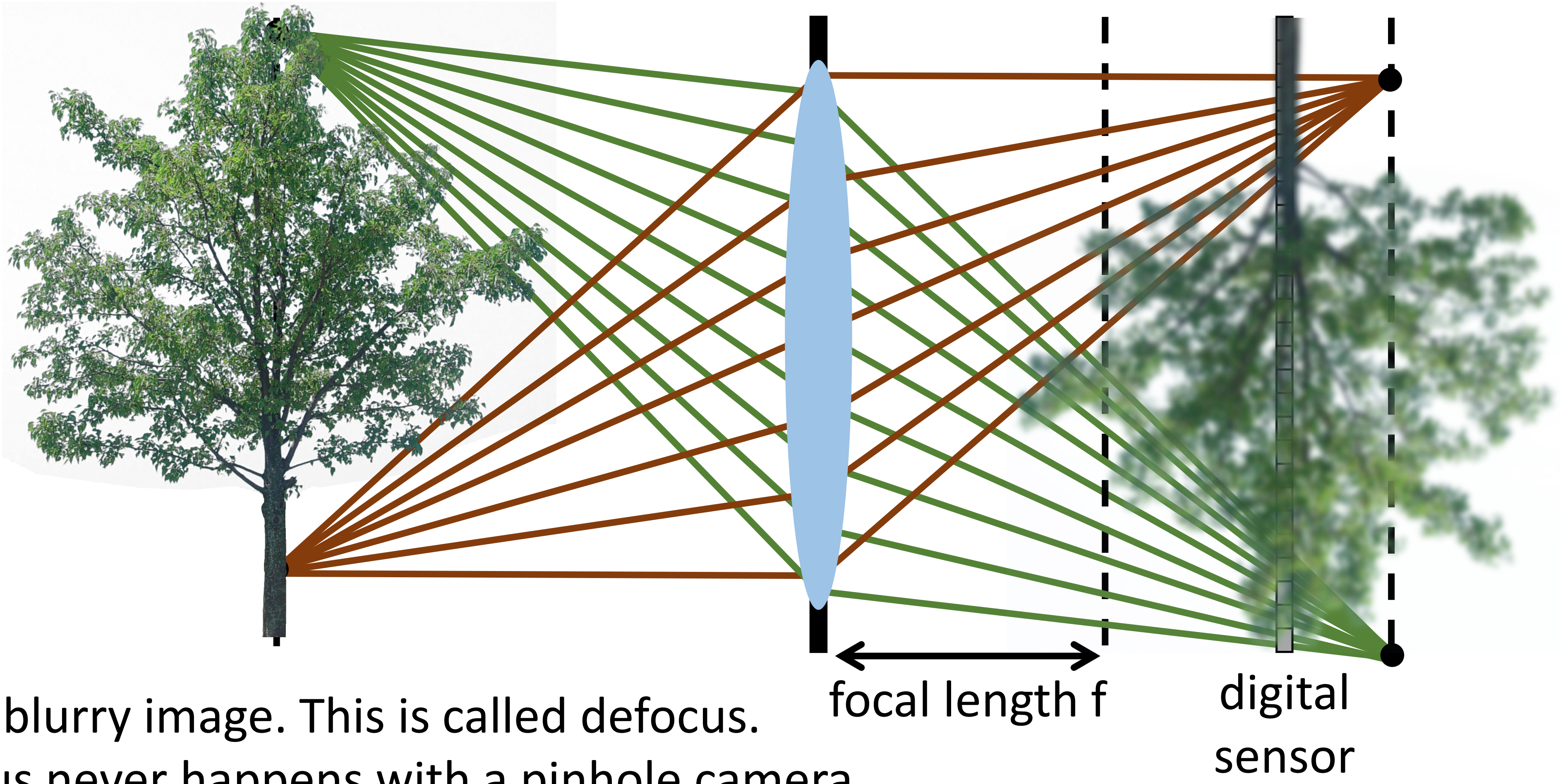


We get a blurry image. This is called defocus.

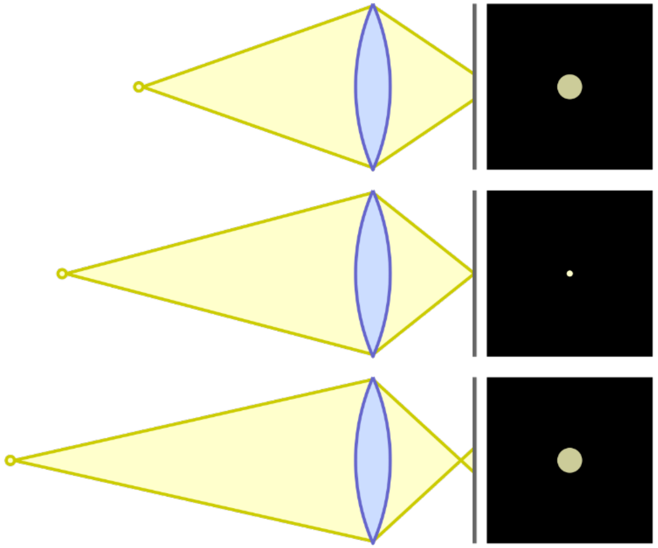
- Defocus never happens with a pinhole camera.

Defocus

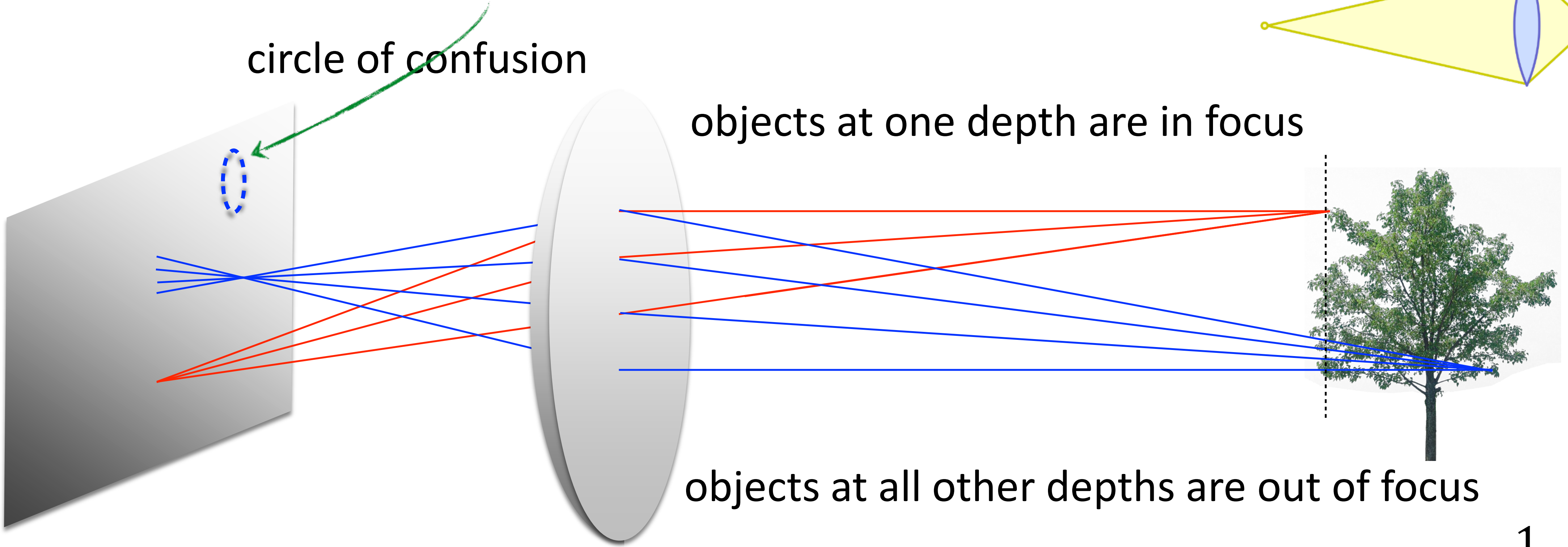
$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

Can't we just move the sensor to the correct distance?

Defocus



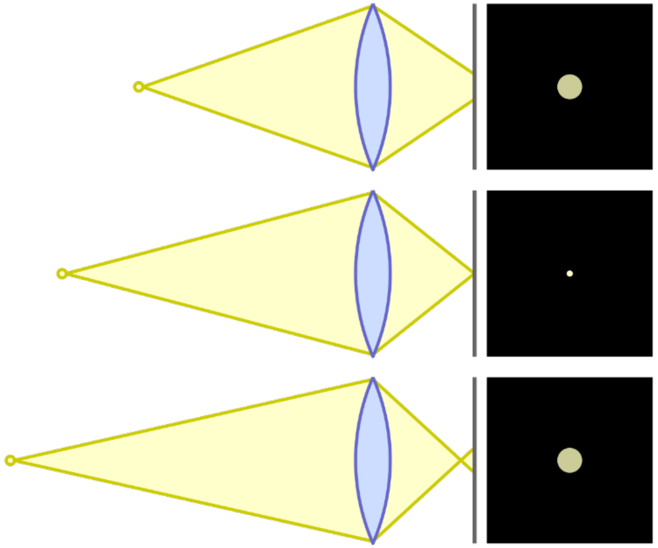
Can’t we just move the sensor to the correct distance?

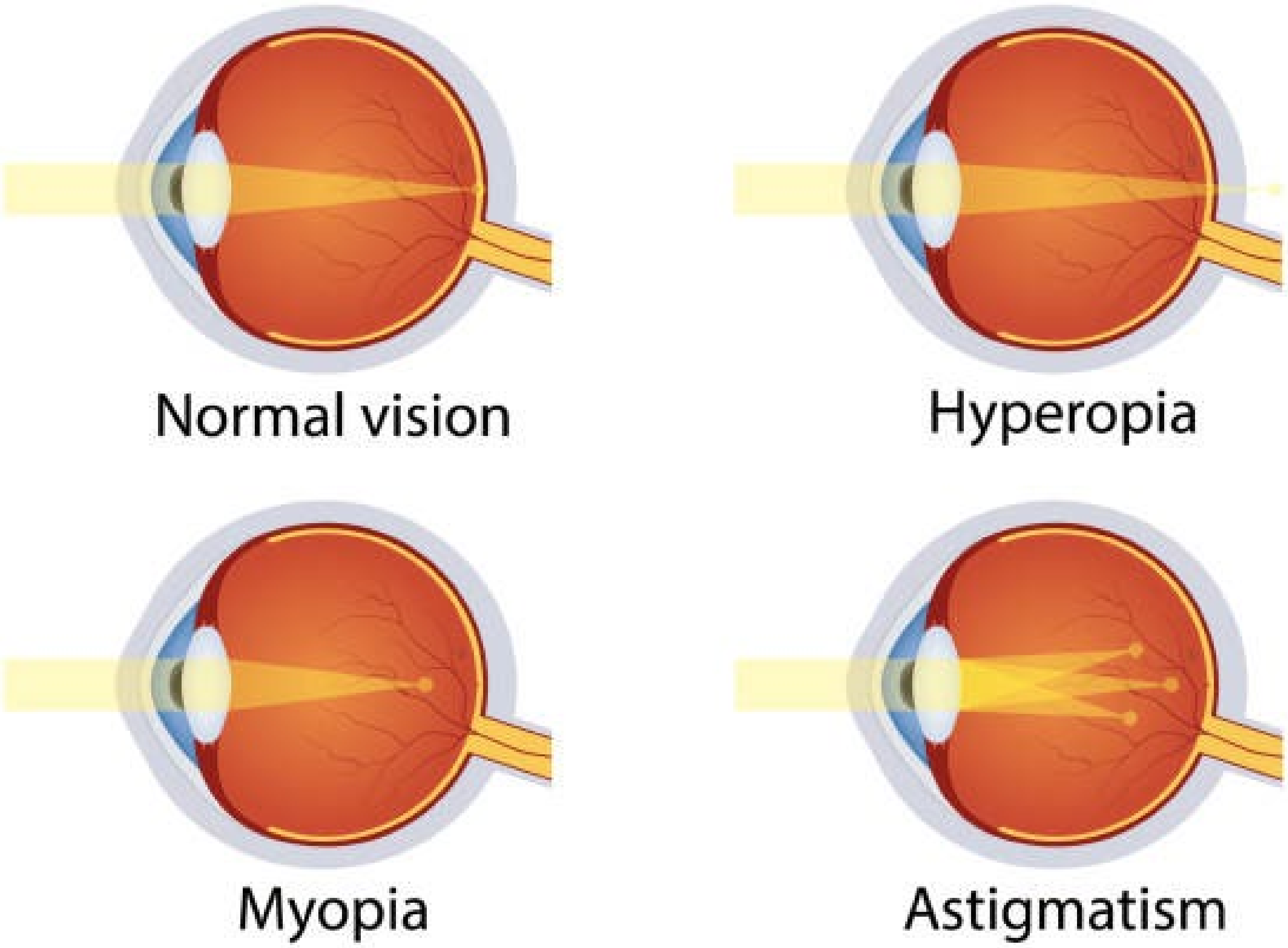


$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

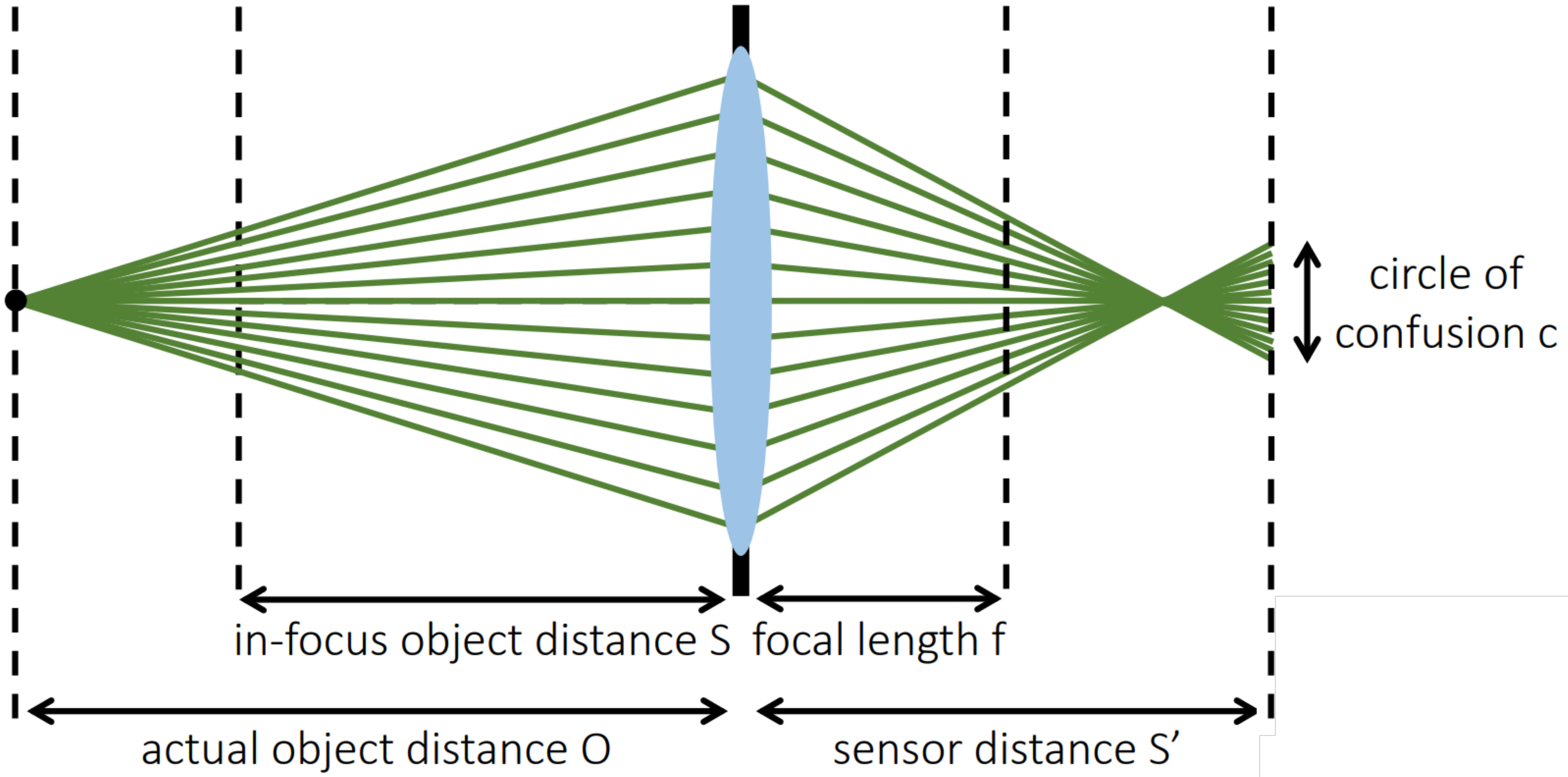
Unless our scene is just one plane, part of it will always be out of focus.

Defocus



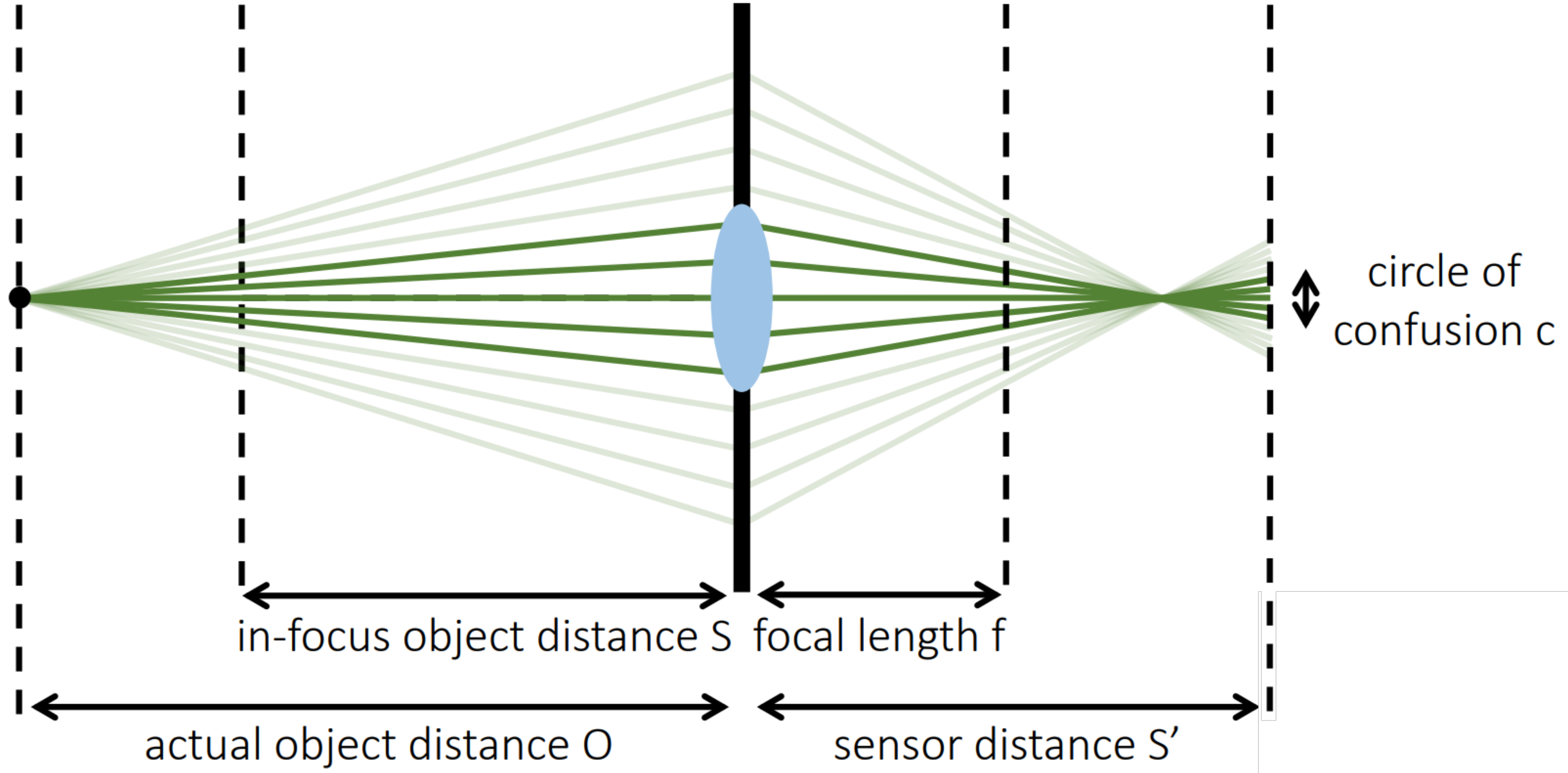


Defocus – Circle of confusion

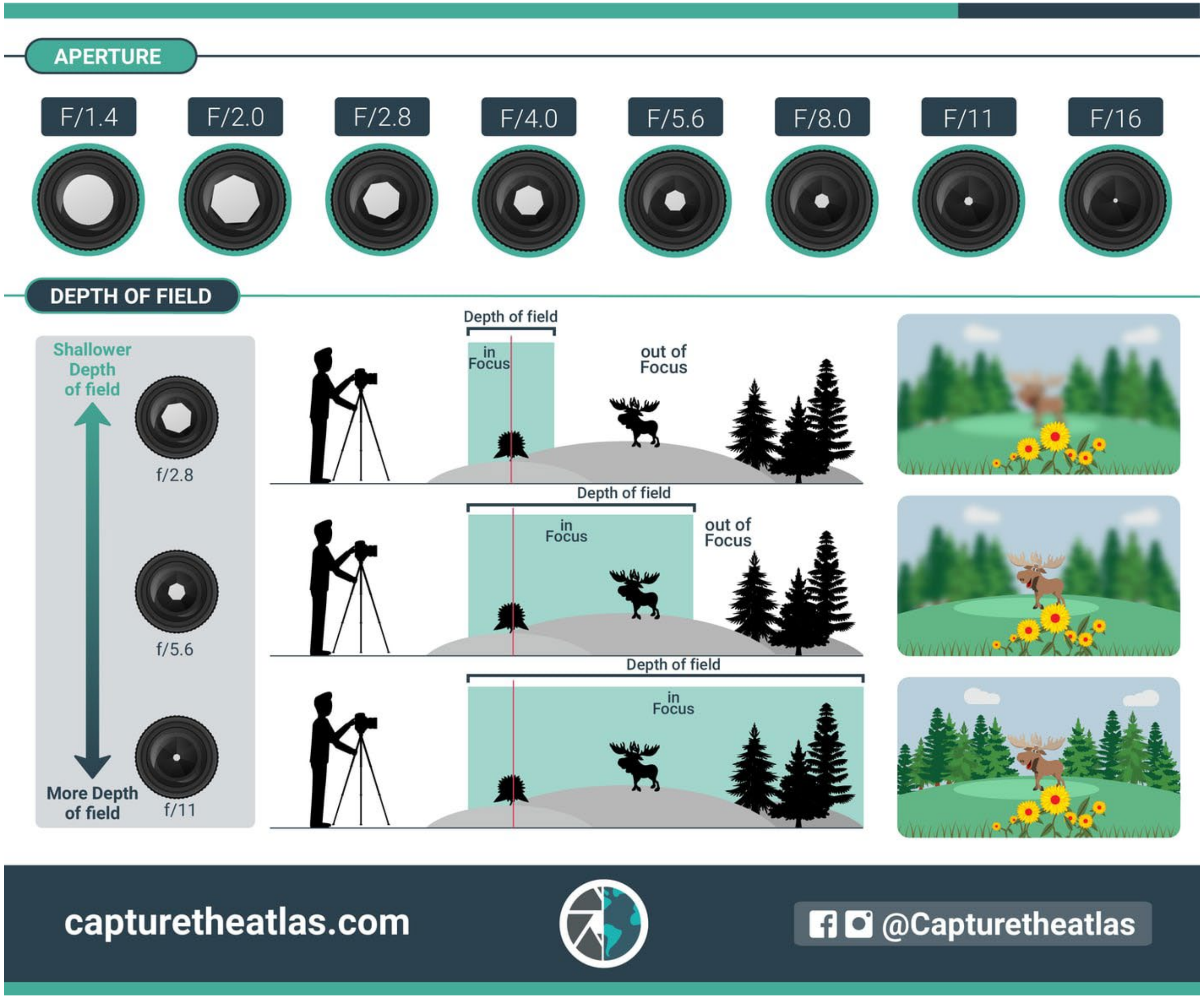


Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Defocus – Circle of confusion

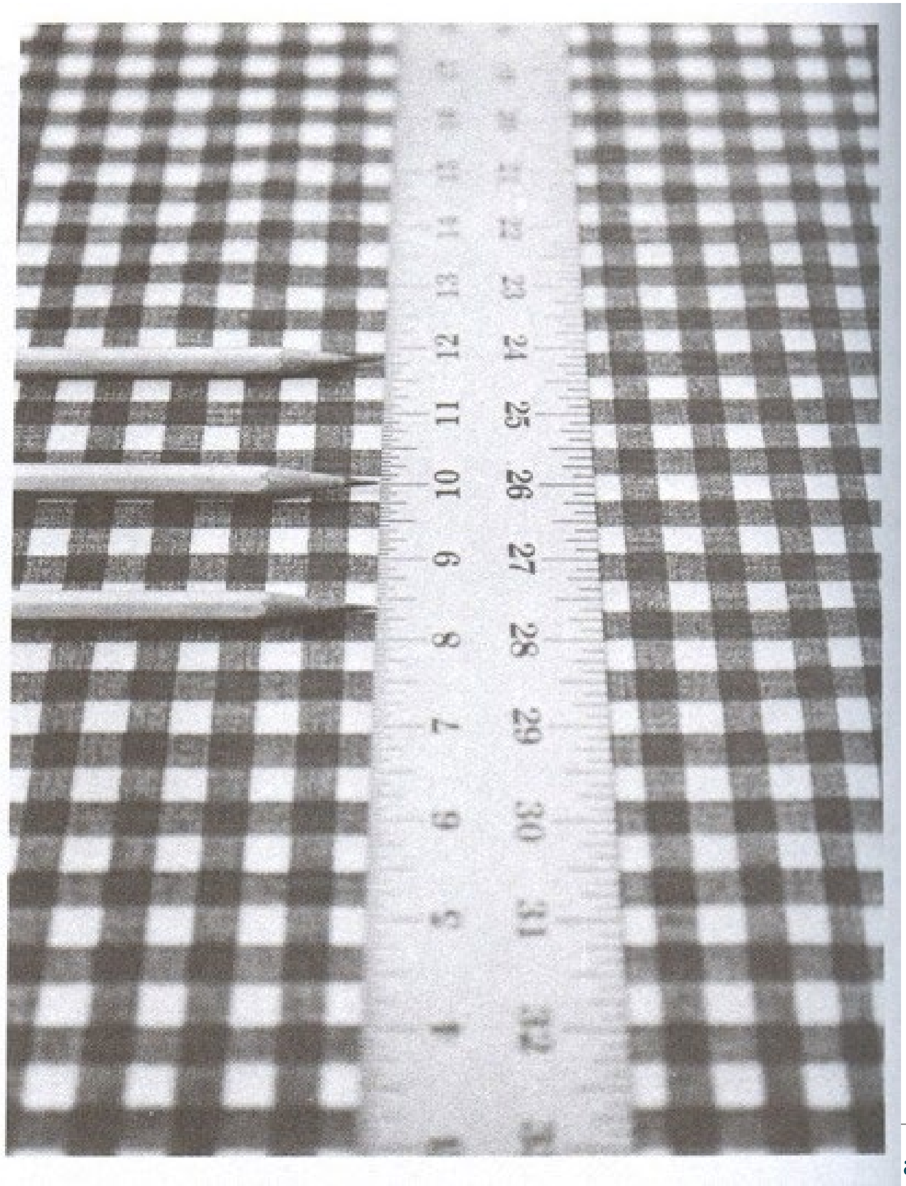


Change of focus for different depths



Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Change of focus for different depths



Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

How do I control what is in focus?

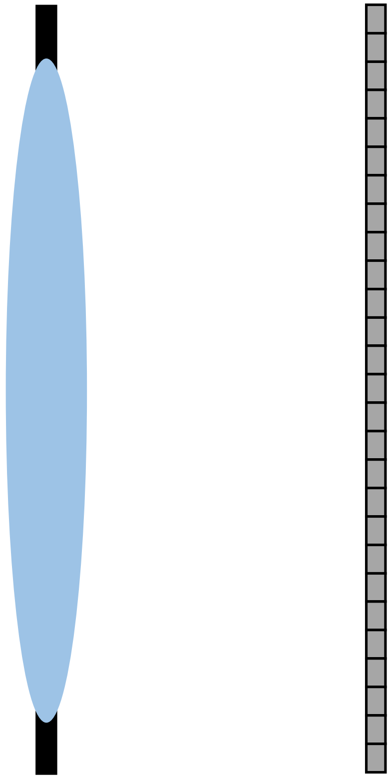
Sources: CMU 15-463, Dartmouth College CS89/189.

How do I control what is in focus?

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

I change the distance between the sensor and the lens

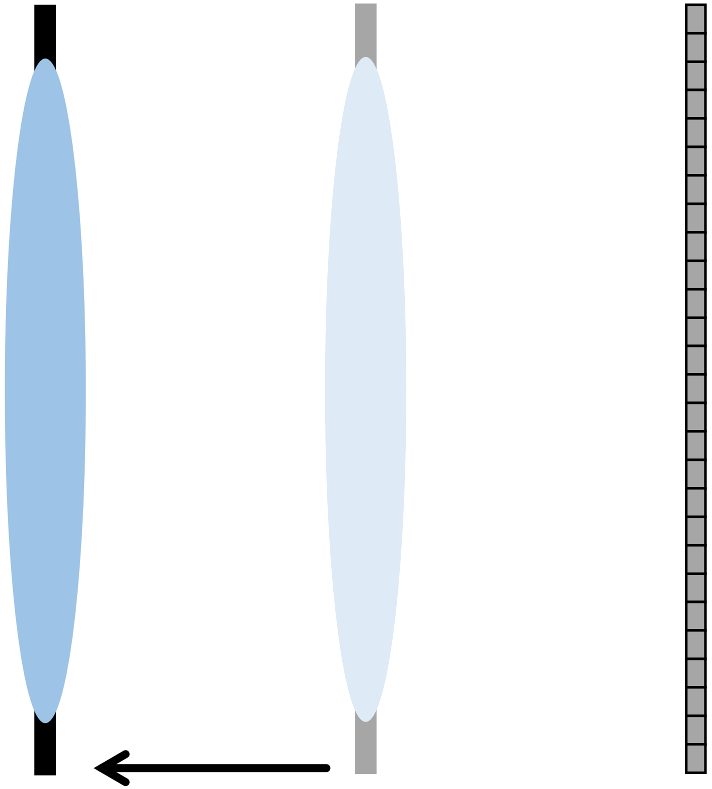
plane in focus



Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

How do I control what is in focus? $\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$

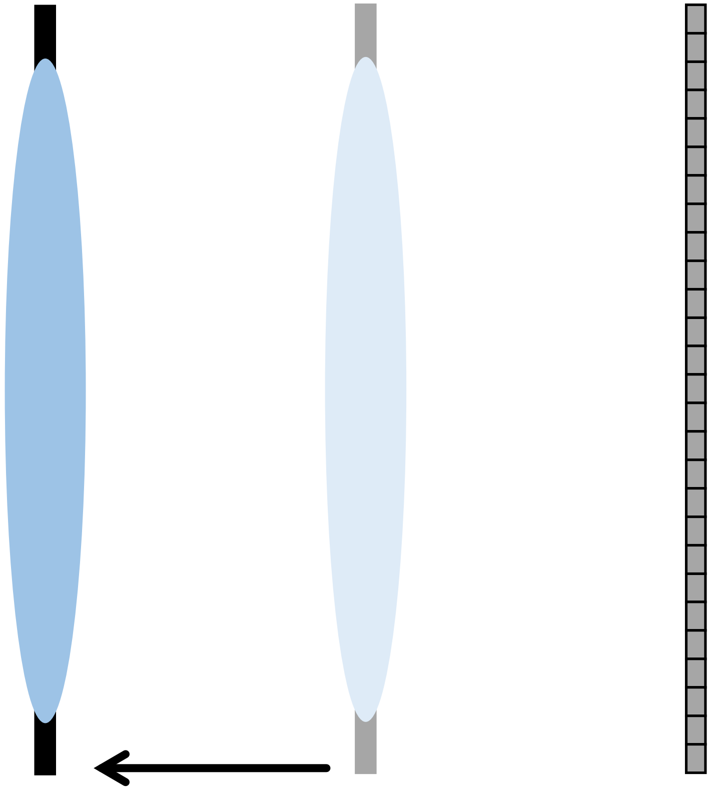
I change the distance between the sensor and the lens



- What happens to plane in focus?

How do I control what is in focus? $\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$

I change the distance between the sensor and the lens



move lens further away from sensor

- Focal plane moves closer

The lens on your camera

Focus ring: controls distance of lens from sensor

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Sequence of images at different focus settings

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

So far…

$$\frac{1}{D'} + \frac{1}{D} = \frac{1}{f}$$

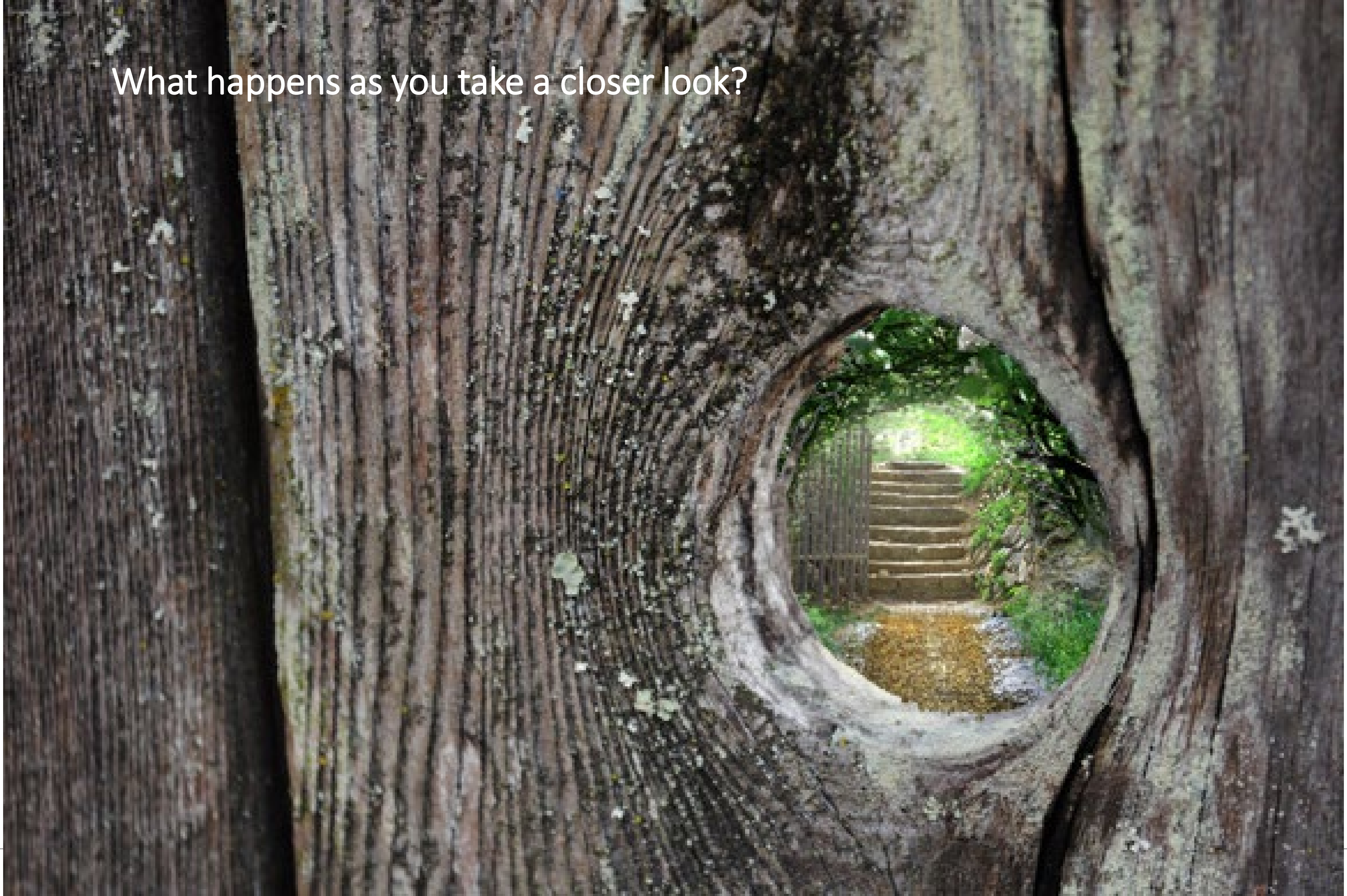
- Pinhole camera
- Thin lens model
- Relationship between scene and captured image
  - Plane in focus (D) depends on the focal length (f) and the sensor distance (D’)
  - Usually not all my scene can be in focus for a fixed focus lens
  - Focal length depends on my lens (material and shape): To change the focus plane (D) I need to change the distance between lens and sensor (D’)
  - ↓ Aperture size ↓ Incoming light in the sensor
  - ↓ Aperture size ↑ Depth of field
  - ↑ Focal length ↑ Magnification (magnified circle of confusion)
  - ↑ Focal length ↓ Depth of field
  - ↑ Focal length ↓ Field of view

Small focal lengths (wide-angle lenses) have other drawbacks related with the shape of the lens: barrel distortion, perspective distortion, vignetting…

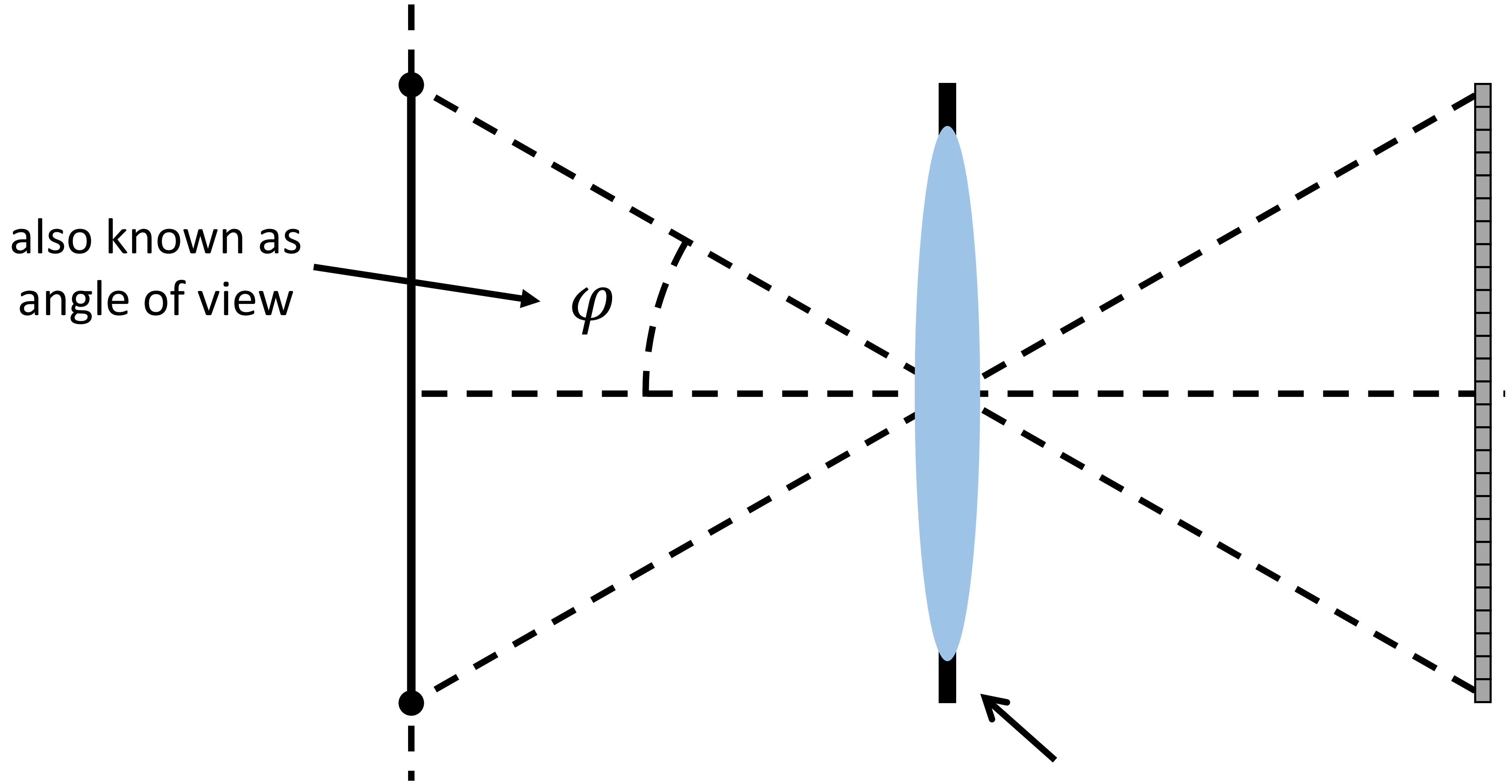
Field of view

Sources: CMU 15-463, Dartmouth College CS89/189.

What happens as you take a closer look?

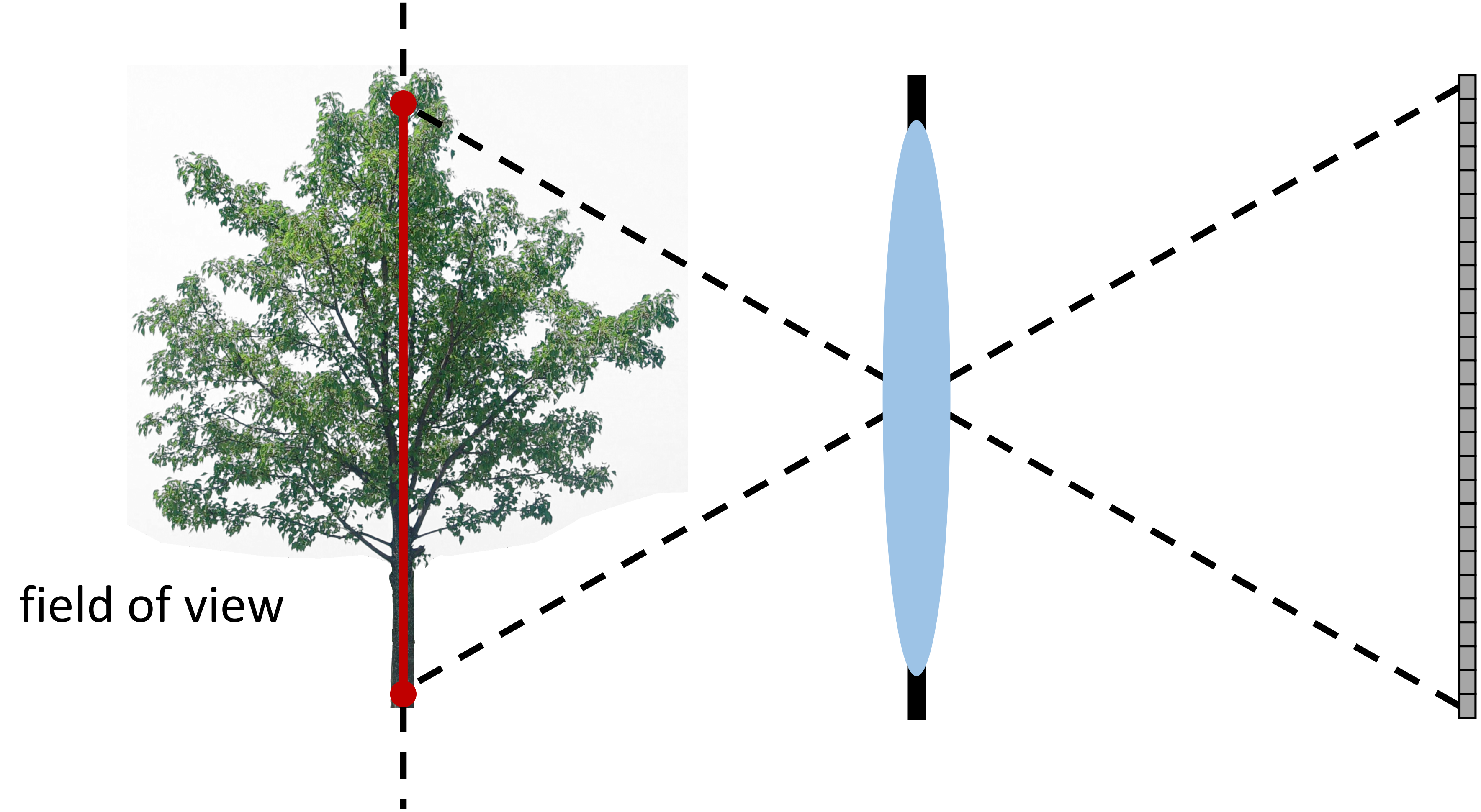


Field of view



Field of view

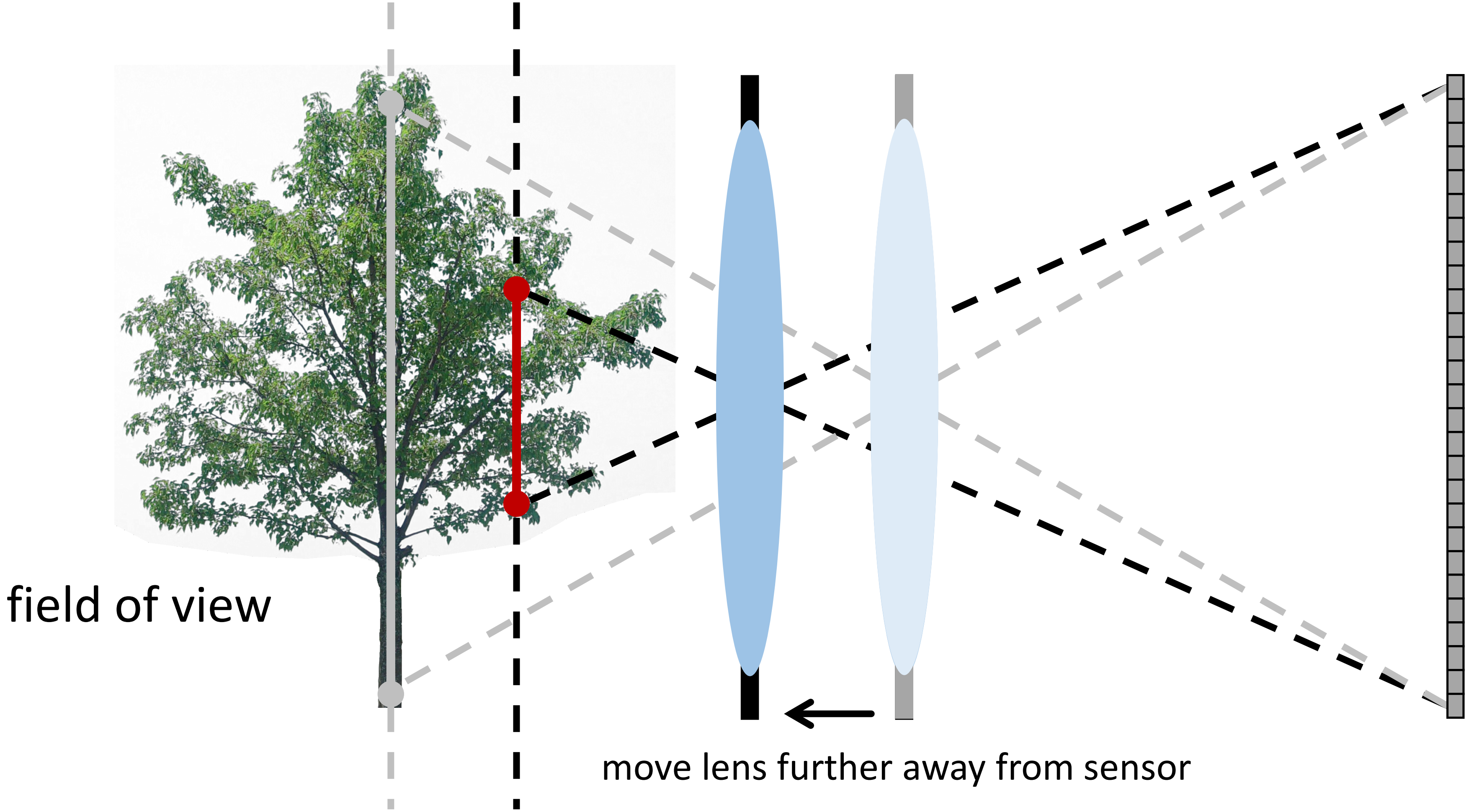
The part of the in-focus plane that gets mapped on the sensor



- What happens to field of view as we focus closer?

Field of view

The part of the in-focus plane that gets mapped on the sensor

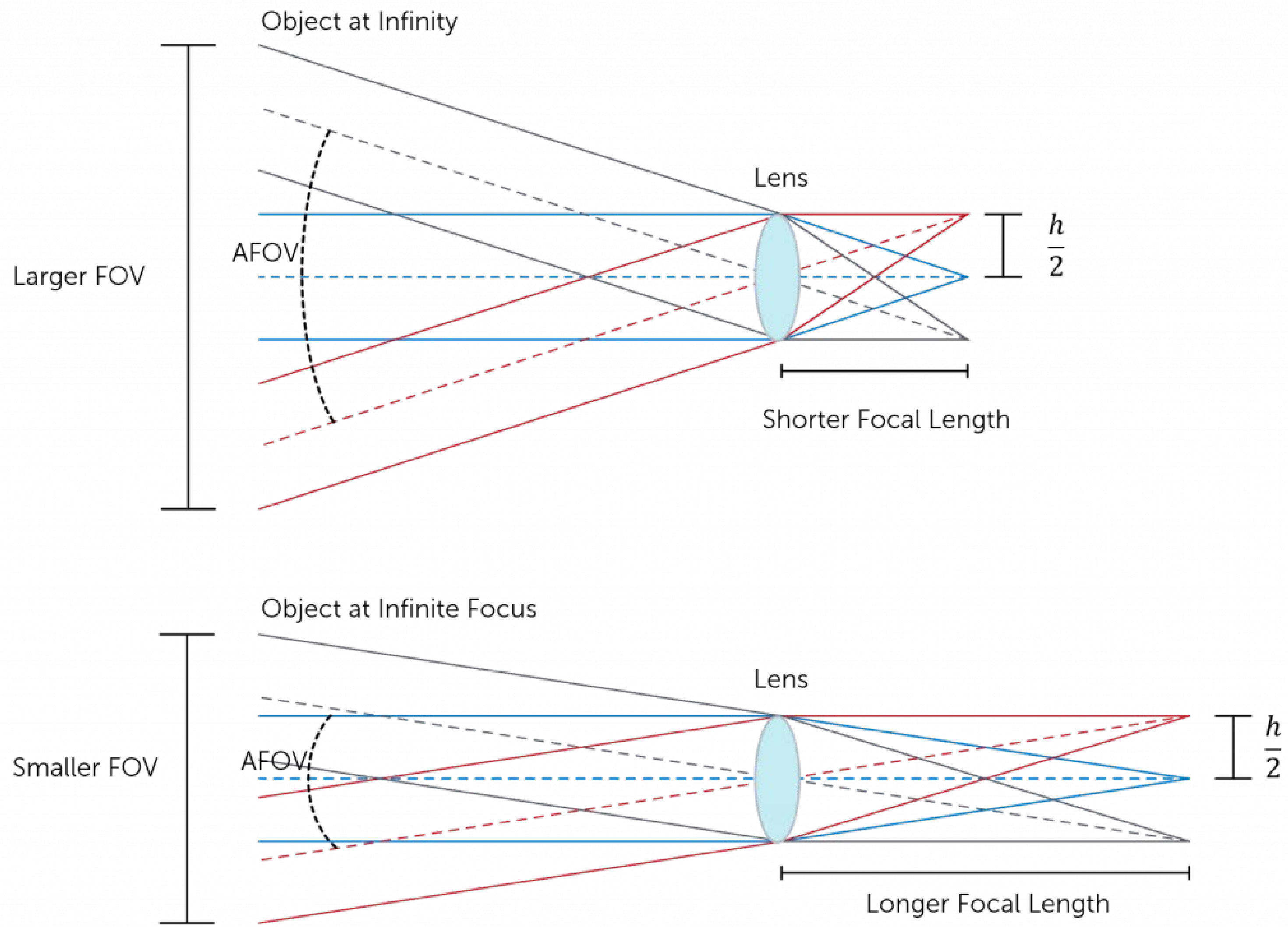


- What happens to field of view as we focus closer? → It becomes smaller.

Sources: CMU 15-463, Dartmouth College CS89/189.

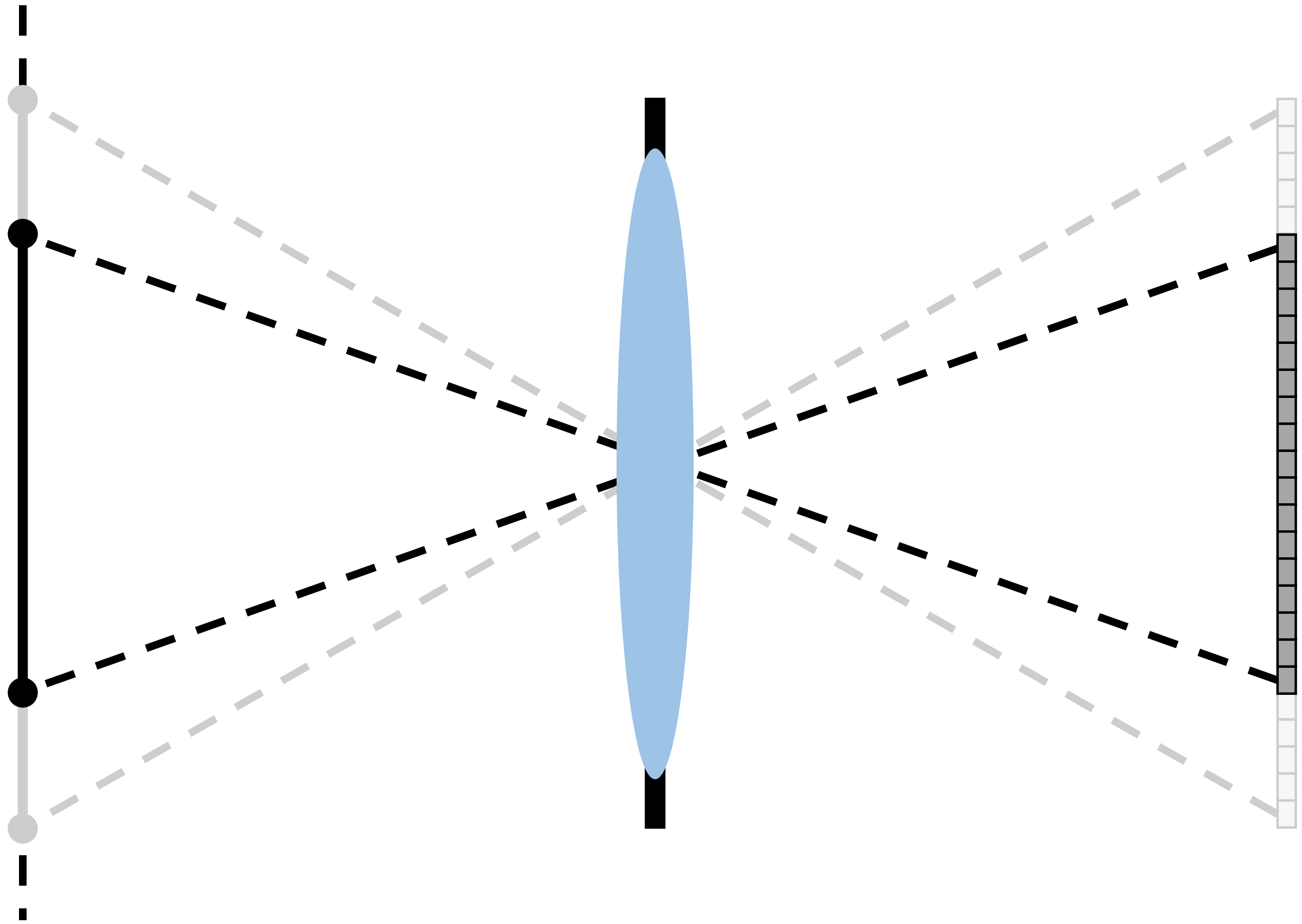
Field of view

The part of the in-focus plane that gets mapped on the sensor



Sources: CMU 15-463, Dartmouth College CS89/189.

Field of view also depends on sensor size



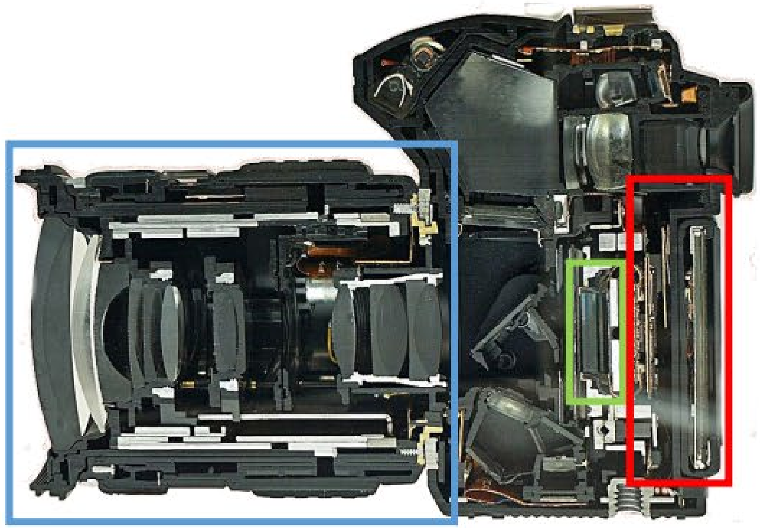
- What happens to field of view when we reduce sensor size? → It decreases.

Digital photography pipeline

Sources: CMU 15-463, Dartmouth College CS89/189.

The modern photography pipeline

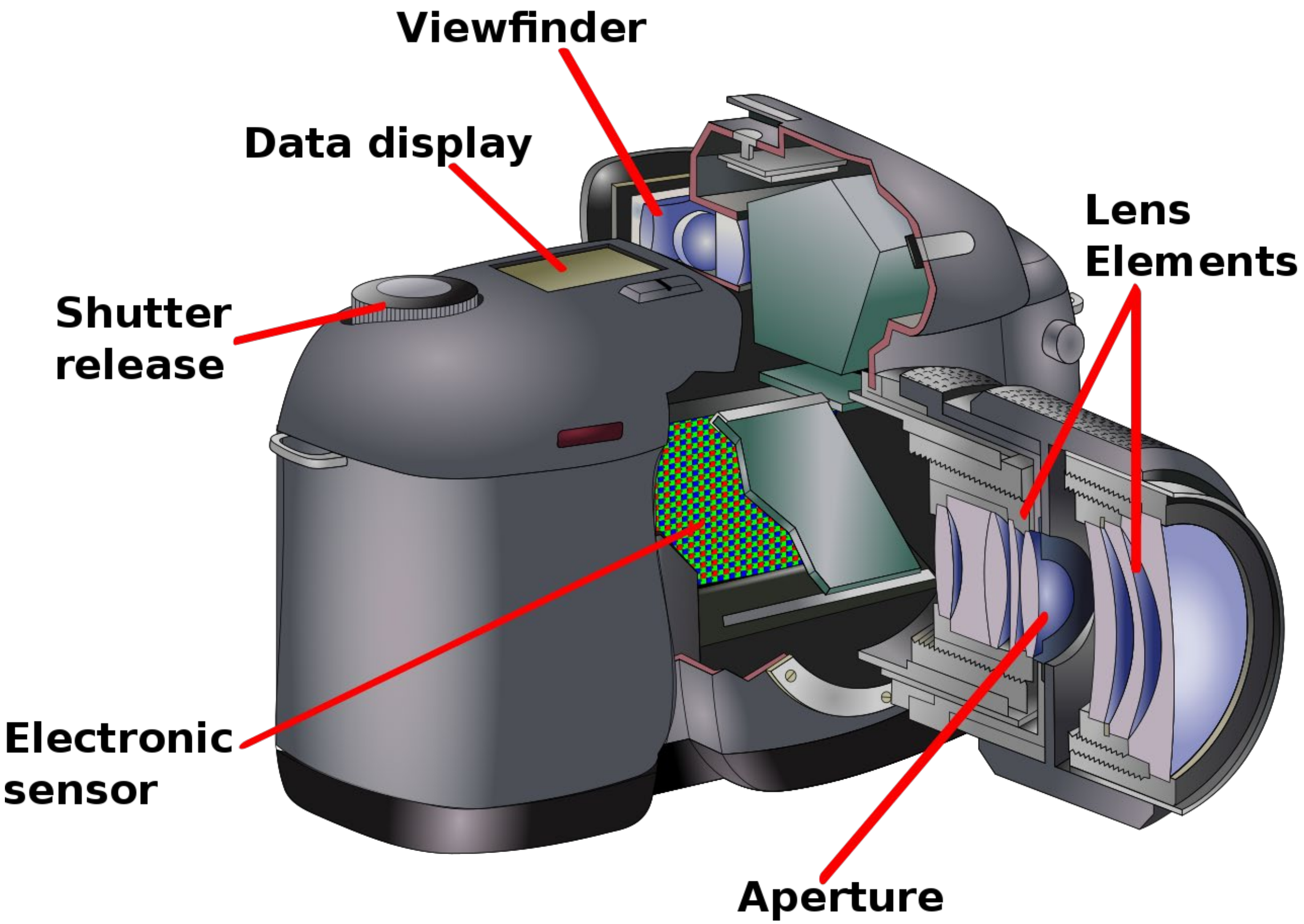
post-capture processing
(lectures 5-10)



optics and optical controls → sensor, analog front-end, and color filter array → in-camera image processing pipeline

Digital camera

Raster input device

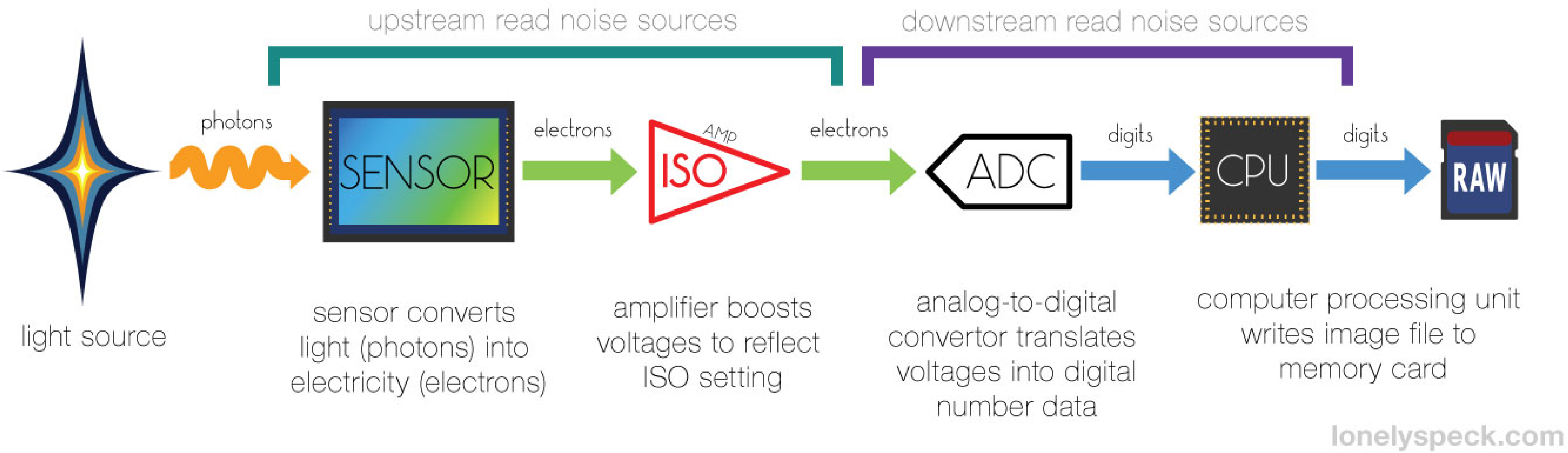


Discretizes incoming light with a 2D array of photosensors

Canon 6D sensor

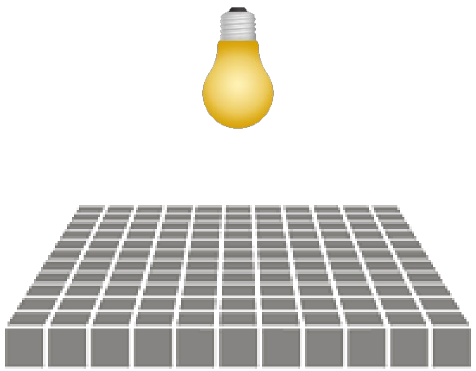
Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Digital camera



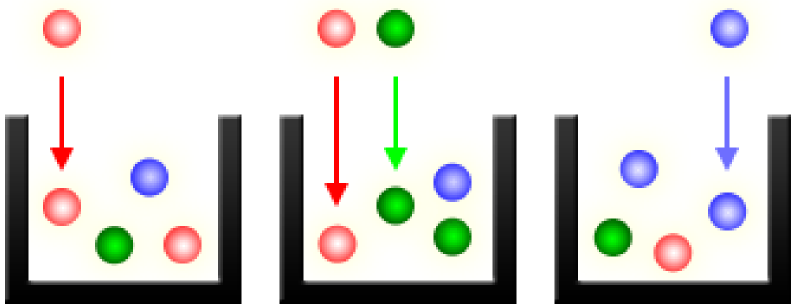
What does an imaging sensor do?

When the camera shutter opens…
… exposure begins…



array of photon buckets

photons



close-up view of photon buckets

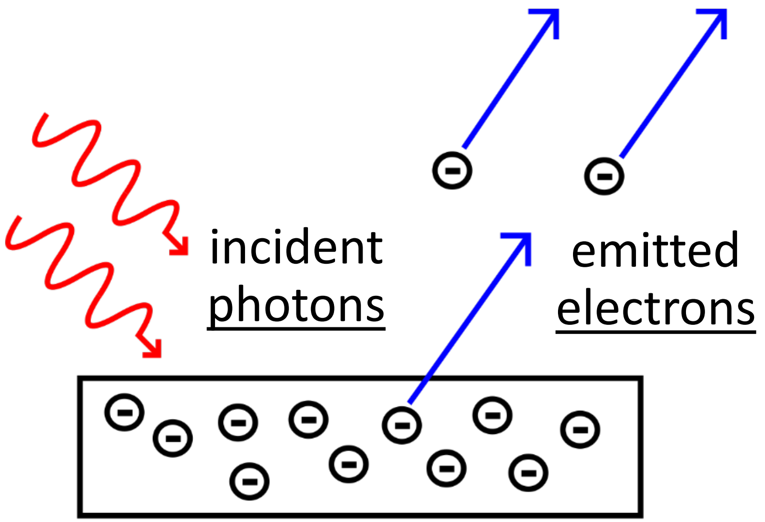
… photon buckets begin to store photons...

… until the camera shutter closes. Then, they convert stored photons to intensity values.

Photodiode quantum efficiency (QE)

How many of the incident photons will the photodiode convert into electrons?

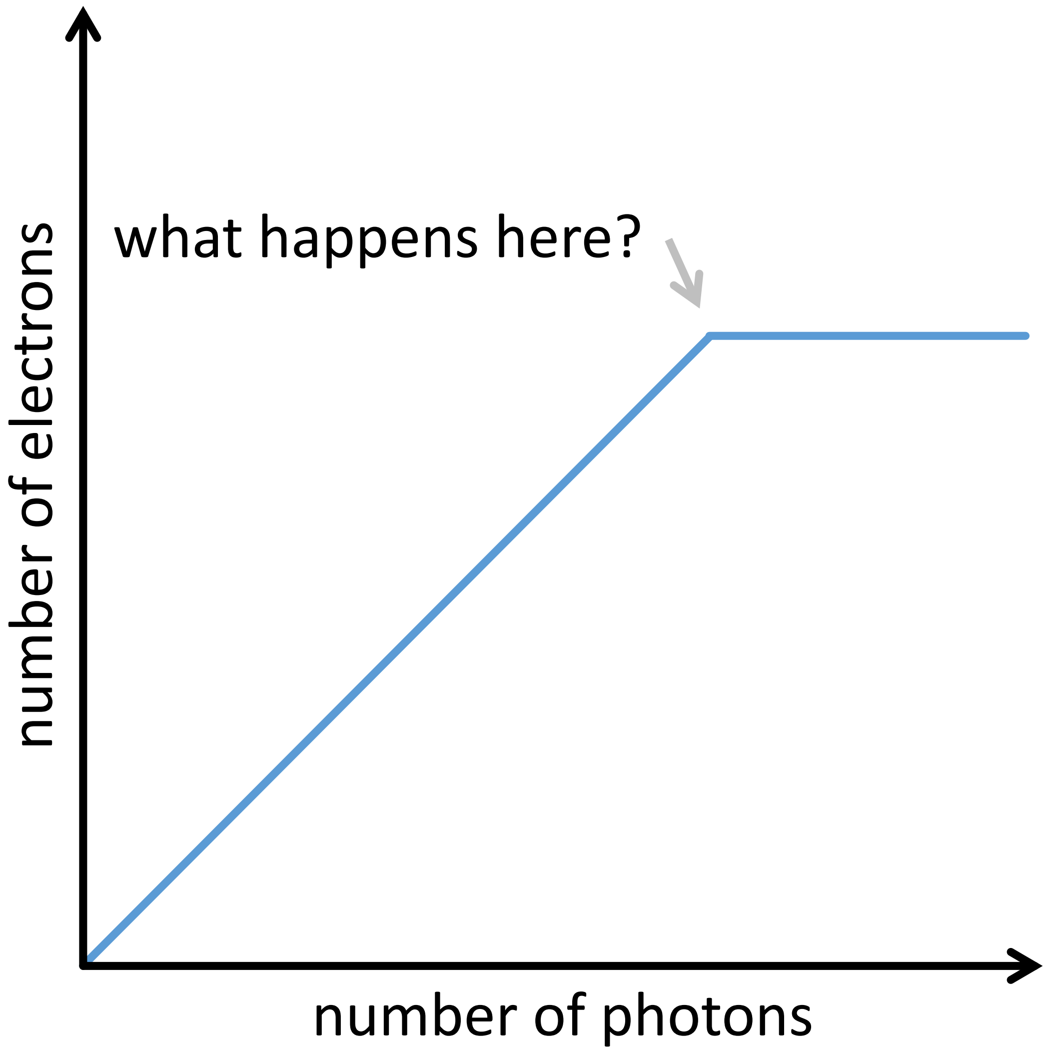
$$QE = \frac{\#\ \text{electrons}}{\#\ \text{photons}}$$



- Fundamental optical performance metric of imaging sensors.
- Not the only important optical performance metric!

Photodiode response

The photodiode response is mostly *linear*



Photodiode response

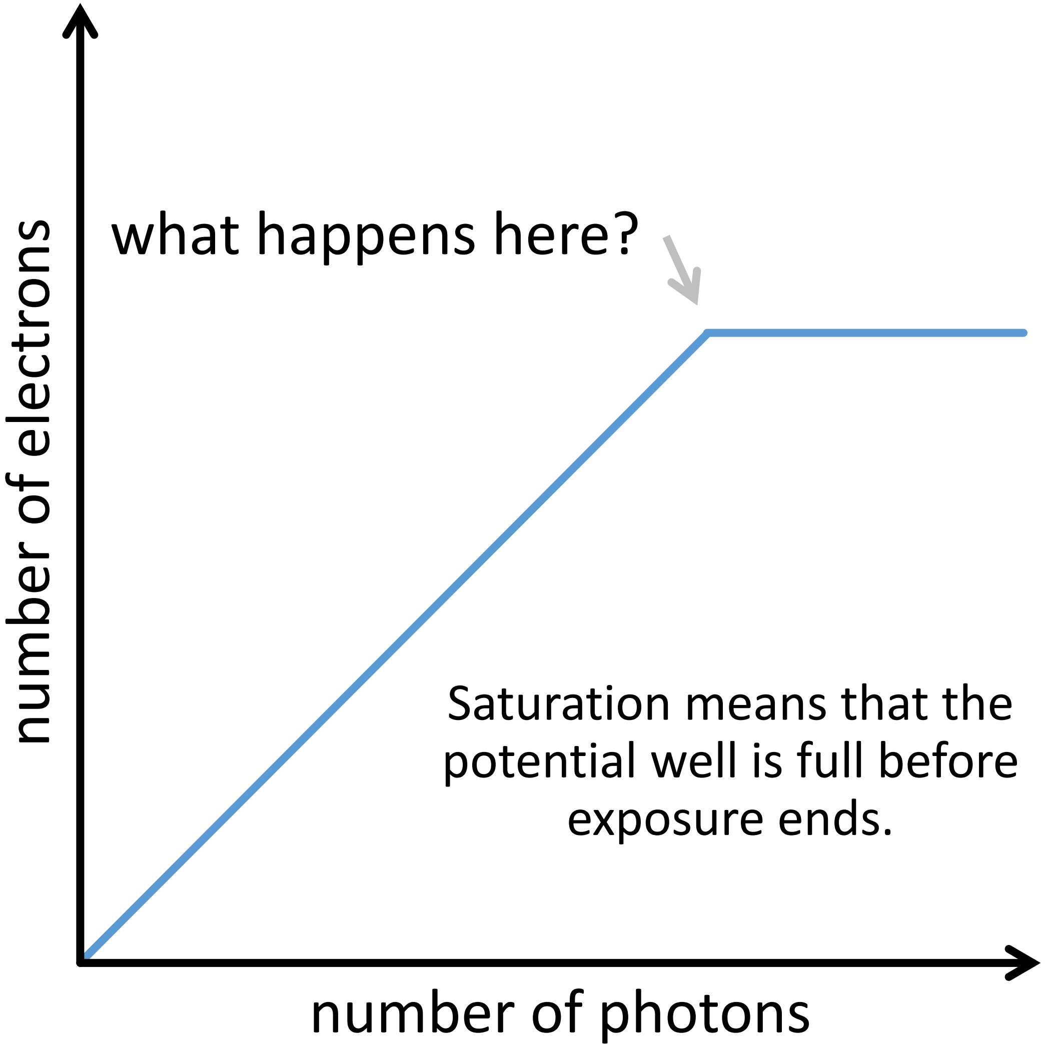
The photodiode response is <u>mostly</u> linear, but:

- non-linear when potential well is saturated (over-exposure)
- non-linear near zero (due to noise)

High-dynamic-range imaging: Later in the course!

under-exposure (non-linearity due to sensor noise)

over-exposure (non-linearity due to sensor saturation)



Exposure

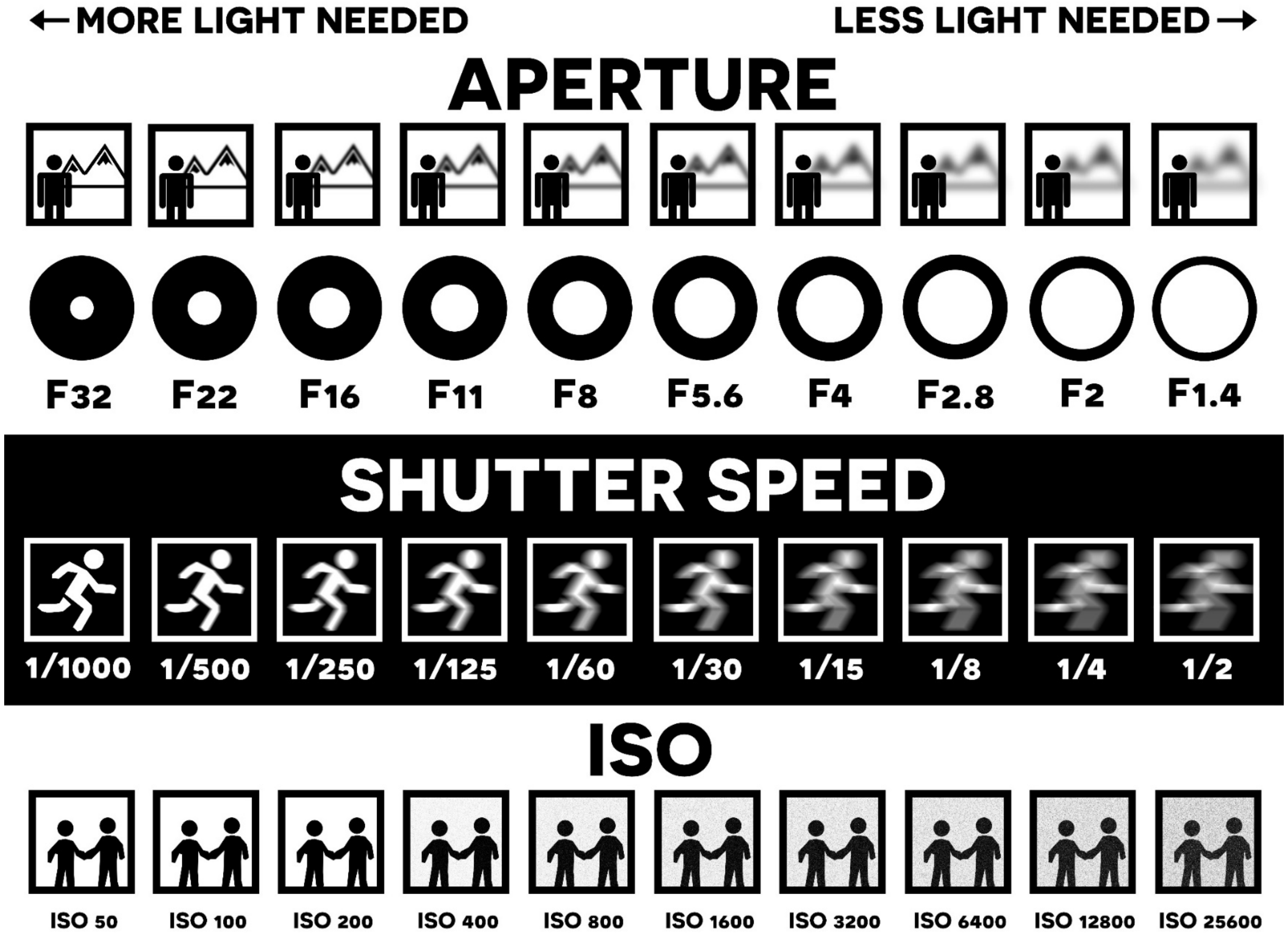
Get the right amount of light to sensor/film

Three main parameters:

- Shutter speed
- Aperture (area of lens)
- Sensor/film sensitivity (ISO)

Sources: CMU 15-463, Dartmouth College CS89/189.

Exposure



Sources: CMU 15-463, Dartmouth College CS89/189.

Exposure: Aperture

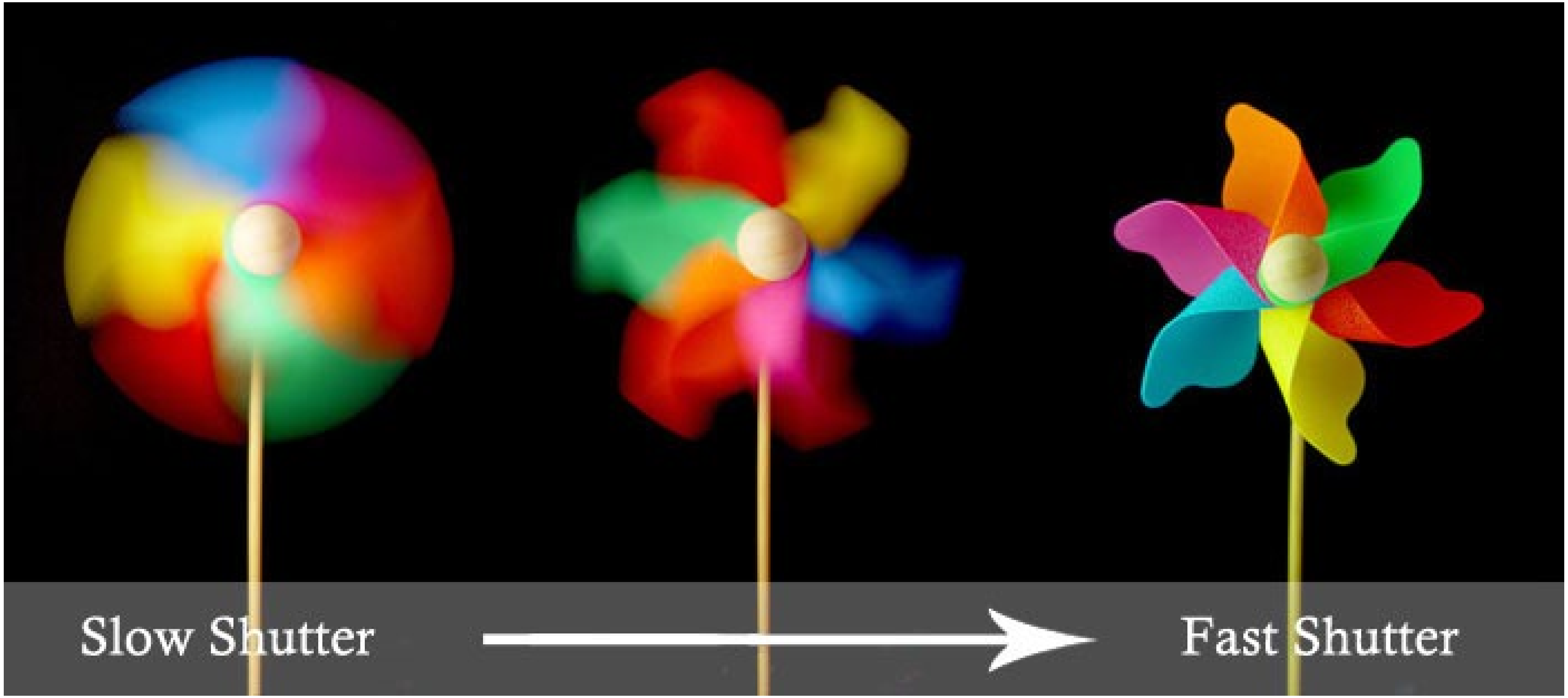
At f/2.0 focus is clearly only on the top statue

At f/5.6 the adjacent statues creep into focus

At f/22 all statues are in focus

Sources: CMU 15-463, Dartmouth College CS89/189.

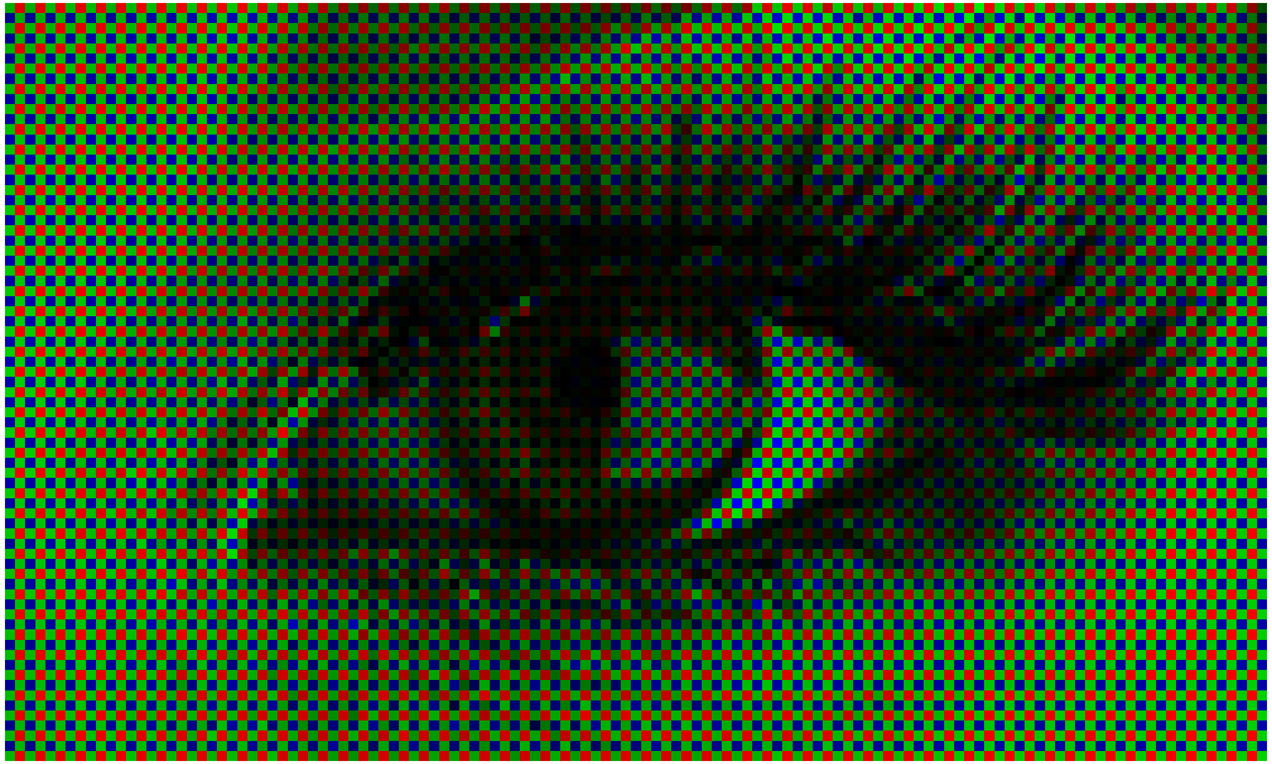
Exposure: Shutter speed

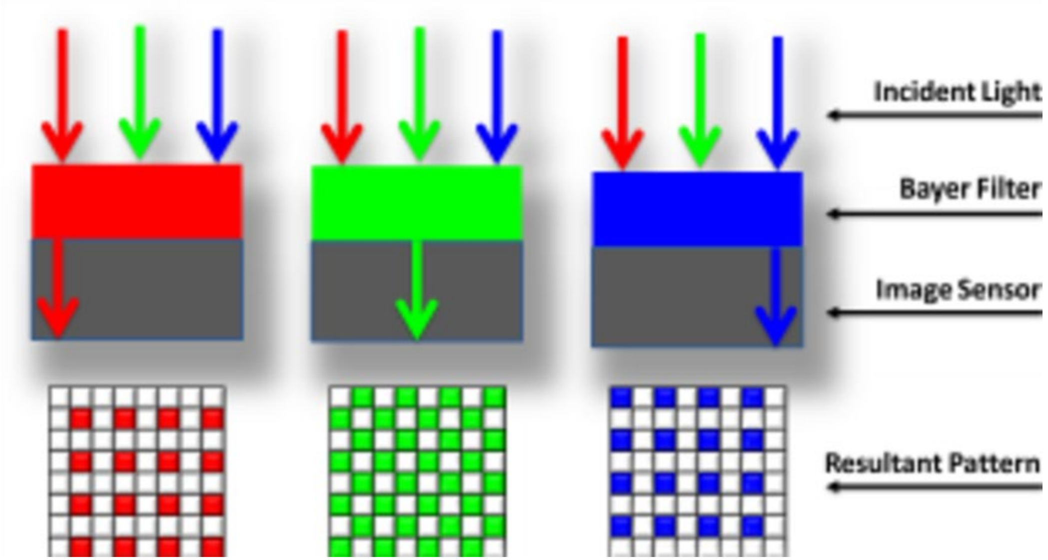


Exposure: ISO

Sources: CMU 15-463, Dartmouth College CS89/189.

How do we capture color?

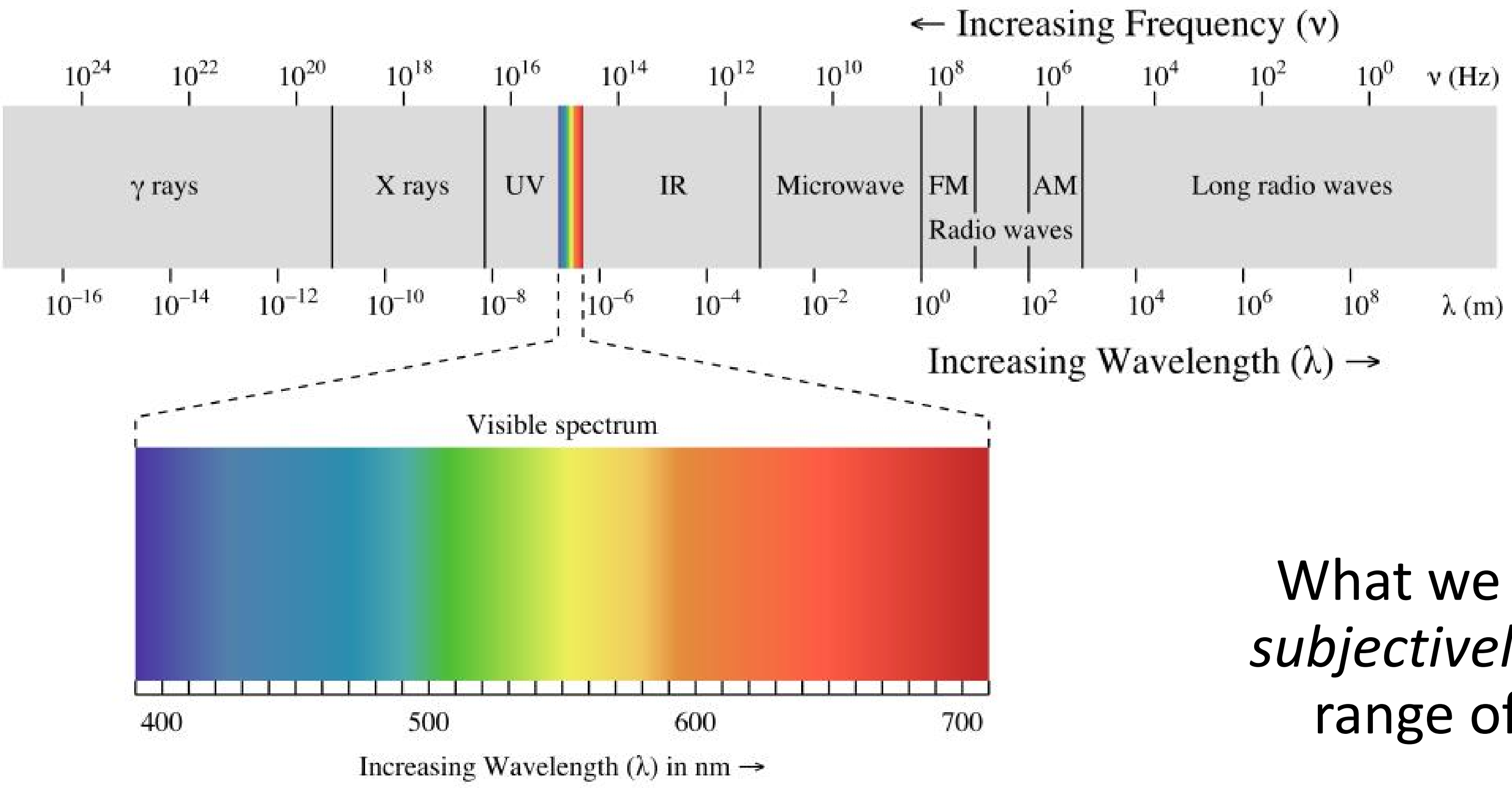




Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Color characterization

- “Color” is not an *objective* physical property of light (electromagnetic radiation).
- Instead, light is characterized by its wavelength.



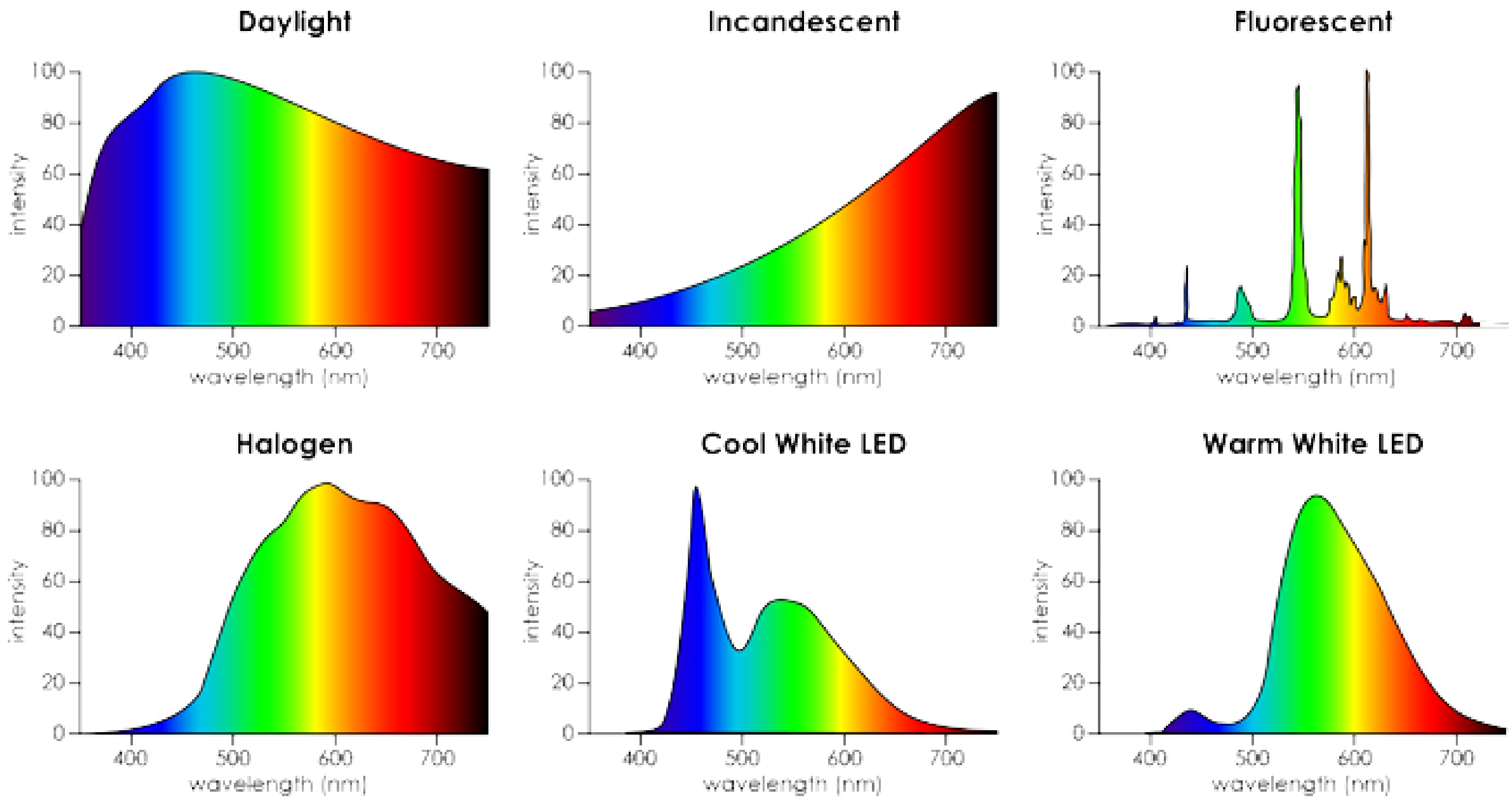
electromagnetic spectrum

What we call “color” is how we *subjectively* perceive a very small range of these wavelengths.

Spectral Power Distribution (SPD)

- Most types of light “contain” more than one wavelength.
- We can describe light based on the distribution of power over different wavelengths.

We call our sensation of all of these distributions “white”.



Sources: CMU 15-463, Dartmouth College CS89/189.

Spectral Sensitivity Function (SSF)

- *Any* light sensor (digital or not) has different sensitivity to different wavelengths.
- This is described by the sensor's *spectral sensitivity function* $f(\lambda)$.
- When measuring light of some SPD $\Phi(\lambda)$, the sensor produces a *scalar* response:

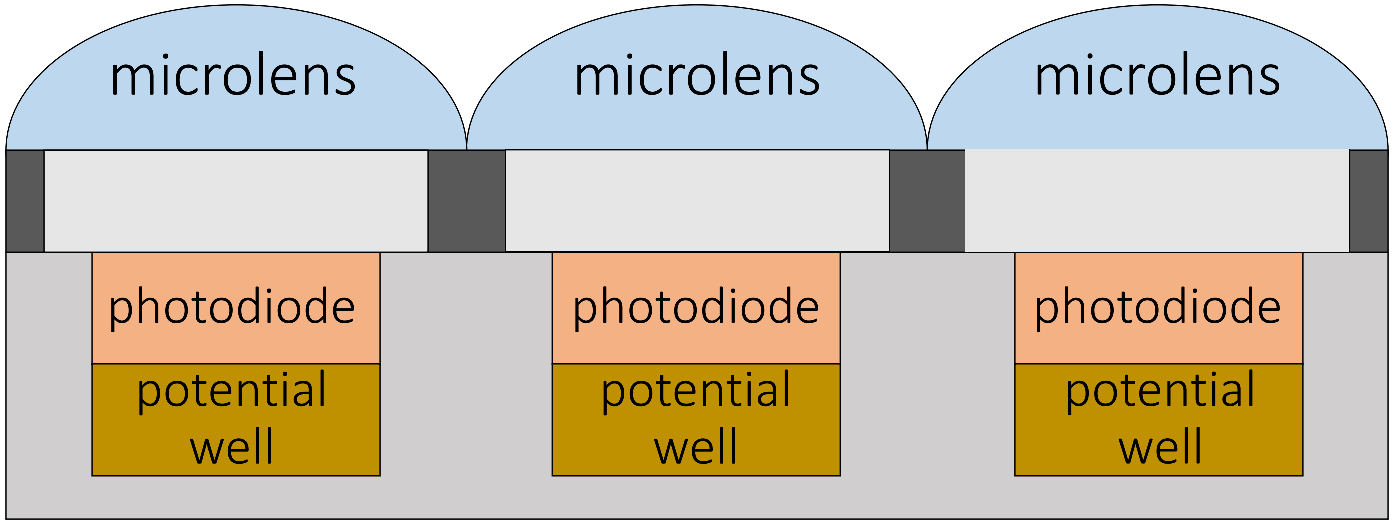
$$\underbrace{R}_{\text{sensor response}} = \int_\lambda \underbrace{\Phi(\lambda)}_{\text{light SPD}} \underbrace{f(\lambda)}_{\text{sensor SSF}} d\lambda$$

Weighted combination of light's SPD: light contributes more at wavelengths where the sensor has higher sensitivity.

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

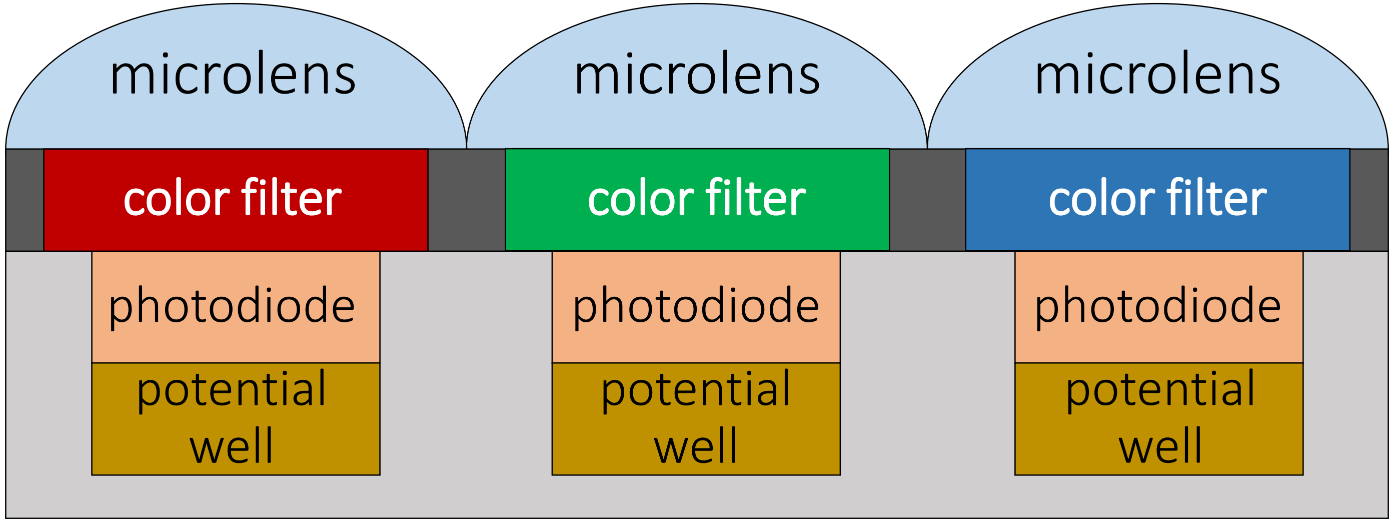
Color filter arrays (CFA)

- To measure color with a digital sensor, mimic cone cells of human vision system.



Color filter arrays (CFA)

- To measure color with a digital sensor, mimic cone cells of human vision system.
- "Cones" correspond to pixels that are covered by different color filters, each with its own spectral sensitivity function.



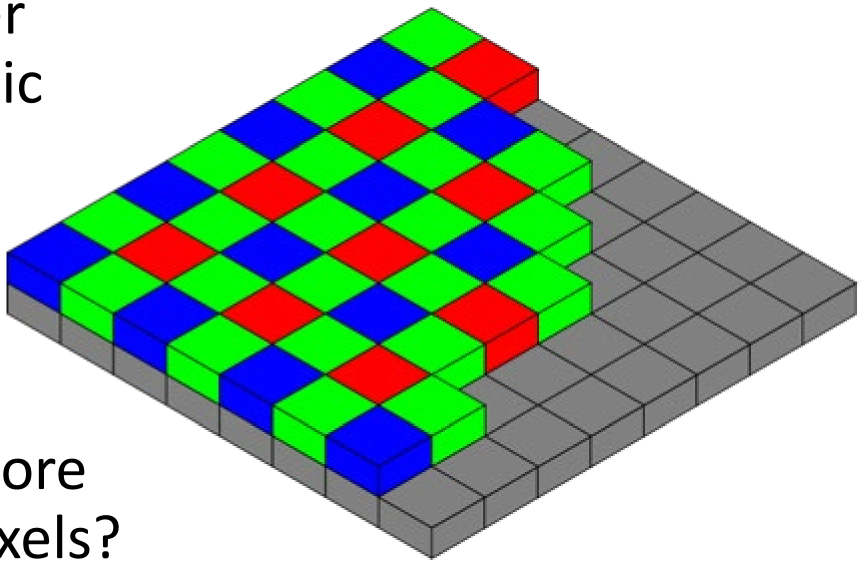
Sources: CMU 15-463, Dartmouth College CS89/189.

What color filters to use?

Two design choices:

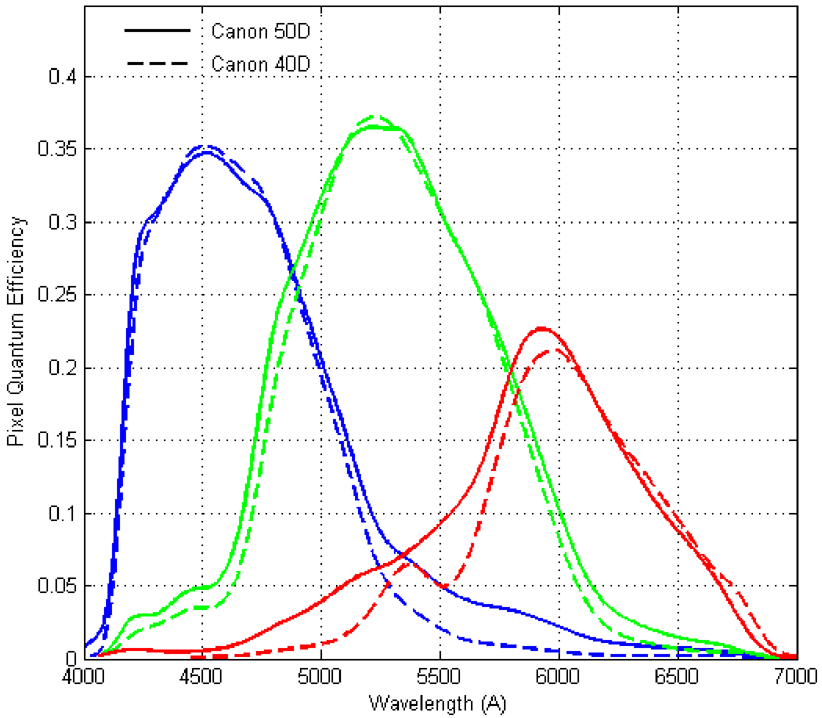
- What spectral sensitivity functions $f(\lambda)$ to use for each color filter?
- How to spatially arrange ("mosaic") different color filters?

Bayer mosaic



Why more green pixels?

SSF for Canon 50D



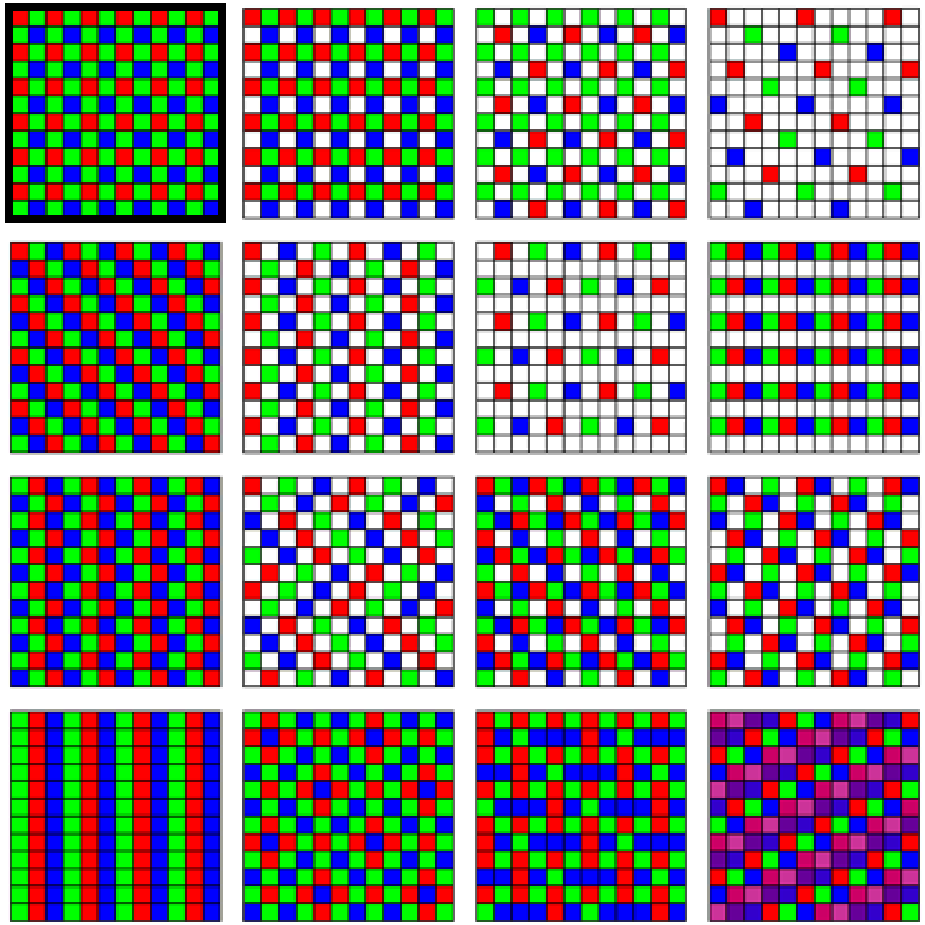
Generally do not match human LMS.

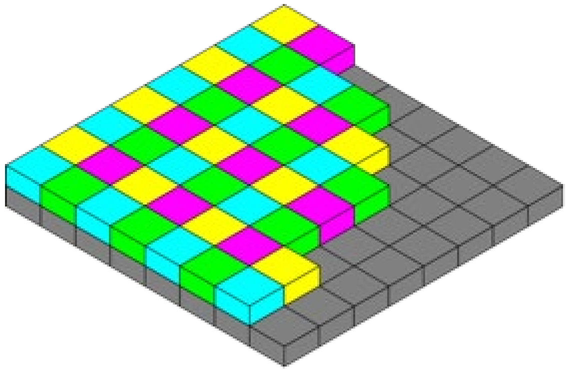
$f(\lambda)$

Sources: CMU 15-463, Dartmouth Co... 189.

Many different CFAs

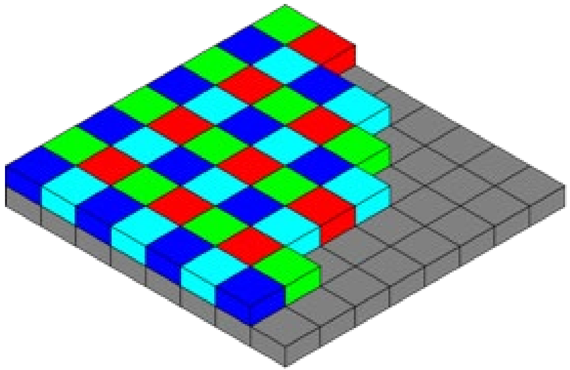
Finding the "best" CFA mosaic is an active research area.





CYGM

Canon IXUS, Powershot



RGBE

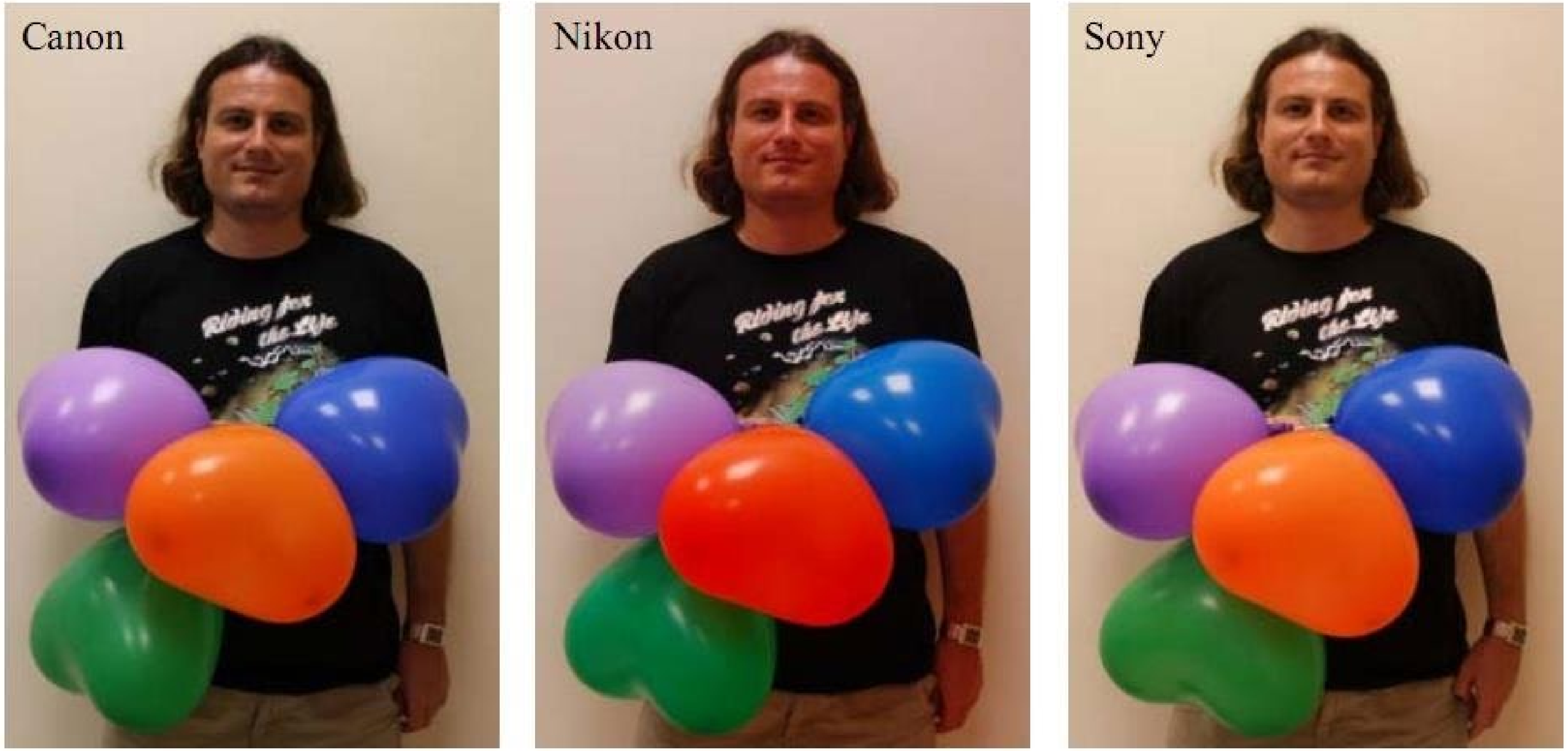
Sony Cyber-shot

How would you go about designing your own CFA? What criteria would you consider?

Many different spectral sensitivity functions

Each camera has its more or less unique, and most of the time *secret*, SSF.

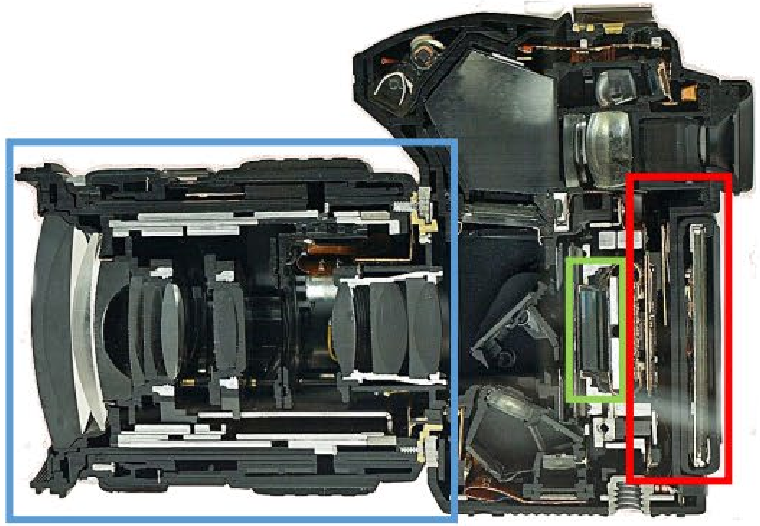
- Makes it very difficult to correctly reproduce the color of sensor measurements.



Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

The modern photography pipeline

post-capture processing
(lectures 5-10)



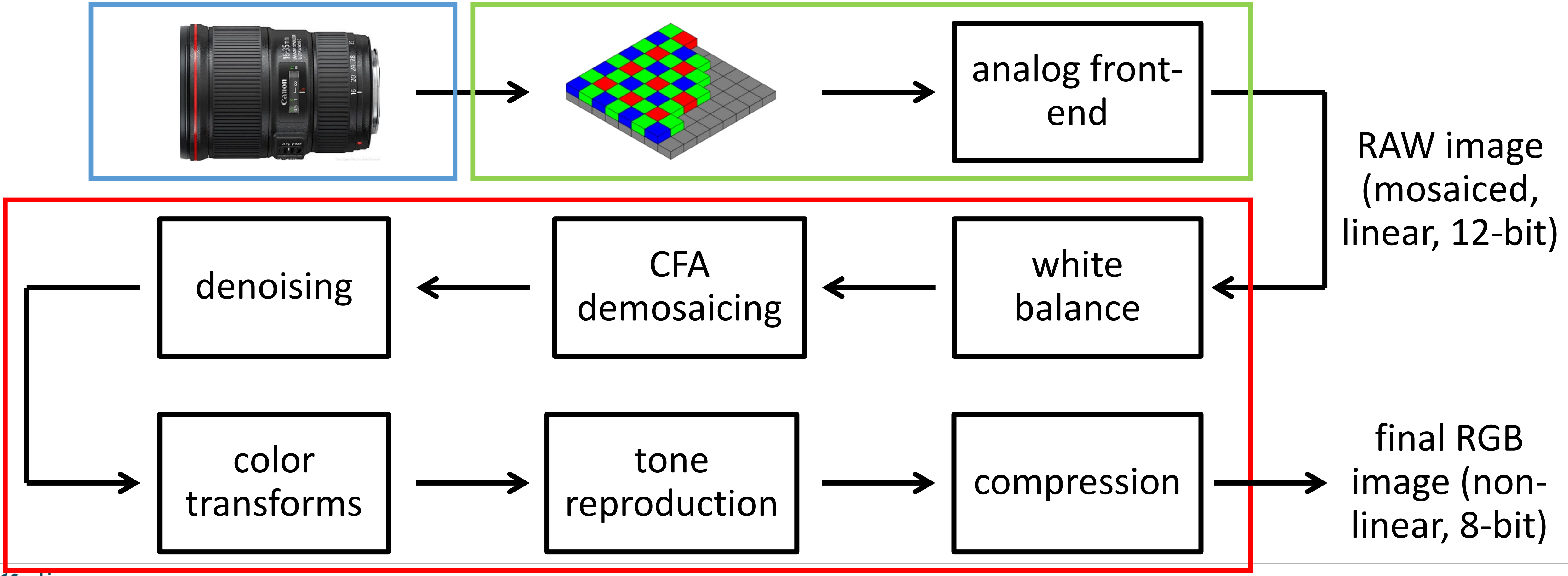
optics and optical controls → sensor, analog front-end, and color filter array → in-camera image processing pipeline

The (in-camera) image processing pipeline

Sources: CMU 15-463, Dartmouth College CS89/189.

The (in-camera) image processing pipeline

The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



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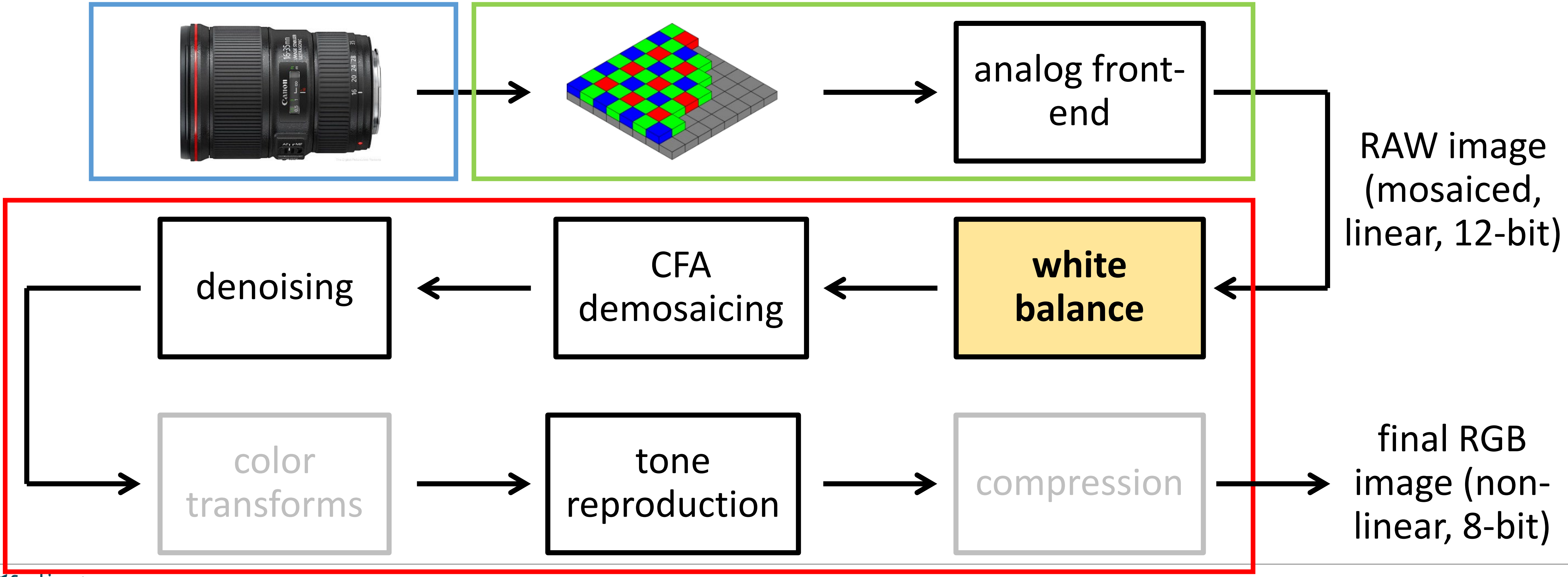


White balancing

Human visual system has *chromatic adaptation*:
- We can perceive white (and other colors) correctly under different light sources.

Retinal vs
perceived color.

[Slide credit: Todd Zickler]

White balancing

Human visual system has *chromatic adaptation*:

- We can perceive white (and other colors) correctly under different light sources.
- Cameras cannot do that (there is no “camera perception”).

White balancing: The process of removing color casts so that colors that we would *perceive* as white are *rendered* as white in final image.

different whites

image captured under fluorescent

image white-balanced to daylight

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

White balancing presets

Cameras nowadays come with a large number of presets: You can select which light you are taking images under, and the appropriate white balancing is applied.

WB SETTINGS	COLOR TEMPERATURE	LIGHT SOURCES
	10000 - 15000 K	Clear Blue Sky
	6500 - 8000 K	Cloudy Sky / Shade
	6000 - 7000 K	Noon Sunlight
	5500 - 6500 K	Average Daylight
	5000 - 5500 K	Electronic Flash
	4000 - 5000 K	Fluorescent Light
	3000 - 4000 K	Early AM / Late PM
	2500 - 3000 K	Domestic Lightning
	1000 - 2000 K	Candle Flame

Manual vs automatic white balancing

Manual white balancing:

- Select a camera preset based on lighting.

Sources: CMU 15-463, Dartmouth College CS89/189.

Manual vs automatic white balancing

Manual white balancing:
- Select a camera preset based on lighting.
- **Manually select object in photograph that is color-neutral and use it to normalize.**

Grey card:
1. Take a picture of a neutral object (white or gray)
2. Deduce the weight of each channel
3. If the object is recoded as r_w, g_w, b_w use weights for each channel such that $k_r = k/r_w$, $k_g = k/g_w$, $k_b = k/b_w$ where k controls the exposure

Manual vs automatic white balancing

Manual white balancing:

- Select a camera preset based on lighting.
- Manually select object in photograph that is color-neutral and use it to normalize.

Automatic white balancing:

- Grey world assumption: force average color of scene to be grey.
- White world assumption: force brightest object in scene to be white.
- Sophisticated histogram-based algorithms (what most modern cameras do).

Sources: CMU 15-463, Dartmouth College CS89/189.

Automatic white balancing

Grey world assumption:

- Compute per-channel average.
- Normalize each channel by its average.
- Normalize by green channel average.

white-balanced RGB → $\begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} = \begin{bmatrix} G_{avg}/R_{avg} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & G_{avg}/B_{avg} \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$ ← sensor RGB

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Automatic white balancing

Grey world assumption:

- Compute per-channel average.
- Normalize each channel by its average.
- Normalize by green channel average.

white-balanced RGB $\rightarrow$ $$\begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} = \begin{bmatrix} G_{avg}/R_{avg} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & G_{avg}/B_{avg} \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$ $\leftarrow$ sensor RGB

White world assumption:

- Compute per-channel maximum.
- Normalize each channel by its maximum.
- Normalize by green channel maximum.

white-balanced RGB $\rightarrow$ $$\begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} = \begin{bmatrix} G_{max}/R_{max} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & G_{max}/B_{max} \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$ $\leftarrow$ sensor RGB

Automatic white balancing example

input image

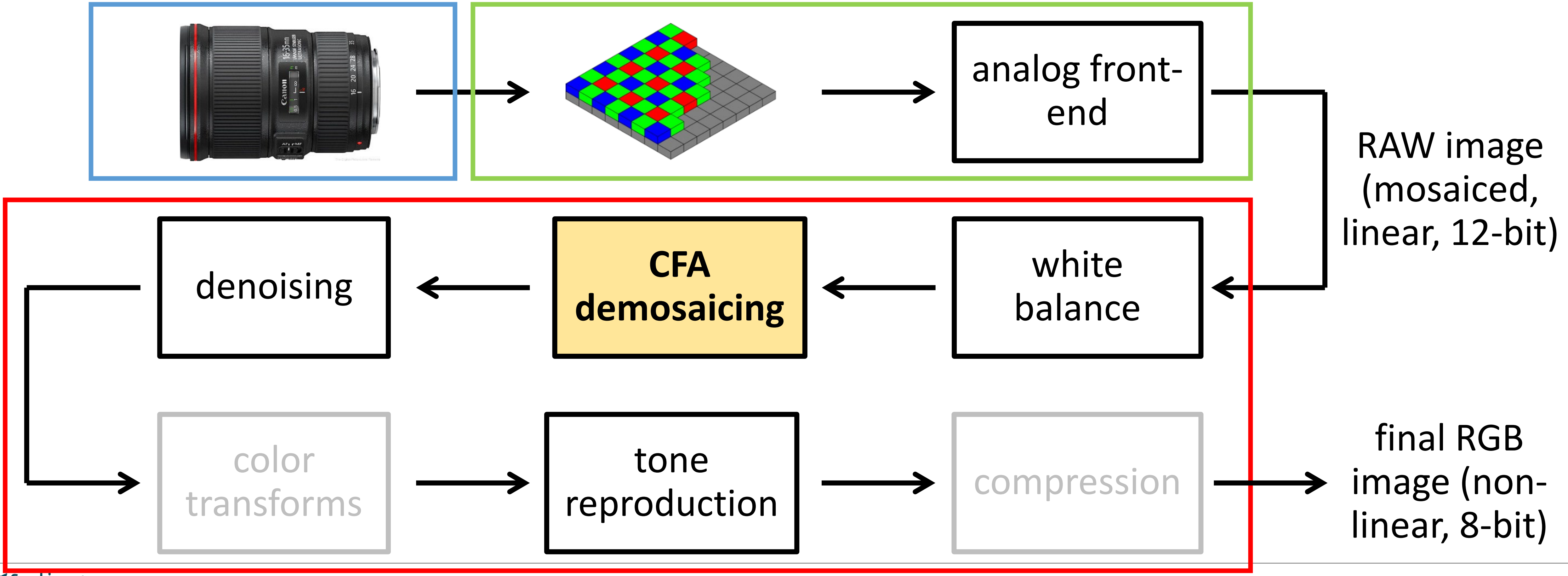
grey world

white world

Sources: CMU 15-463, Dartmouth College CS89/189.

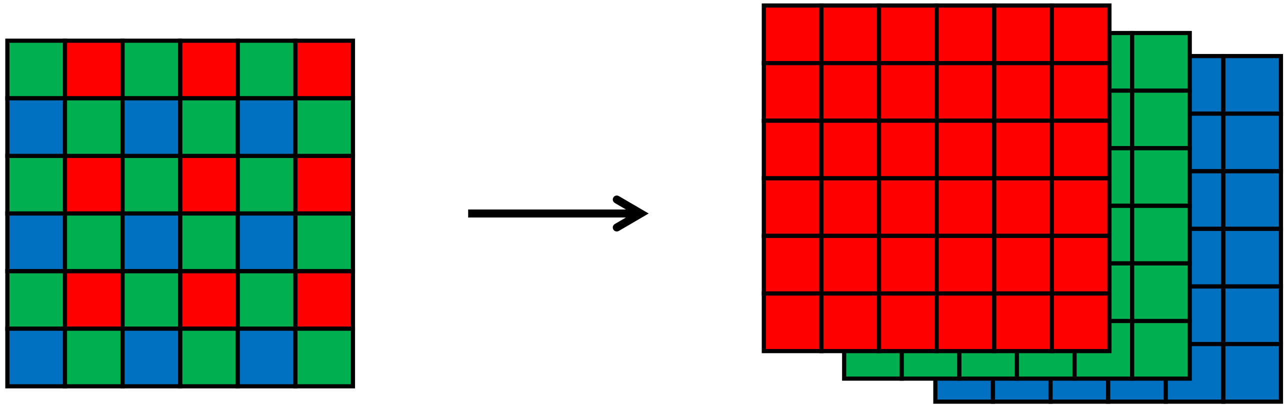
The (in-camera) image processing pipeline

The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



CFA demosaicing

Produce full RGB image from mosaicked sensor output.

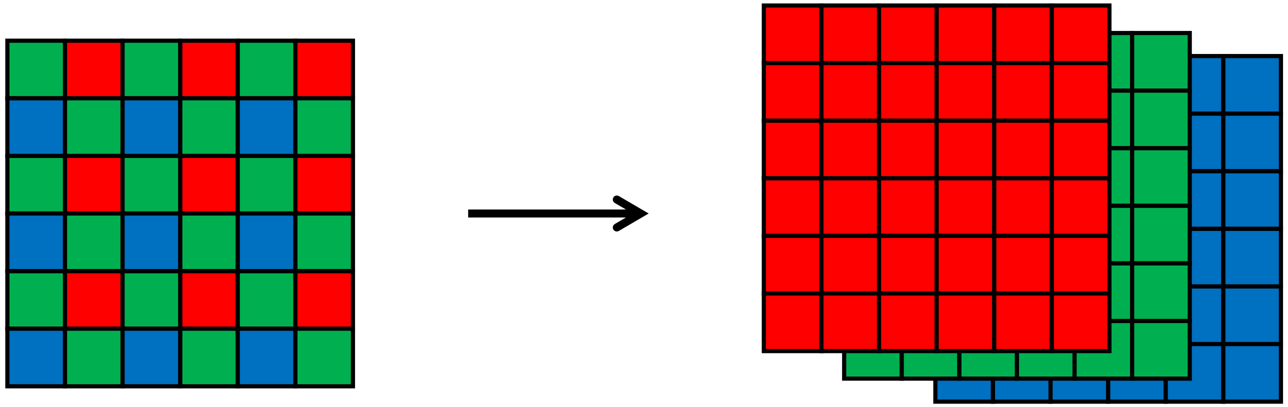


Any ideas on how to do this?

Sources: CMU 15-463, Dartmouth College CS89/189.

CFA demosaicing

Produce full RGB image from mosaicked sensor output.

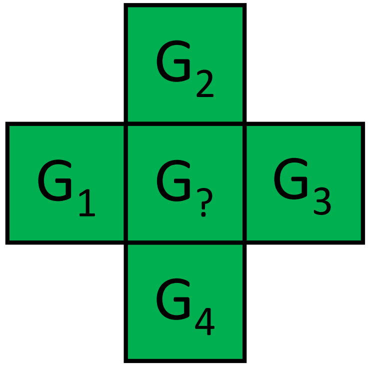


Interpolate from neighbors:

- Bilinear interpolation (needs 4 neighbors).
- Bicubic interpolation (needs more neighbors, may overblur).
- Edge-aware interpolation.

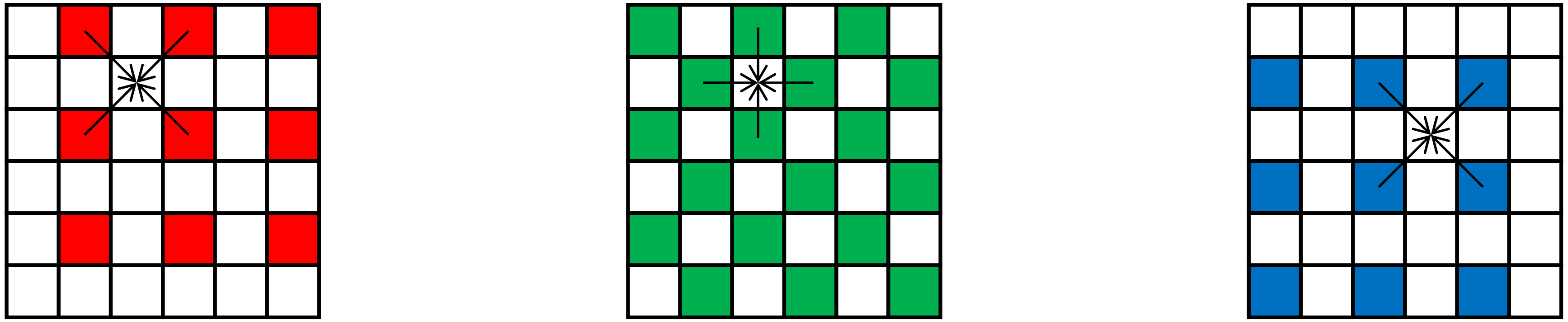
Demosaicing by bilinear interpolation

Bilinear interpolation: Simply average your 4 neighbors.



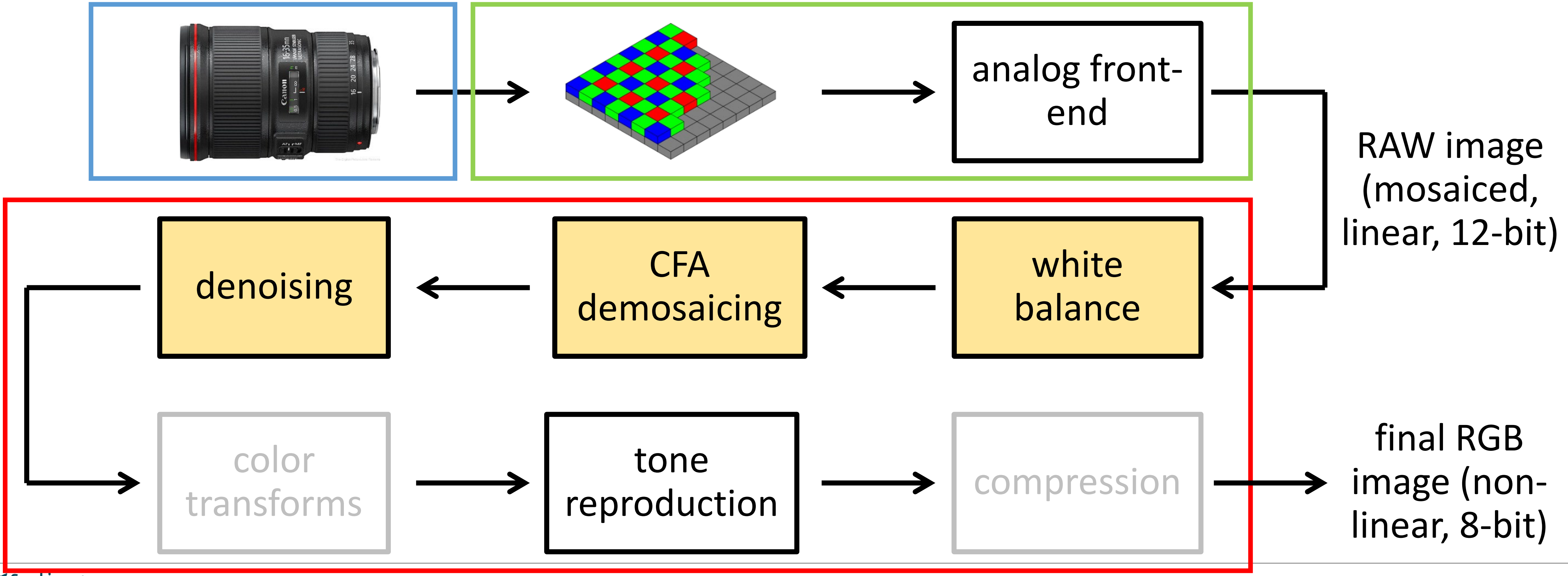
$$G_? = \frac{G_1 + G_2 + G_3 + G_4}{4}$$

Neighborhood changes for different channels:



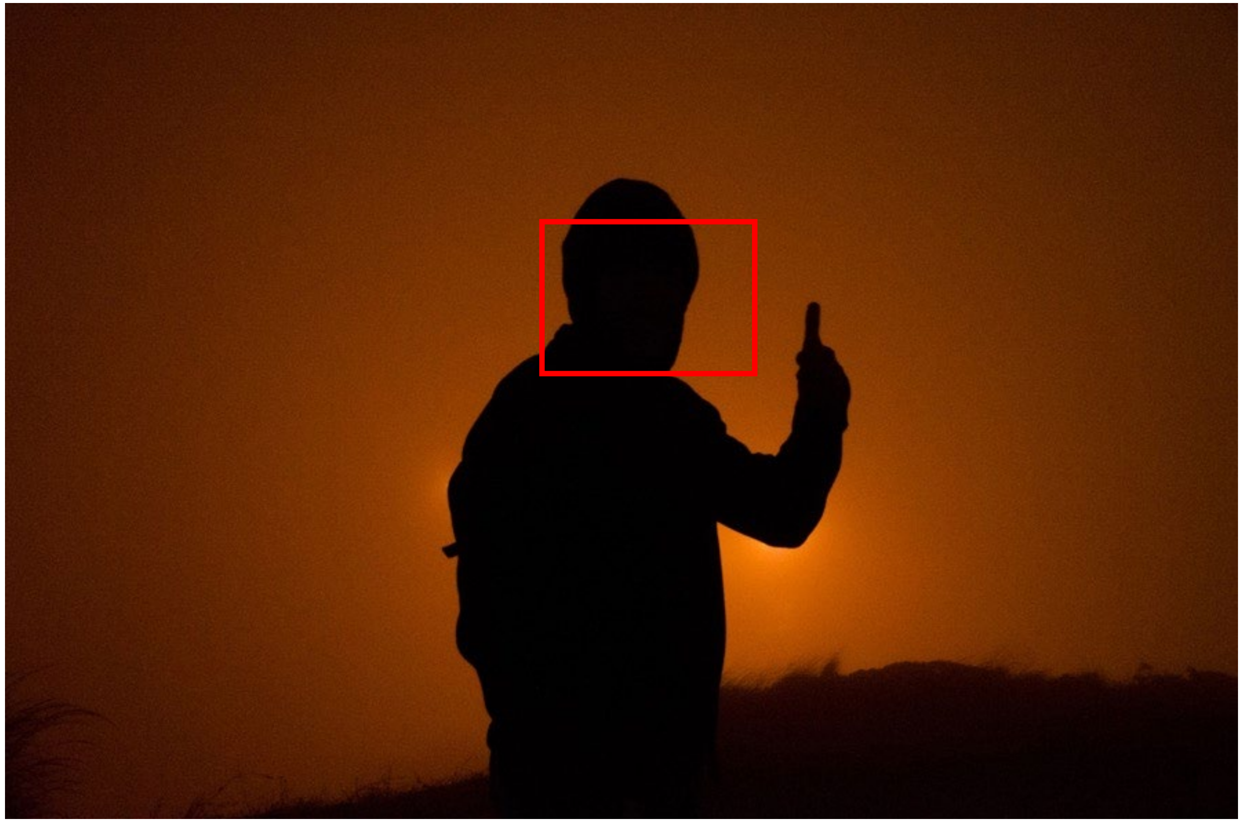
The (in-camera) image processing pipeline

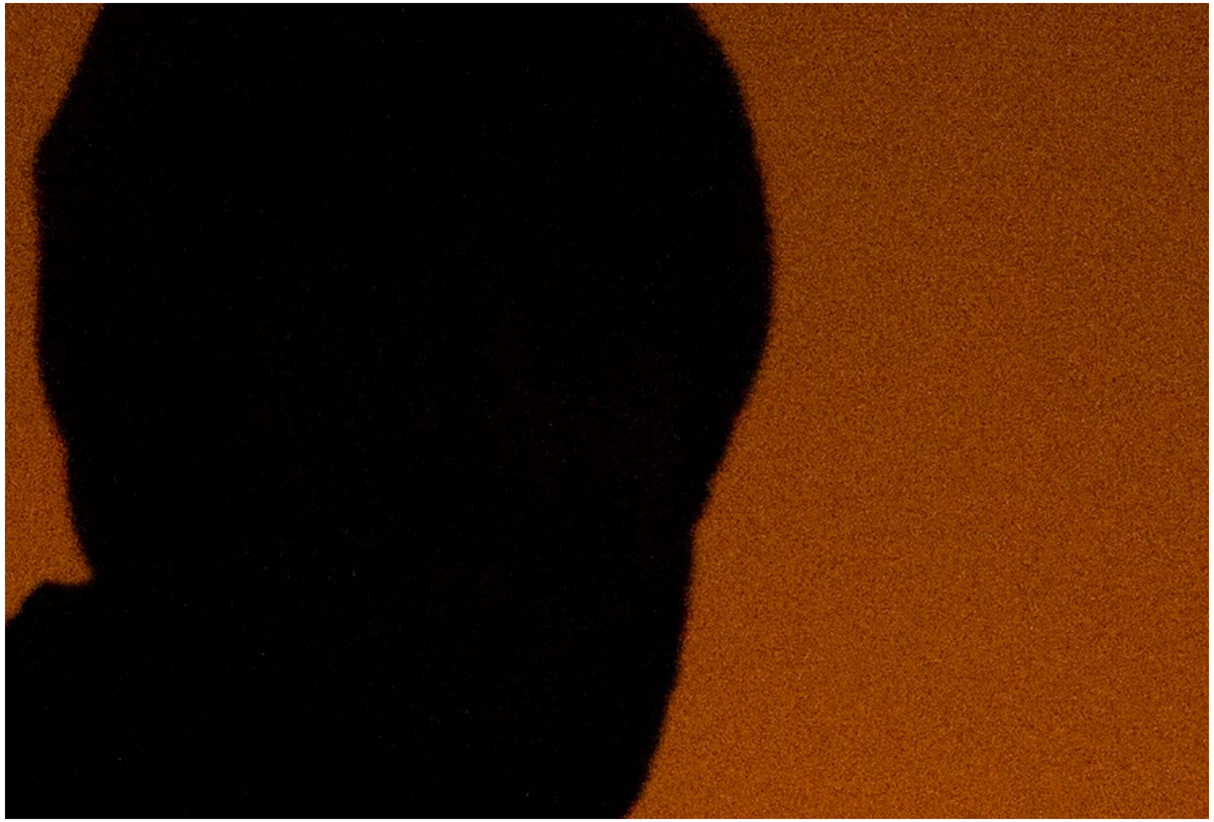
The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



Noise in images

Can be very pronounced in low-light images.





Sources: CMU 15-463, Dartmouth College CS89/189.

Three types of sensor noise

Actually many more: https://www.stemmer-imaging.com/en/knowledge-base/sensor-noise/

1) (Photon) shot noise:

- Photon arrival rates are a random process (Poisson distribution).
- The brighter the scene, the larger the variance of the distribution.

2) Dark-shot noise:

- Emitted electrons due to thermal activity (becomes worse as sensor gets hotter.)

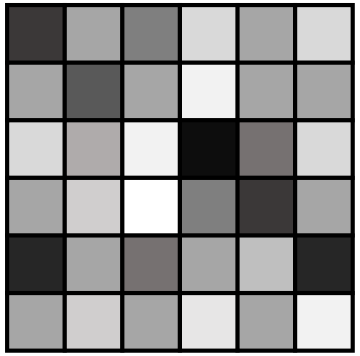
3) Read noise:

- Caused by read-out and electronics (e.g., gain, A/D converter).

Bright scene and large pixels: photon shot noise is the main noise source.

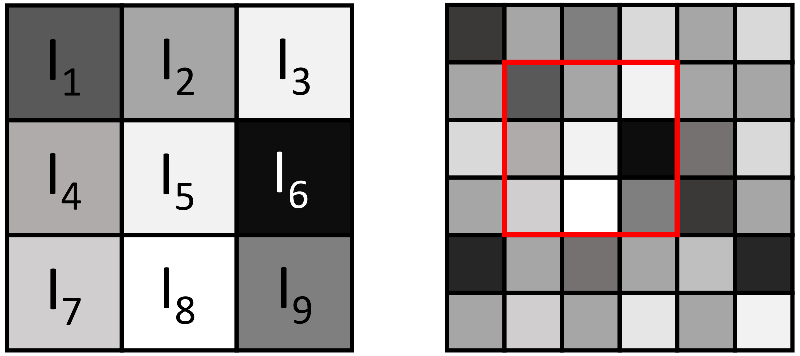
Sources: CMU 15-463, Dartmouth College CS89/189.

How to denoise?



How to denoise?

Look at the neighborhood around you.



- Mean filtering (take average):

$$I'_5 = \frac{I_1 + I_2 + I_3 + I_4 + I_5 + I_6 + I_7 + I_8 + I_9}{9}$$

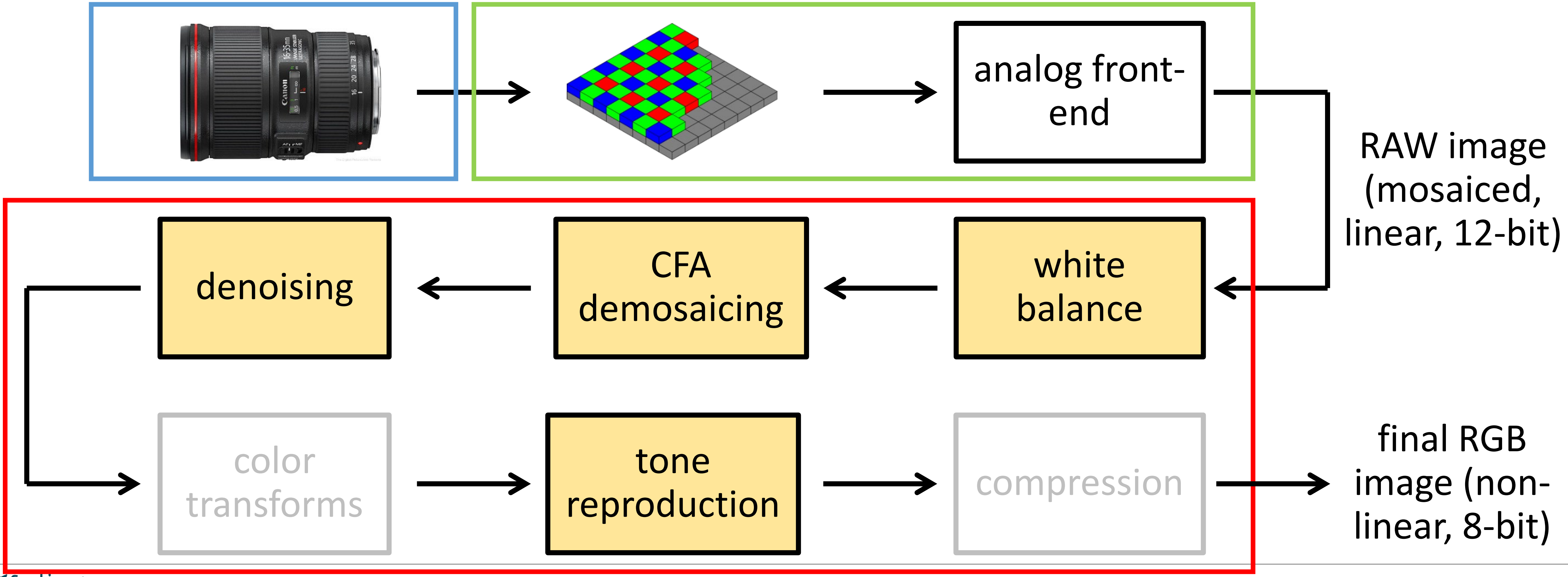
- Median filtering (take median):

$$I'_5 = \text{median}(I_1, I_2, I_3, I_4, I_5, I_6, I_7, I_8, I_9)$$

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

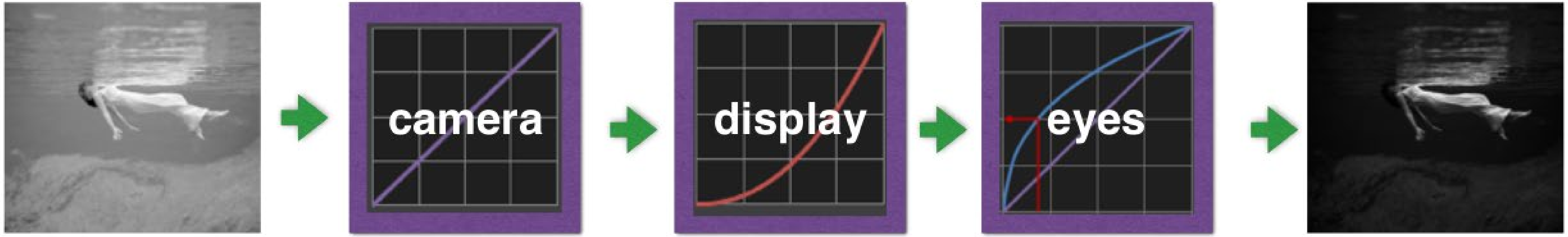
The (in-camera) image processing pipeline

The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



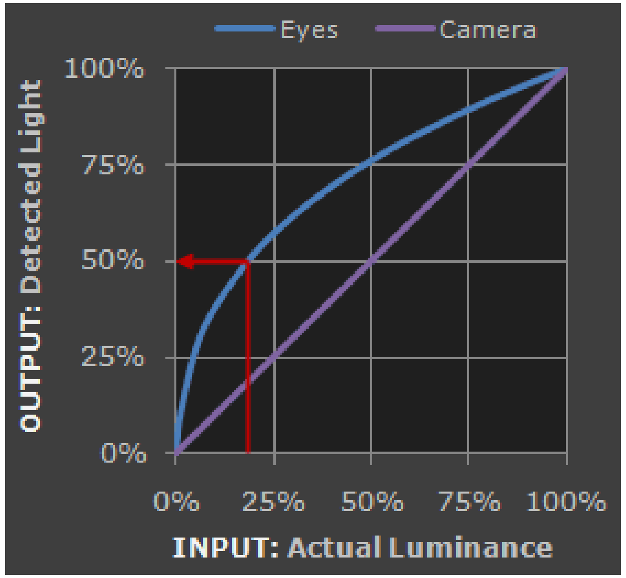
Tone reproduction

- Also known as gamma encoding (and erroneously as gamma correction).
- Without tone reproduction, images look very dark.



Why does this happen?

Perceived vs measured brightness by human eye



We have already seen that sensor response is linear.

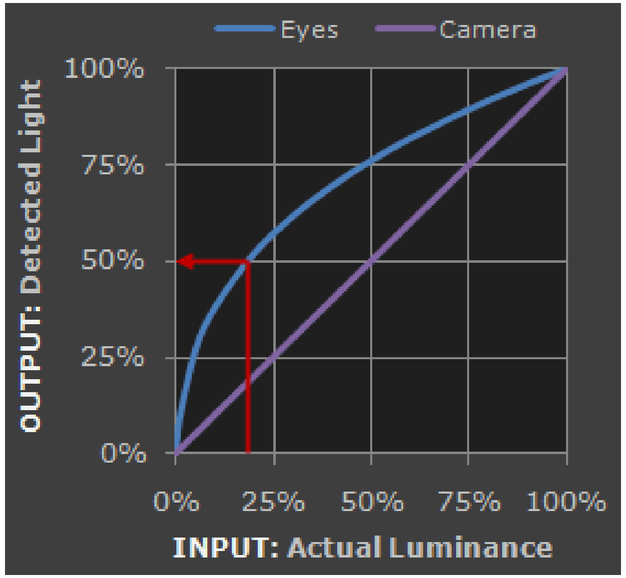
Human-eye *response* (measured brightness) is also linear.

However, human-eye *perception* (perceived brightness) is *non-linear*:

- More sensitive to dark tones.
- Approximately a Gamma function.

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Perceived vs measured brightness by human eye



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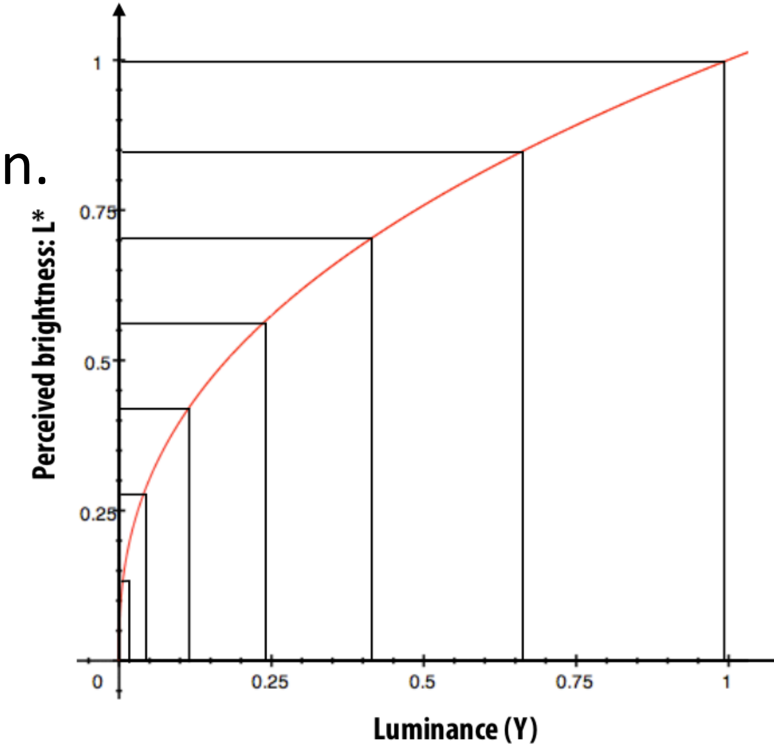
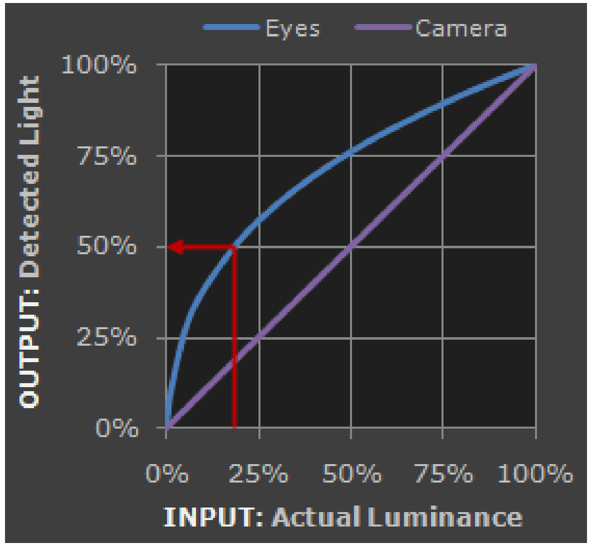
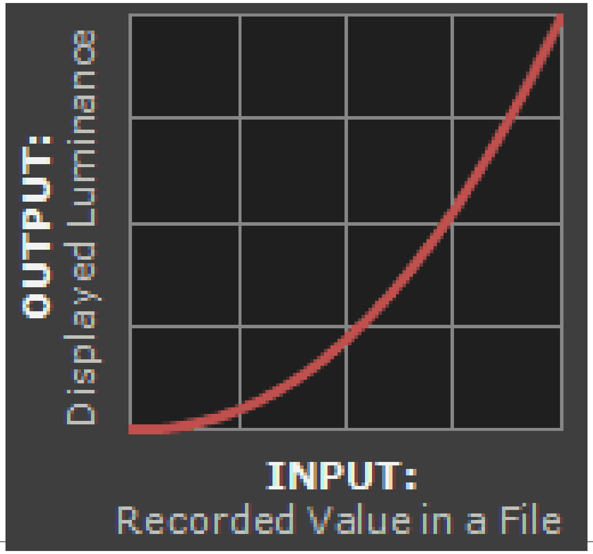


Sources: CMU 15-463, Dartmouth College CS89/189.

What about displays?





We have already seen that sensor response is linear.

Human-eye *response* (measured brightness) is also linear.

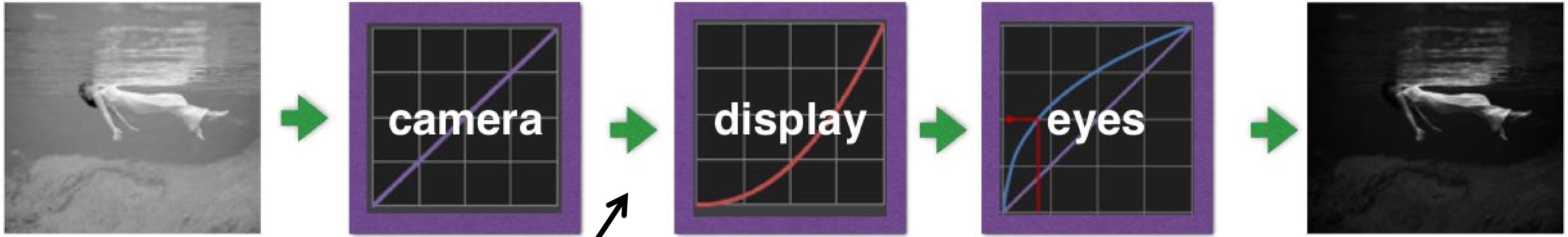
However, human-eye *perception* (perceived brightness) is *non-linear*:

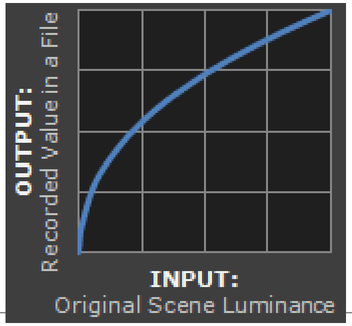
- More sensitive to dark tones.
- Approximately a Gamma function.

Displays have a response opposite to that of human perception.

Tone reproduction

- Because of mismatch in displays and human eye perception, images look very dark.





- Pre-emptively cancel-out the display response curve.
- Add inverse display transform here.
- This transform is the tone reproduction or gamma encoding.

Sources: CMU 15-463, Dartmouth College CS89/189.

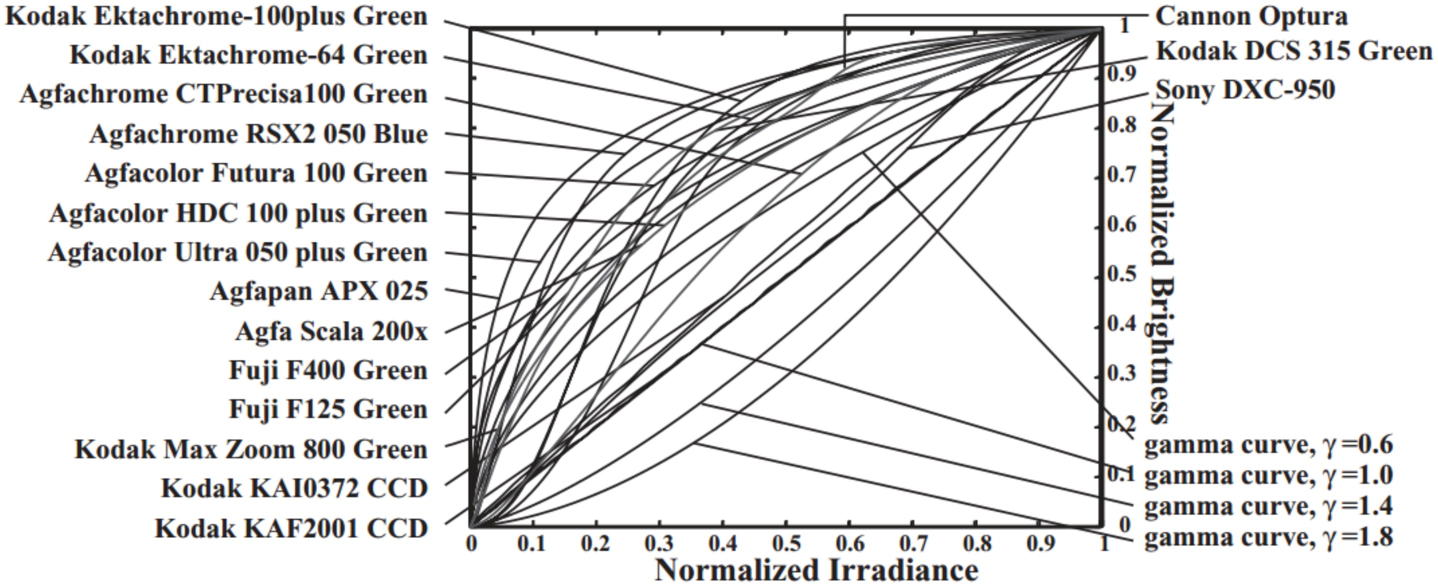
Tone reproduction curves

The exact tone reproduction curve **depends on the camera**.

- Often well approximated as L^{γ}, for different values of the power γ ("gamma").
- A good default is $\gamma = 2.2$.

before gamma

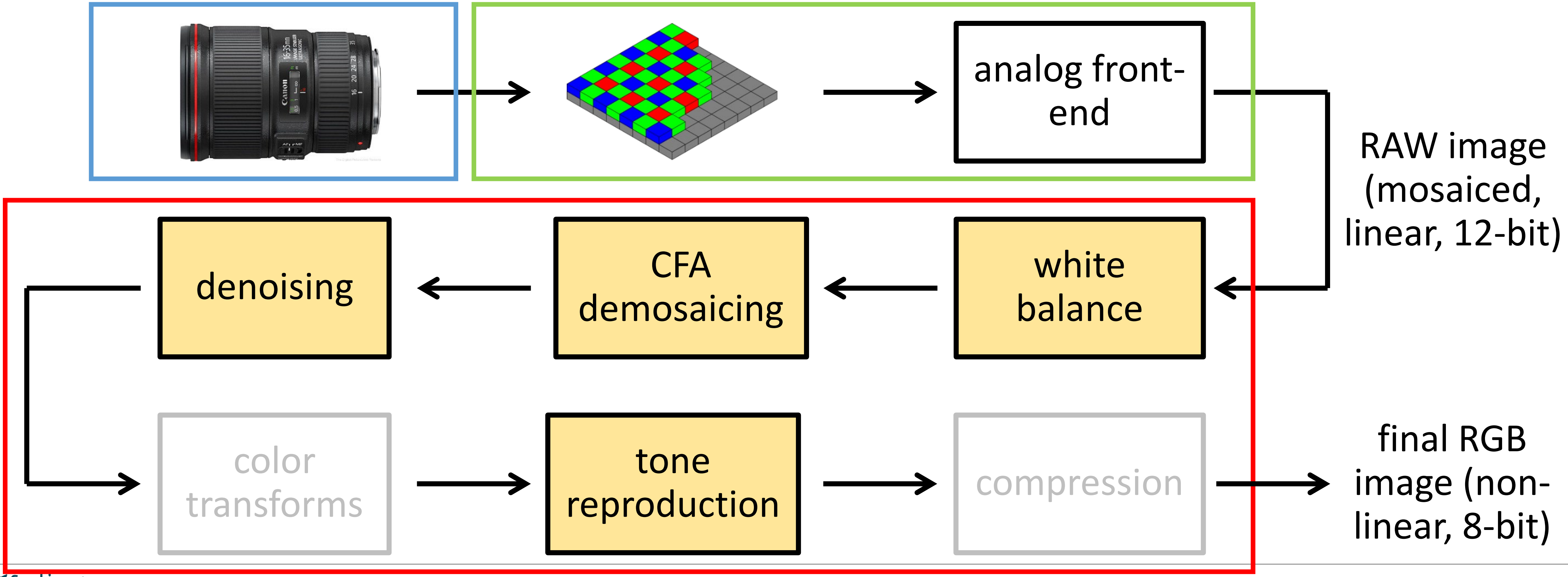
after gamma



Sources: CMU 15-463, Dartmouth College CS89/189.

The (in-camera) image processing pipeline

The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



Some general thoughts on the image processing pipeline

Do I ever need to use RAW?

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

Do I ever need to use RAW?

Emphatic yes, and that’s what you’ll use in the first lab session!

- Every time you use a physics-based computer vision algorithm, you *need linear measurements of radiance*.
- Examples: photometric stereo, shape from shading, image-based relighting, illumination estimation, anything to do with light transport and inverse rendering, etc.
- Applying the algorithms on non-linear (i.e., not RAW) images will produce completely invalid results.

Sources: CMU 15-463, Dartmouth College CS89/189.

What if I don't care about physics-based vision?

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

What if I don’t care about physics-based vision?

You often still *want* (rather than need) to use RAW!

- If you like re-finishing your photos (e.g., on Photoshop), RAW makes your life much easier and your edits much more flexible.

Is it even possible to get access to RAW images?

Sources: CMU 15-463, Dartmouth College CS89/189.

Is it even possible to get access to RAW images?

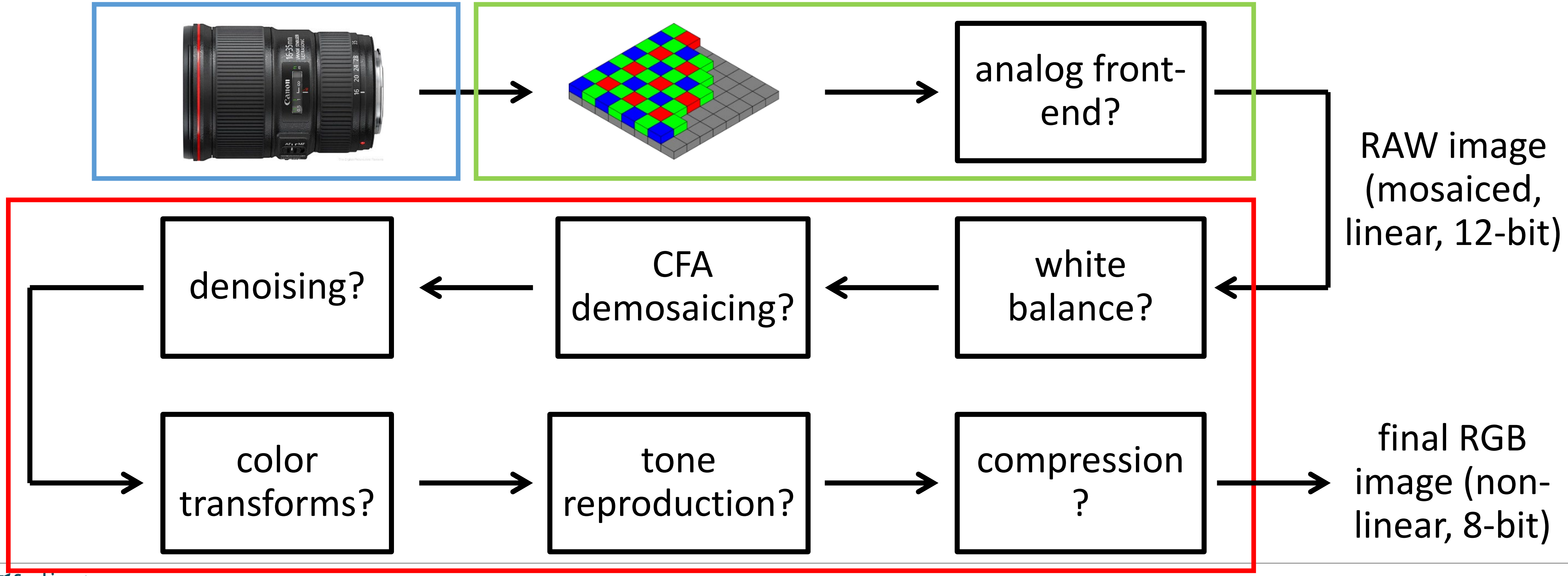
Quite often yes!

- Most high-end cameras provide an option to store RAW image files.
- Certain phone cameras allow, directly or indirectly, access to RAW.
- Sometimes, it may not be “fully” RAW. The Lightroom app provides images after demosaicking but before tone reproduction.

Sources: [CMU 15-463](), [Dartmouth College CS89/189]().

The hypothetical image processing pipeline

The sequence of image processing operations applied by the camera's image signal processor (ISP) to convert a RAW image into a "conventional" image.



Take-home messages

The values of pixels in a photograph and the values output by your camera's sensor are two very different things.

The relationship between the two is complicated and unknown.

There are many limitations in the pipeline that can be overcome with computational imaging methods.

References

Basic reading:

- Szeliski textbook, Section 2.3.
- Michael Brown, "Understanding the In-Camera Image Processing Pipeline for Computer Vision," CVPR 2016, slides available at: http://www.comp.nus.edu.sg/~brown/CVPR2016_Brown.html

Additional reading:

- Adams et al., "The Frankencamera: An Experimental Platform for Computational Photography," SIGGRAPH 2010.
  The first open architecture for the image processing pipeline, and precursor to the Android Camera API.
- Heide et al., "FlexISP: A Flexible Camera Image Processing Framework," SIGGRAPH Asia 2014.
  Discusses how to implement a single-stage image processing pipeline.
- Buckler et al., "Reconfiguring the Imaging Pipeline for Computer Vision," ICCV 2017.
- Diamond et al., "Dirty Pixels: Optimizing Image Classification Architectures for Raw Sensor Data," arXiv 2017.
  Both papers discuss how to adaptively change the conventional image processing pipeline so that it is better suited to various computer vision problems.
- Chakrabarti et al., "Rethinking Color Cameras," ICCP 2014.
  Discusses different CFAs, including ones that have white filters, and how to do demosaicing for them.
- Gunturk et al., "Demosaicking: Color Filter Array Interpolation," IEEE Signal Processing Magazine 2005
  A nice review of demosaicing algorithms.
- Chakrabarti et al., "Probabilistic Derendering of Camera Tone-mapped Images," PAMI 2014.
  Discusses how to (attempt to) derender an image that has already gone through the image processing pipeline of some (partially calibrated) camera.